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# FROM SET-VALUED TO VECTOR SPACE-VALUED DIAGRAMS: ON THE ADDITIVE IMAGE OF THE FREE FUNCTOR

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# Abstract

This thesis investigates the **free functor** from set valued representations to representations in the category of vector spaces with motivation rooted in Topological Data Analysis (TDA). The main goal is to study the **additive image** of this functor (closure of the image under direct sums and summands). Of particular interest is the choice of indexing categories for which the additive image of free equals the whole category of representations; in this case, we say that free is additively surjective.

The central focus lies on the study of additively set realizable representations (i.e., in the additive image of free) with particular emphasis on so-called indicator modules, i.e., representations that are entirely determined by their support. We provide main structural theorems characterizing small categories for which all pointwise finite-dimensional indecomposable representations are indicator representations, and those for which all indecomposable additively set realizable representations are indicator representations. Before approaching these theorems, we also study **thread quivers**, exploring their properties as a useful tool for understanding the structure of representations.

Another significant portion of this work is dedicated to **Dynkin quivers**, specifically the **quiver of type** $E_6$. We classify all orientations of $E_6$ for which the free functor is additively surjective, extending previous results to arbitrary fields.

Lastly, we analyze the representations of **rooted tree quivers**, which are directed graphs without cycles and a unique sink. We provide a geometric structure of the additive set realizable representations of such quivers, and using such tools, we give a geometric interpretation of the $E_6$ classification mentioned previously.

## Acknowledgements

I want to thank my supervisors, professors Magnus Botnan and Francesca Arici, not only for their scientific guidance but also for the great human support throughout this journey. Working on very recent and emerging topics made the research more open and flexible, with many possible directions to explore. I am grateful for the freedom they gave me to develop the thesis in the way I found most meaningful.

A special thanks goes to Gabriele, who shared most of the writing days with me and spent a lot of time answering whenever I asked, “Have you ever heard of this topic?” or “Can you listen to see if what I’m doing makes any sense?”

Thanks also to all the people I studied with at Science Park, and to my beach volleyball group, who, perhaps unknowingly, helped me stay sane during the most stressful stages of thesis writing.

Finally, I want to thank my mother, who visits me every year wherever I am in Europe, even though about Netherlands she said "Ma è troppo diverso qua" (here is too different from Italy).

Last but not least, I am deeply grateful to my girlfriend, whose constant support, despite the distance, has been indispensable to me.

# Contents

# Introduction

## Motivation rooted in Topological Data Analysis

In recent years, **topological data analysis** (TDA) has established itself as a powerful framework to analyze complex data, extracting qualitative geometric features by employing methods from topology and homological algebra. Among the tools developed within this framework, the most widely used is **persistent homology**, which allows one to study the evolution of topological features across different scales.

The *persistent homology pipeline* can be generally understood in three steps:

1. **From data to topological spaces:** From a given dataset, often equipped with a distance, one constructs a commutative diagram of topological spaces, which, in the simplest case, is just a filtration, i.e., a family of spaces

$$X_1 \hookrightarrow X_2 \hookrightarrow \cdots \hookrightarrow X_n$$

linked by inclusion maps. These are typically built from point cloud data using simplicial complexes such as Rips complexes (defined below in Definition 2). In Fig. 1, there is an example of such filtrations.

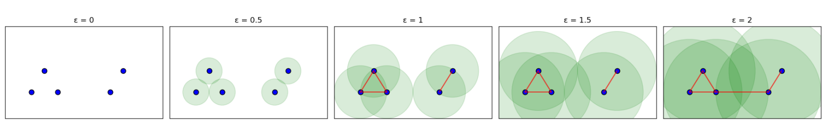



Figure 1: A simple example of a Vietoris–Rips filtration. As the scale parameter $\varepsilon$ increases (from left to right), we draw a red edge between any two points whose distance is less than or equal to $\varepsilon$. The green circles represent this distance: each circle has radius $\varepsilon$, centered at a data point. A red edge is drawn between two points precisely when one point lies inside the circle centered at the other. This process helps us track how the shape of the data changes across different scales. Image produced using Python.

2. **From topological spaces to vector spaces:** A homology functor $H_p(-;\mathbb{F})$, with coefficients in a field $\mathbb{F}$, is then applied to the commutative diagram of topological spaces. In the case of a filtration, it yields a sequence of vector spaces connected by linear maps:

$$H_p(X_1;\mathbb{F}) \to H_p(X_2;\mathbb{F}) \to \cdots \to H_p(X_n;\mathbb{F}).$$

   For $p = 0$, this process captures the evolution of connected components of the dataset, and the result is a linear representation of the commutative diagram, i.e., in the diagram, each vertex (i.e., topological space) becomes a vector space, and each arrow (i.e., inclusion) becomes a linear map. For example, applying $H_0(\mathbb{F})$ to the filtration in Fig. 1, we obtain the following linear representation.

$$\mathbb{F}^5 \xrightarrow{\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}} \mathbb{F}^5 \xrightarrow{\begin{bmatrix}1&1&1&0&0\\0&0&0&1&1\end{bmatrix}} \mathbb{F}^2 \xrightarrow{\begin{bmatrix}1&0\\0&1\end{bmatrix}} \mathbb{F}^2 \xrightarrow{\begin{bmatrix}1&1\end{bmatrix}} \mathbb{F}$$

3. **From vector spaces to descriptors:** Finally, from the resulting sequence, one extracts **invariants** (e.g., barcodes in the case of filtrations) describing the birth and death of topological features (connected components, cycles, voids, etc.) as the filtration progresses. These descriptors are then used in tasks such as classification or visualization.

To better understand the homology pipeline, we recall the definition of a simplicial complex.

**Definition 1.** *A $k$-**simplex** is the convex hull of $k+1$ affinely independent points in a Euclidean space. For instance:*

- *a $0$-simplex is a point,*
- *a $1$-simplex is a line segment,*
- *a $2$-simplex is a filled triangle,*
- *a $3$-simplex is a filled tetrahedron.*

*A **face** of a $k$-simplex is any convex hull generated by a nonempty subset of its $k+1$ vertices. For example, a triangle has edges and vertices as its faces.*

*A **simplicial complex** is a finite collection of simplices that satisfies the following two properties:*

1. *Every face of a simplex in the complex is also in the complex;*

2. *The intersection of any two simplices is either empty or a face common to both.*

Simplicial complexes allow us to model spaces by gluing together simplices along shared faces. In topological data analysis, they are typically constructed from point cloud data using distance based rules, such as in the Rips filtrations.

Summarizing, simplicial complexes are generalized graphs that, apart from vertices and edges, can include also triangles, tetrahedrons and simplices of higher dimensions. In Fig. 2 we can see examples of 0-simplex, 1-simplex, 2-simplex and 2-simplicial complex.

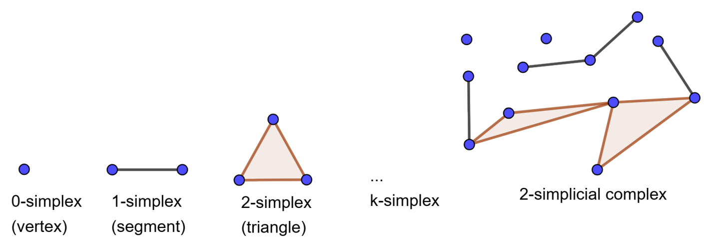



Figure 2: Examples of simplices of dimensions 0, 1 and 2, and example of simplical complex of dimension 2. Image produced with GeoGebra.

To transform a data set into simplicial complexes, one of the most used construction is the Rips filtration, which is defined as follows.

**Definition 2.** *Let $X \subset \mathbb{R}^n$ be a finite set of points equipped with the Euclidean distance $d$, and let $r \in [0, \infty)$. The* ***Vietoris–Rips complex*** $\mathrm{VR}_r(X)$ *is the simplicial complex whose $k$-simplices are the convex hulls of $(k+1)$-tuples of points $\{x_0, \ldots, x_k\} \subseteq X$ such that*

$$d(x_i, x_j) \le r \quad \text{for all } i, j.$$

*The* ***Rips filtration*** *is the family of simplicial complexes $\left(\mathrm{VR}_r(X)\right)_{r\ge 0}$, ordered by inclusion as $r$ increases.*

To visualize better this process, consider a finite filtration of simplicial complexes:

$$K_1 \subseteq K_2 \subseteq \cdots \subseteq K_m.$$

By applying homology in degree zero with field coefficients, we obtain a sequence of vector spaces:

$$H_0(K_1; \mathbb{F}) \to H_0(K_2; \mathbb{F}) \to \cdots \to H_0(K_m; \mathbb{F}).$$

Such one parameter linear representations decomposes into a direct sum of interval modules, that are representations of the form

$$0 \longrightarrow 0 \longrightarrow \mathbb{F} \xrightarrow{Id} \mathbb{F} \xrightarrow{Id} \mathbb{F} \dashrightarrow \mathbb{F} \longrightarrow 0 \dashrightarrow 0 \ .$$

Each such module corresponds to a topological feature persisting across a contiguous range of indices. These intervals are commonly represented using a **barcode**, a collection of horizontal bars where each bar is born at the index where the feature appears (i.e., the summand becomes non-zero), and dies where it vanishes again. In Fig. 3, we see an example of filtration and the corresponding barcode. The filtration used in the figure is the Rips filtration.

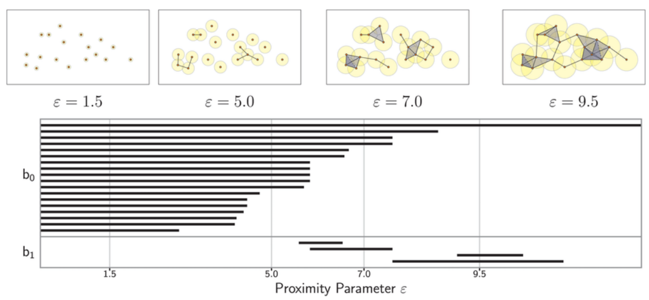



Figure 3: An example of barcodes representing the persistent modules for $H_0$ and $H_1$. The data set (left, at filtration parameter =1.5) is filtered using the Rips filtration. At this stage, the parameter is too low to form any simplices of dimension higher than 0. Image from ([1, Introduction])

This is the classical one-parameter case. However, many real-world datasets exhibit more complexity and require *multi-parameter filtrations*, such as when the data is filtered simultaneously by time and density, or by multiple scalar functions. In such cases, the filtration becomes indexed by a poset $P$ that is not totally ordered, but a product of total orders $T_1 \times T_2 ... \times T_n$, leading to **multidimensional persistence modules**, for which we refer to [12] for a more detailed discussion.

A significant complication arises when passing from one to multiple parameters: unlike the one-dimensional case, the category of representations of $P$ over a field $\mathbb{F}$ often has **wild representation type**, meaning that no complete classification of indecomposables exists. As shown by Carlsson and Zomorodian ([6, Theorem 2]), even in degree one homology, any representation of a grid-shaped poset can arise from applying homology in degree 1 to a bifiltration of spaces.

In contrast, homology in degree zero behaves more tamely. Indeed, as observed in [20, Proposition 3.1], any representation arising from homology in degree 0 factor through the category of sets and vice versa. More formally, if $D\colon P \to \mathrm{Top}$ is a diagram of topological spaces, then $H_0(D;\mathbb{F})$ factors as $\mathrm{free} \circ \pi_0(D)$, where $\pi_0$ assigns to each space the set of its path components, and free assigns to a set $T$ the free vector space with basis $T$. On the other hand, if $D : P \to \mathrm{set}$ is a diagram of sets, then $\mathrm{free}(D)$ factors as $H_0 \circ \mathrm{disc}(D)$, where disc is the functor taking a set to the corresponding discrete topological space.

In addition to the filtrations arising from simplicial complexes, such as the Rips one, another natural and widely used source of filtered diagrams comes from **hierarchical clustering methods**. We refer to [19] for a more detailed discussion. Clustering methods are techniques that group data points based on their pairwise similarities, typically encoded via a distance on a finite set $M$ of data points.

One of the most common examples is **single linkage clustering**, which proceeds as follows: for each scale parameter $\varepsilon \geq 0$, connect all pairs of points whose distance is at most $\varepsilon$. The resulting graph has a number of connected components, which define the clusters at that scale. As $\varepsilon$ increases, clusters merge, and this process yields a nested sequence of partitions of the dataset.

This construction is naturally functorial: one obtains a functor

$$C(M,d)\colon [0,\infty) \to \mathrm{Set}$$

which assigns to each scale $\varepsilon$ the set of clusters at that scale, and to each inclusion $\varepsilon \leq \varepsilon'$ the map that merges clusters accordingly. In other words, $C(M,d)$ is a diagram of sets describing how clusters evolve across scales. Post-composing this diagram with the free functor, yields

$$\mathrm{free} \circ C(M,d)\colon [0,\infty) \to \mathrm{Vect}_{\mathbb{F}},$$

which is exactly the same as computing $H_0$ of the corresponding filtered simplicial complex (in this case, the Rips complex) over the field $\mathbb{F}$. This reinforces the interpretation of clustering as a topological operation detecting connected components.

We now illustrate this procedure with an explicit example: we start with five points and a distance function, compute the single linkage clustering, and apply free to obtain a linear representation.

**Example 1.** *In this example, we analyze a single linkage clustering on five points in a metric space. The pairwise distances between these points are shown in the following table and serve as input data for the clustering procedure.*

| | A | B | C | D | E |
|---|---|---|---|---|---|
| A | 0 | 1 | 5 | 6 | 7 |
| B | 1 | 0 | 4 | 6 | 7 |
| C | 5 | 4 | 0 | 2 | 3 |
| D | 6 | 6 | 2 | 0 | 1 |
| E | 7 | 7 | 3 | 1 | 0 |

Table 1: Distance matrix between points $A, B, C, D, E$

*The next diagram, called* ***dendrogram****, graphically represents the outcome of the single linkage clustering procedure. At each increasing scale parameter $\varepsilon$, clusters whose minimal distance is at most $\varepsilon$ are merged. The horizontal lines indicate the merging events between clusters, while vertical lines connect individual points or clusters. The labels $F, G, H, I$ denote the clusters formed progressively: first $A$ and $B$ at the same time of $D$ and $E$, next $C$ joins the cluster $G$, and finally all clusters merge into $I$.*

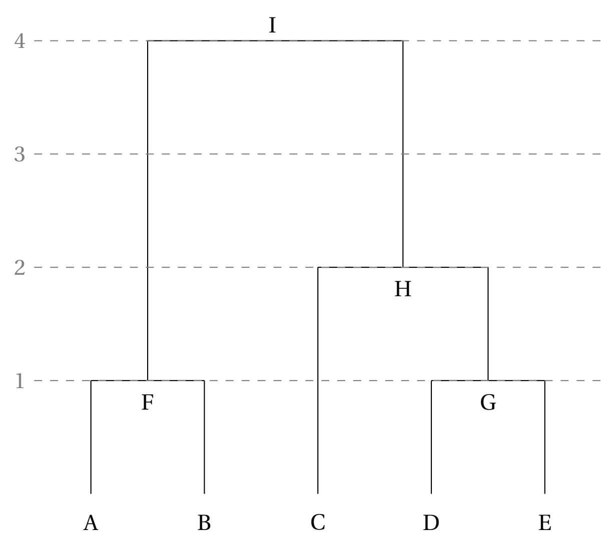



Figure 4: Dendrogram representing the result of a simple linkage clustering

*This nested sequence of clusters defines a functor*

$$C(M,d)\colon [0,\infty) \to \mathrm{Set},$$

*which assigns to each scale the corresponding set of clusters. Composing this with the* free *functor yields a filtration of vector spaces*

$$\mathrm{free} \circ C(M,d) \colon [0,\infty) \to \mathrm{Vect}_{\mathbb{F}},$$

*corresponding to the zero-th homology filtration of the five points* $A, B, C, D$ *and* $E$ *built using the Rips complexes. In the diagram below, we illustrate this filtration by restricting to a finite set of scale values* $\{0,1,2,3,4\} \subset [0,\infty)$*, corresponding to the natural numbers up to the final merging event.*

$$\mathbb{F}^5 \xrightarrow{\begin{bmatrix}1&1&0&0&0\\0&0&1&0&0\\0&0&0&1&1\end{bmatrix}} \mathbb{F}^3 \xrightarrow{\begin{bmatrix}1&0&0\\0&1&1\end{bmatrix}} \mathbb{F}^2 \xrightarrow{\begin{bmatrix}1&0\\0&1\end{bmatrix}} \mathbb{F}^2 \xrightarrow{\begin{bmatrix}1&1\end{bmatrix}} \mathbb{F}$$

This framework can be extended to include additional parameters. For instance, in *density-aware clustering*, one may first filter the data by a density threshold $\delta$ before applying the clustering algorithm at each scale $\varepsilon$. The resulting bifiltration gives rise to a diagram

$$C(M,d) \colon [0,\infty)^2 \to \mathrm{Set},$$

which, after linearization, yields a two-parameter persistence module. The structure maps often reflect the fact that clustering maps are surjective (clusters merge but do not split), which translates algebraically into morphisms that are surjective in one or both directions.

Both factorizations, from clustering methods or from $H_0$ computed as result of the homology pipeline, imply that studying the image of $H_0(-;\mathbb{F})$ reduces to studying the functor

$$\mathrm{free} \colon \mathrm{Rep}(P, \mathrm{Set}) \to \mathrm{Rep}(P, \mathrm{Vect}_{\mathbb{F}}),$$

which allows for a categorical and algebraic approach to persistent homology. In this thesis, we focus on this categorical viewpoint: understanding which representations of posets (or more generally, quivers (directed graph) and small categories) can arise from zero-degree persistent homology. In particular, by analyzing the essential image of the free functor and its closure with respect to direct sums and direct summands (the so called additive image), we aim to characterize which linear representations can arise from filtered diagrams of spaces through $H_0$. We will specifically focus on classifying small categories based on structural properties that affect the behavior of indicator representations (analogs of interval modules), with the goal of extending barcode-based descriptions. Specific attention will also be given to Dynkin quivers, thread quivers (quivers where arrows are replaced by totally ordered sets), and categories admitting a terminal object.

## Contributions and structure

In the **first chapter**, we establish the notation and present the theoretical framework needed for the rest of the thesis. We begin by recalling the definitions of **quivers**, i.e., directed graphs, and small categories, and by introducing the concept of connectedness within a category. We also provide a categorical perspective on quivers and **posets**. We then define, given a small category $X$, the **notion of representation as a functor** from $X$ to a category $C$, with a focus on $C$ equal to the category of sets or vector spaces. However, some results will concern specifically **pointwise finite-dimensional representations**, i.e., representations of a category $X$ in the category of finite-dimensional vector spaces. This is followed by the introduction of the central notion of **additive realizability**, along with some preliminary results and examples. We define the **additive image** of an additive functor as the closure of the image under direct sums and direct summands. We present the **functor** free from representations in the category of sets to representations in the category of vector spaces over $\mathbb{F}$ as the functor that maps each set $S$ in a vector space with basis $S$ and we do some first examples of representations in the additive image of free (said **additive set realizable**).

The preliminaries continue with a discussion of the **left Kan extension**, which will be used to relate representations of a subcategory to those of a larger one. We conclude with an overview of Dynkin quivers, recalling their role in the classification of indecomposable representations and anticipating some examples where the free functor either fails or succeeds in being **additively surjective** (i.e., all objects of the target category are in the additive image).

In the **second chapter**, we study **thread quivers**, which are categories obtained from quivers by replacing arrows with totally ordered sets. We will show that, when restricting our attention to pointwise finite-dimensional representations, the representations of a thread quiver defined using only finite total orders *well describe*, in a sense made precise in the chapter, the representations of thread quivers defined with arbitrary total orders.

In particular, we prove in Theorem 2.2.1 that if a representation $R$ of a thread quiver $X$ is indecomposable, it is well described by a subcategory $X'$ of $X$ and $R|_{X'}$ is indecomposable too. Moreover, we provide the following new result regarding the additive set realizability of thread quiver representations.

**Result A.** *(Proposition 2.2.6) Let $R$ be a vector space representation of a thread quiver $X$ that is well described by a subcategory $X'$. Then, $R$ is additive set realizable if and only if $\mathrm{res}|_{X'}(R)$ is additive set realizable.*

These results are particularly useful: in many situations, studying the representations of a thread quiver $C$ is equivalent to studying the representations of all its discrete subcategories. Indeed, they will play a key role later, in the discretization of $G$-posets and $B$-posets in the proof of Theorem 3.2.1.

In the third chapter, we prove two important results of this thesis regarding the **indicator representations**, i.e., representations that assign a 1-dimensional vector space (often the base field $k$) to each object, and either the identity map (when it is possible) or the zero map to each arrow. Below, we can see an example with a quiver $Q$, one of its indicator representations $R$, and one of its indecomposable representations that is not an indicator representation. The quiver in this example is a $D_4$-quiver (a type of Dynkin quiver which appears frequently throughout this thesis). The first result characterizes the class of small categories for which every pointwise finite-dimensional category is an indicator representation.

$$\begin{tikzcd} Q & \bullet & \\ \bullet \arrow[r] & \bullet \arrow[u] \arrow[r] & \bullet \end{tikzcd}$$

$$\begin{tikzcd} R & 0 & \\ 0 \arrow[r, "0"'] & \mathbb{F} \arrow[u, "0"'] \arrow[r, "\mathrm{Id}"'] & \mathbb{F} \end{tikzcd} \qquad \begin{tikzcd} S & \mathbb{F} & \\ \mathbb{F} \arrow[r, "\begin{bmatrix}1\\0\end{bmatrix}"'] & \mathbb{F}^2 \arrow[u, "\begin{bmatrix}1 & 1\end{bmatrix}"] \arrow[r, "\begin{bmatrix}0\\1\end{bmatrix}"'] & \mathbb{F} \end{tikzcd}$$

**Result B.** *(Theorem 3.2.1) Given $X$ a small category, every pointwise finite-dimensional indecomposable representation of $X$ on $\mathrm{vect}_{\mathbb{F}}$ is an indicator representations if and only if $X$ is a disjoint union of $G$-posets, a class of poset categories that generalize the* *structure of quivers of type $A_n$, and $B$-posets, posets with the following shape:*

$$\begin{tikzcd} M_0 \sim M_1 \\ m_0 \sim m_1 \arrow[u, bend left, "X_0"] \arrow[u, bend right, "X_1"'] \end{tikzcd},$$

*where the arrows $X_0$ and $X_1$ are totally ordered sets.The definitions of both types of posets are in Section 3.1.*

The second result classifies all small categories $C$ such that every pointwise finite-dimensional additive set realizable indecomposable representation is an indicator representation. Extending the result by Bauer, Botnan, Oppermann, Steen [20, Theorem 7.1]to the infinite setting.

**Result C.** *(Theorem 3.2.2) Let $X$ be a small category. Then all pointwise finite-dimensional indecomposable additively* ***Set*** *realizable representations are indicator representations if and only if $X$ is equivalent to a poset category consisting of a disjoint union of*

1. *$G-posets$,*

2. *posets with a single maximum $M$, such that the poset without the maximum is a disjoint union of $G$-posets, and for every $G$-poset $C \subseteq X \setminus \{M\}$, if $C$ has a minimum, then the minimum is unique (i.e., $C$ is a total order, or $C = X_1 \cup X_2$ with $X_1$ and $X_2$ being total orders with the same minimum).*

These results are obtained using the decomposition into interval representations for total orders representations, structural obstructions (such as $\widetilde{A}_n$, $D_4$ and Kronecker quiver configurations) and tools including Kan extensions and results about thread quivers. This chapter contributes to a deeper understanding of indicator representations of categories of arbitrary cardinality.

In the fourth chapter, we investigate the additive realizability of representations arising from Dynkin quivers, focusing in particular on the $E_6$ case. While the additively surjective behavior of the free functor is already known for Dynkin quivers of type $A_n$ and $D_n$, the situation for $E$-type quivers is still partially open. We provide the following complete classification of the orientations of $E_6$ for which the free functor is additively surjective, generalizing a known result for the field $\mathbb{F}_2$.

**Result D.** *(Theorem 4.1.1) If $X$ is a quiver which underlying graph is $E_6$, the functor* free *is additively surjective if and only if $X$ has one of the following structures:*

- *One arm of length two is completely oriented outwards,*
- *the arm of length one is oriented outwards, and at least one additional arrow is oriented outwards.*

The proof makes use of:

- an explicit analysis of the indecomposable representations corresponding to the real positive roots of $E_6$;
- dimension bounds at terminal objects (via Lemma 1.3.5);
- the fact that free remains additive surjective acting on the quiver through arrow reversals from sources of low degree and edge contractions;
- computational checks for the indecomposable representations with the largest dimensions.

This classification resolves the $E_6$ case. In addition to this specific result, the methods developed, including combinatorial tools for edge contraction and arrow reversal, representation-theoretic arguments, and computational checks for largest roots-offer a practical framework for analyzing more intricate cases, including $E_7$ and $E_8$.

Given that the classification is already known for all orientations of $A_n$ and $D_n$, this result contributes to proving the following conjecture.

**Conjecture**: For every Dynkin quiver $Q$, the orientations for which the free functor from sets representations to vector spaces representations is additively surjective does not depend on the base field $\mathbb{F}$.

We will see more about this conjecture in *Further directions* at the end of the thesis.

In the **last chapter**, we deal with posets with a terminal object with a special focus on rooted tree quivers- quivers with a terminal object whose underlying graph is a tree. We recall some results from [17] and explain how they can be applied to classify the additive image of free for such quivers. This results give an explicit construction with a geometric interpretation for the additive set realizable representation of such quiver. At the end of the chapter, we present explicit examples using specific orientations of $E_6$ and $E_7$, giving a graphical point of view to the proof of Theorem 4.1.1 for a specific orientation of $E_6$.

# Chapter 1

# Preliminaries/ Setting the stage

In this chapter, we present the main tools used throughout this thesis. We analyze some important categories, the concept of representations, and the notion of additive realizability of a representation. We then proceed with the study of the left Kan extension with a special attenction to inclusion of small categories, through which we obtain an important theorem concerning additive realizability, and Dynkin quivers.

## 1.1 Quivers, small categories and posets

In this section we will establish the framework of this thesis and present some preliminary results. We consider quivers, small categories and posets.

**Definition 1.1.1.** *A* ***quiver*** *$Q$ consists of:*

- *a set of vertices $Q_0$;*
- *a set of arrows $Q_1$;*
- *a function $s : Q_1 \to Q_0$ giving the starting point of each arrow;*
- *a function $t : Q_1 \to Q_0$ giving the target point of each $\alpha$.*

Quivers are simply directed multigraphs, possibly with multiple arrows between vertices and loops. In Fig. 1.1, we can see a simple example of a quiver with three vertices and multiple arrows between two of them.

**Definition 1.1.2.** *A category $C$ is called* ***small*** *if both the collection of objects and morphisms are sets.*

$$\begin{tikzcd} 1 \arrow[r, bend left, "\alpha"] \arrow[r, bend right, "\beta"'] & 2 \arrow[r, "\gamma"] & 3 \end{tikzcd}$$

Figure 1.1: An example of a quiver with three vertices and three arrows

For the purposes of this work, it is helpful to think of small categories as those whose objects can be regarded as mere points, without internal structure.

**Remark 1.1.3.** *Notice that any quiver can be seen as a small category. The objects are the vertices. The morphisms are all the directed paths in the quiver, i.e. sequences of consecutive arrows where the target of one arrow matches the source of the next and the composition of morphisms is given by concatenation of paths. Each object has an identity morphism, corresponding to the lazy path (the path which does nothing).*

We now define a new small category whose objects are quivers and whose morphisms are structure-preserving maps between them.

**Definition 1.1.4.** *The category* Quiv *is defined as follows:*

- $\mathrm{Obj}(\mathrm{Quiv})$ *are all quivers;*
- *for any pair of quivers $Q = (Q_0, Q_1, s, t)$ and $Q' = (Q'_0, Q'_1, s', t')$ the morphisms from $Q$ to $Q'$ are pairs*

$$(f_0 : Q_0 \to Q'_0,\ f_1 : Q_1 \to Q'_1) \text{ such that } s' \circ f_1 = f_0 \circ s \text{ and } t' \circ f_1 = f_0 \circ t$$

**Definition 1.1.5.** *Let $X$ be a category and $r$ and $s$ two its objects, we say that $r, s$ are* ***connected*** *if there exists a finite sequence of objects*

$$r = x_0, x_1, \ldots, x_n = s$$

*such that for each $i = 0, \ldots, n-1$, there exists a morphism*

$$f_i : x_i \to x_{i+1} \quad \text{or} \quad f_i : x_{i+1} \to x_i$$

*in $X$. Hence, we say that a category is connected if any two objects are connected. Moreover, for any $r \in X$, we define the* ***connected component*** *of $r$ as the full subcategory composed by all objects connected with $r$.*

This notion generalizes the concept of connectedness in a quiver (or directed graph): if we view a quiver as a category, then two vertices are connected in the usual graph-theoretic sense if and only if they are connected in the categorical sense defined above.

**Definition 1.1.6.** *Let $X$ be a small category and $r$ and $s$ two its objects, we say that $r, s$ are **strongly connected** if at least one of the following condition holds:*

- $\mathrm{Hom}_X(r,s) \neq \emptyset$,
- $\mathrm{Hom}_X(s,r) \neq \emptyset$.

The next proposition shows that a category is well described by its connected components.

**Proposition 1.1.7.** *Given a small category $X$, it is possible to choose a set $J \subset \mathrm{Obj}(X)$ such that*

$$\mathrm{Obj}(X) = \bigsqcup_{j \in J} \mathrm{Obj}(C_j)$$

*where $C_j$ denotes the full subcategory of $X$ corresponding to the connected component containing the object $j$.*

*Proof.* It is easy to check that being connected is an equivalence relation. Hence, if $r$ and $s$ are connected $C_r = C_s$ and if $r$ and $s$ are not connected $C_r \cap C_s = \emptyset$.
By this observation, it is enough to construct $J$ as a maximal subset of $\mathrm{Obj}(X)$ such that all objects in it are pairwise not connected. □

We conclude this section with a remark and two definitions concerning **partially ordered sets**, or *posets*, as they will play a central role in the classification results of Chapter 3 ( Theorem 3.2.1 and Theorem 3.2.2).

**Definition 1.1.8.** *A* partially ordered set *(or* poset*) is a set $P$ equipped with a binary relation $\leq$ satisfying the following properties for all $x, y, z \in P$:*

- ***Reflexivity:*** *$x \leq x$,*
- ***Antisymmetry:*** *if $x \leq y$ and $y \leq x$, then $x = y$,*
- ***Transitivity:*** *if $x \leq y$ and $y \leq z$, then $x \leq z$.*

For finite posets, we introduce the concept of a Hasse diagram. A **Hasse diagram** is a graphical representation of a finite poset. In a Hasse diagram, elements of the poset are represented as vertices, and a line segment is drawn upwards from $x$ to $y$ if $x \leq y$ and there is no element $z$ such that $x \leq z \leq y$ (i.e., $y$ covers $x$). All other relations are implied by transitivity and are not explicitly drawn.

**Remark 1.1.9.** *Any poset $P$ can be viewed as a category, which we will call its* associated poset-category, *in the following way:*

- *the objects are the elements of $P$,*
- *for $x, z \in P$, the morphism set is*

$$\mathrm{Hom}(x,z) = \begin{cases} \{\alpha_{xz}\} & \text{if } x \leq z, \\ \emptyset & \text{otherwise.} \end{cases}$$

*This category has at most one morphism between any two objects, and the composition of morphisms corresponds to transitivity of the order.*

*Conversely, any category $C$ satisfying the following conditions can be regarded as a poset:*

- *for any two objects $A, B \in C$, there is at most one morphism between them, i.e.,* $|\mathrm{Hom}(A,B)| \leq 1$,
- *if there exist morphisms $A \to B$ and $B \to C$, then there is a morphism $A \to C$ (by composition).*

*We call such a category a* poset-category, *and we define a partial order on its objects by declaring $A \leq B$ if* $\mathrm{Hom}(A,B) \neq \emptyset$.

**Examples** of posets relevant to this thesis include:

- **totally ordered sets**, which correspond to sets where for each pair $x, y$, $x < y$ or $y < x$ like $\mathbb{Z}$ and $\mathbb{R}$;
- **rooted tree quivers**, where the ancestor relation in the tree gives the partial order and there is a unique maximal element. For example, consider the quiver below.

$$\begin{tikzcd}
 & & r & \\
 & a \arrow[ur] & & b \arrow[ul] \\
c \arrow[ur] & & d \arrow[ul] &
\end{tikzcd}$$

Here, the partial order on vertices is given by $x \leq y$ if there exists a (directed) path from $x$ to $y$ in the quiver. For example, $c \leq r$ since there is a path $c \to a \to r$.

Rooted tree quivers of this form appear frequently throughout the thesis. The final chapter is devoted to the study of their associated poset-categories and the classification of their additive set realizable representations (see Definition 1.2.2).

Another important example of poset is the complete total orders, i.e. totally ordered set that has no "gaps".

Formally, a totally ordered set $S$ is **complete** if every non-empty subset of $S$ that has an upper bound also has a least upper bound (supremum) within $S$. This property ensures there are no "holes" in the ordering. For example, while the set of rational numbers $\mathbb{Q}$ is a total order, it is not complete because subsets like $\{q \in \mathbb{Q} \mid q^2 < 2\}$ have upper bounds but no least upper bound *in* $\mathbb{Q}$ (i.e., $\sqrt{2}$ is missing). The set of real numbers $\mathbb{R}$, on the other hand, is a complete total order. We recall that any totally ordered set can be embedded in a complete totally order set, using the so called *Dedekin-MacNeille Completion.*

## 1.2 Representations

For a small category $X$ and a category $C$, a **representation** of $X$ in $C$ is a functor $R: X \to C$. To denote the image of any object $x$ and morphism $\alpha$ in $X$, we will use $R_x$ and $R_\alpha$. Moreover, if $X$ is a poset-category and $x \leq y$, we will denote the unique morphism between $R_x$ and $R_y$ as $R_{xy}$.

We let $\mathrm{Rep}(X, C)$ denote the associated functor category, where the objects are the representations of $X$ in $C$ and morphisms are natural transformations of functors. Concretely, for any pair of representations $M$ and $N$ a morphism from $M$ to $N$ is given by a set of morphisms $\{f_x : M_x \to N_x \quad \forall\, x \in X\}$ in $C$ such that $\forall \alpha \in \mathrm{Hom}_X(x, y) \forall x, y \in \mathrm{Obj}(X)$ the following diagram commutes.

$$
\begin{tikzcd}
M_x \arrow[r, "M_\alpha"] \arrow[d, "f_x"'] & M_y \arrow[d, "f_y"] \\
N_x \arrow[r, "N_\alpha"'] & N_y
\end{tikzcd}
\tag{1.1}
$$

It is easy to check that if the map $f_x : M_x \to N_x$ is an invertible morphism for any $x$, the diagram above commutes also for $g$ defined with $\{f_x^{-1} : N_x \to M_x \quad \forall\, x \in X\}$; this means that $g$ is a morphism from $N$ to $M$ and that $g = f^{-1}$. Thus, $f$ is an isomorphism if and only if it is an isomorphism point-wise. In the thesis, the category $C$ will always be one of the following:

- Set: the category of sets;
- set: the full subcategory of Set whose objects are all finite sets;
- $\mathrm{Set}_*$: the category of pointed sets (sets with a distinguished element), with morphisms being set maps that send distinguished points to distinguished points;
- $\mathrm{set}_*$: the full subcategory of $\mathrm{Set}_*$ consisting of finite pointed sets;
- $\mathrm{Vect}_{\mathbb{F}}$: the category of vector spaces over the field $\mathbb{F}$;
- $\mathrm{vect}_{\mathbb{F}}$: the full subcategory of $\mathrm{Vect}_{\mathbb{F}}$ consisting of finite-dimensional vector sapces.

In the case of vector space representations, the field $\mathbb{F}$ will be specified only when it plays a role.

In this thesis, our main focus will be on the classification of indecomposable representations. We begin by recalling the following definitions concerning direct sums and indecomposability.

**Definition 1.2.1.** *Let $R$ and $S$ be two representations of a small category $X$ in a category $C$ (e.g., vector spaces or sets). The **direct sum** representation $R \oplus S$ is defined pointwise as follows:*

- *for each vertex $x \in \mathrm{Obj}(X)$,*

$$(R \oplus S)_x := R_x \oplus S_x,$$

*where $\oplus$ denotes the coproduct in $C$ (e.g., direct sum of vector spaces, disjoint union for sets);*

- *for each morphism $\alpha : x \to y$,*

$$(R \oplus S)_\alpha := R_\alpha \oplus S_\alpha : R_x \oplus S_x \to R_y \oplus S_y,$$

*defined componentwise.*

**Definition 1.2.2.** *A representation $R \in \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$ is called **indecomposable** if whenever*

$$R \cong M \oplus N,$$

*then either $M$ or $N$ is the **trivial representation**,that is a representation that assigns the zero vector space at every vertex.*

In Chapter 3, special emphasis will be placed on **indicator (or thin) representations**, those representations $R$ in $\mathrm{Vect}_{\mathbb{F}}$ for which there is a subset $S \subset Obj(X)$ such that

$$R_x = \begin{cases} \mathbb{F} & x \in S \\ 0 & x \notin S \end{cases} \qquad \forall f : x \to y \quad R(f) = \begin{cases} \mathrm{Id}_{\mathbb{F}} & x, y \in S \\ 0 & \text{otherwise} \end{cases} .$$

Interval representations are a standard example of indicator representation is in the following definition.

**Definition 1.2.3.** *([12, Definition 2.3]) An **interval** $I$ in a poset $P$ is a non-empty subset $I$ of $P$ satisfying the two following conditions:*

1. *If $s, t \in I$ and $s \le u \le t$, then $u \in I$,*
2. *If $s, t \in I$, then there are $s = u_0, \ldots, u_m = t \in I$ such that $u_i$ and $u_{i+1}$ are comparable for all $0 \le i < m$.*

*Given an interval $I$, the **interval representation** $\mathbb{F}_I$ is its indicator representation.*

**Definition 1.2.4.** *Let $R$ be a representation of the category $C$ in $\mathrm{Vect}_{\mathbb{F}}$, the **support** of $R$ is $\{c \in C$ such that $R_c \neq 0\}$*

**Definition 1.2.5.** *Let $X$ be a small category, and let $X = \bigsqcup_{j \in J} C_j$ be its decomposition into connected components, which exists by Proposition 1.1.7. For any representation $R \in \mathrm{Rep}(X, C)$, we define the **restriction to the component** $C_j$ as the representation $\mathrm{res}_{C_j}(R)$ given by:*

$$\mathrm{res}_{C_j}(R)_x := \begin{cases} R_x & \text{if } x \in C_j, \\ 0 & \text{otherwise,} \end{cases} \qquad \mathrm{res}_{C_j}(R)_f := \begin{cases} R_f & \text{if } f : x \to y \text{ with } x, y \in C_j, \\ 0 & \text{otherwise.} \end{cases}$$

*This is a well-defined representation since there are no morphisms in $X$ between different connected components.*

The next proposition is very intuitive but nevertheless essential, as it explains why in classifications of indecomposable representations (such as Theorem 3.2.2 and Theorem 3.2.1) it is enough to work with a connected category. Indeed, we will prove that any representation splits as a direct sum of representations supported on connected components. As a consequence, the support of an indecomposable representation in $\mathrm{Rep}(X, D)$ lies in a single connected component of $X$.

**Proposition 1.2.6.** *Let $X = \bigsqcup_{j\in J} C_j$ be the decomposition of a small category into its connected components. Then for any representation $R \in \mathrm{Rep}(X, D)$, there is an isomorphism*

$$R \cong \bigoplus_{j\in J} \mathrm{res}_{C_j}(R).$$

*Proof.* We construct a natural isomorphism between $R$ and the $\bigoplus_{j\in J} \mathrm{res}_{C_j}(R)$. For each object $x \in X$, there exists a unique $j \in J$ such that $x \in C_j$. By definition of $\mathrm{res}_{C_j}(R)$, we have

$$\left(\bigoplus_{j\in J} \mathrm{res}_{C_j}(R)\right)_x = \mathrm{res}_{C_j}(R)_x = R_x,$$

since all other summands are zero at $x$. Define the component of the natural transformation $\Phi$ at each object $x$ as the identity map:

$$\Phi_x : \left(\bigoplus_{j\in J} \mathrm{res}_{C_j}(R)\right)_x \to R_x, \quad \Phi_x = \mathrm{id}_{R_x}.$$

We now check naturality. Let $f : x \to y$ be a morphism in $X$. Then $x, y$ must both belong to the same component $C_j$ for some $j$, otherwise $\mathrm{Hom}_X(x, y) = \emptyset$. Within the component $C_j$, both $R(f)$ and $\mathrm{res}_{C_j}(R)(f)$ coincide, and the morphism $\Phi$ is just the identity, so the diagram

$$\begin{tikzcd}
\left(\bigoplus_{j\in J} \mathrm{res}_{C_j}(R)\right)_x \arrow[r, "\Phi_x"] \arrow[d, "(\bigoplus \mathrm{res}_{C_j}(R))(f)"'] & R_x \arrow[d, "R(f)"] \\
\left(\bigoplus_{j\in J} \mathrm{res}_{C_j}(R)\right)_y \arrow[r, "\Phi_y"] & R_y
\end{tikzcd}$$

commutes, since the vertical arrows are equal to $R(f)$ and the horizontal ones are the identity. This proves that $\Phi$ is a natural transformation.

It remains to show that $\Phi$ is an isomorphism but, for each object $x$, the component $\Phi_x$ is the identity on $R_x$, so $\Phi$ is an isomorphism of functors.

Hence, $R$ is isomorphic to the direct sum of its restrictions to the connected components of $X$. □

## 1.3 Additive realizability

We now introduce **additive realizability**, which constitutes the central notion of this thesis. We also recall some related properties presented in [20], which provides a broad and structured overview of the topic and serves as a general reference throughout this work.

**Definition 1.3.1.** *For a functor $T : A \to B$, the* ***essential image*** *is*

$$\{b \in B \text{ such that } T(a) \cong b \text{ for some } a \in A\}.$$

*Furthermore, for $B$ additive, the* ***additive image*** *of $T$ is the closure of the essential image under direct summands and direct sums. Consequently, we say that a functor $T : A \to B$ is* ***essentially (additively) surjective*** *if its essential (additive) image is $B$.*

We will focus on studying the additive image of

$$\mathrm{free} : \mathrm{Rep}(C, \mathrm{Set}) \to \mathrm{Rep}(C, \mathrm{Vect}_{\mathbb{F}}) \text{ and } \mathrm{free} : \mathrm{Rep}(C, \mathrm{set}) \to \mathrm{Rep}(C, \mathrm{vect}_{\mathbb{F}})$$

for $C$ small categories, where we consider the functor free between representations induced by the well known $\mathrm{free} : \mathrm{Set} \to \mathrm{Vect}_{\mathbb{F}}$.

We also consider the pointed version of this functor, denoted by $\mathrm{free}_* : \mathrm{Set}_* \to \mathrm{Vect}_{\mathbb{F}}$ (with the finite version $\mathrm{free}_* : \mathrm{set}_* \to \mathrm{vect}_{\mathbb{F}}$), which sends a pointed set $(S, s_0)$ to the vector space with basis $S \setminus \{s_0\}$, and the distinguished point $s_0$ is mapped to zero. This induces a functor (denoted $\mathrm{free}_*$) between representation categories over pointed sets and over vector spaces:

$$\mathrm{free}_* : \mathrm{Rep}(C, \mathrm{Set}_*) \to \mathrm{Rep}(C, \mathrm{Vect}_{\mathbb{F}})$$

, which acts on a representation $R$ by applying $\mathrm{free}_*$ objectwise. For example, let $C$ be the category with one object and one identity morphism. A representation $R \in \mathrm{Rep}(C, \mathrm{Set}_*)$ is just a pointed set $(S, s_0)$. Then $\mathrm{free}_*(R)$ is the vector space with basis $S \setminus \{s_0\}$, and the only morphism is the identity map.

Concretely, for a representation $R$ on Set, $\mathrm{free}(R)_x = \mathrm{free}(R_x)$ and $\mathrm{free}(R)_\alpha = \mathrm{free}(R_\alpha)$. We say that $R \in \mathrm{Rep}(C, \mathrm{Vect}_{\mathbb{F}})$ is **additively Set realizable** (**additively set realizable**) if it is in the additive image of free from $\mathrm{Rep}(C, \mathrm{Set})$ ($\mathrm{Rep}(C, \mathrm{set})$).

In case we consider a representation on vector spaces, we have the following notion of basis.

**Definition 1.3.2.** *Let $R$ be a representation in$\mathrm{Rep}(C, \mathrm{Vect}_{\mathbb{F}})$. A* ***basis*** *of $R$ is a family $\{B_c\}_{c \in \mathrm{Obj}(C)}$, where for each object $c \in \mathrm{Obj}(C)$, the set $B_c$ is a basis of the vector space $R_c$.*

*In other words, a basis of $R$ consists of a choice of basis for each vector space in the representation.*

The next proposition presents a very useful criterion to establish if a representation is in the additive image of free.

**Proposition 1.3.3.** *[20, Proposition 3.9 and 3.10] Let $X$ be a small category and $R \in$* $\mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$*. If $R$ has a multiplicative basis, that is a basis such that any arrow maps any basis vector to a new basis vector or zero, then it is in additive image of* free.

*Sketch.* Let us consider $R \in \mathrm{Vect}_{\mathbb{F}}$, It follows by the definition that $\mathrm{free}_*$ that $R$ is in its essential image if and only if it has a multiplicative basis. As proved in [20, Proposition 3.9], the additive image of $\mathrm{free}_*$ and free coincide. Hence, if $R$ has a multiplicative basis, it is additively set realizable. □

In the following proposition, we present a first concrete class of additively set realizable representations: the indicator representations. This property is proved in [20, Corollary 3.11] using a functorial approach, but we will give a more direct one.

**Proposition 1.3.4.** *Any indicator representation is in the additive image of* free.

*Proof.* Let us consider a small category $X$, a subset $S \subset Obj(X)$ and the associated indicator representation $I_S$; we will define a representation $R \in \mathrm{Rep}(X, set)$ such that $\mathrm{free}(R) \cong T \oplus I_S$ for some $T \in \mathrm{Rep}(X, \mathrm{vect}_{\mathbb{F}})$.

$$R_x = \begin{cases} \{1,2\} & x \in S \\ \{1\} & x \notin S \end{cases} \qquad R(x \to y) = \begin{cases} \mathrm{Id}_{\{1,2\}} & x, y \in S \\ \mathrm{Id}_{\{1\}} & x, y \notin S \\ i & x \notin S \quad y \in S \\ j & x \in S \quad y \notin S \end{cases}$$

for any $x, y \in \mathrm{Obj}(X)$ and any morphism from $x$ to $y$. The map $i : \{1\} \to \{1,2\}$ is the obvious inclusion which sends 1 to 1 and $j$ is the unique possible set function from $\{1,2\} \to \{1\}$ which maps both 1 and 2 to 1.

$$\mathrm{free}(R)_x = \begin{cases} \mathbb{F}^2 & x \in S \\ \mathbb{F} & x \in S \end{cases} \qquad \text{and}$$

$$\forall \alpha : x \to y \quad \mathrm{free}(R)(\alpha) = \begin{cases} \mathrm{Id}_{\mathbb{F}^2} & x, y \in S \\ \mathrm{Id}_{\mathbb{F}} & x, y \notin S \\ \begin{bmatrix} 1 \\ 0 \end{bmatrix} & x \notin S \quad y \in S \\ \begin{bmatrix} 1 & 1 \end{bmatrix} & x \in S \quad y \notin S \end{cases}$$

for any $x, y \in Obj(X)$ and any morphism from $x$ to $y$. We define $T$ as the indicator representation associated to all $Obj(X)$ ($T_x = F$ for any $x$ and each map is the identity). Now it is enough to check that $\text{free}(R) \cong T \oplus I_S$. We define the isomorphism $f$ as follows.

$$f_x : \text{free}(R)_x \to (T \oplus I_S)_x = \begin{cases} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} & x \in S \\ \text{Id}_{\mathbb{F}} & x \notin S \end{cases}$$

To prove that $f$ is a morphism of representations, it suffices to check that for every morphism $\alpha : x \to y$ in $X$, $(I_S \oplus T)_\alpha \circ f_x = f_y \circ \text{free}(R)_\alpha$. We consider the possible cases based on whether $x$ and $y$ belong to the subset $S$ or not:

- If $x, y \in S$, then $\text{free}(R)_\alpha$ is the identity on $F$, and so are the components of both $I_S$ and $T$. The diagram commutes trivially.
- If $x \in S$, $y \notin S$, then direct computation verifies that both sides are $\begin{bmatrix} 1 & 1 \end{bmatrix}$.
- If $x \notin S$, $y \in S$, then direct computation verifies that both sides are $\begin{bmatrix} 1 & 1 \end{bmatrix}$.
- If $x, y \notin S$, then all maps are identities, and commutativity is clear.

In each case, the required identity holds, so $f$ is a morphism of representations. Since each $f_x$ is an isomorphism of vector spaces, $f$ is an isomorphism of representations. □

Let $X$ be a small category with a terminal object $t$ and let $R$ be one of its indecomposable pointwise finite-dimensional representations; the next lemma allow us to state that if $\dim(R_t) > 1$, $R$ is not additive realizable.

**Lemma 1.3.5.** *[20, Lemma 6.5] Let $X$ be a small category with a **terminal object** $t$ and let $R$ be one of its pointwise finite-dimensional indecomposable representations. Then* $\dim(R_t) \leq 1$.

*Proof.* Suppose $R$ direct summand of $\text{free}(N)$ with $N$ a set-representation of $X$. We can decompose $N$ as follows.

$$N = \bigcup_{h \in N_t} N^h$$

with $N^h_v = N(v \to t)^{-1}(h)$. Hence, we obtain

$$\text{free}(N) = \bigoplus_h \in \text{free}(N^h).$$

Because $R$ is finite-dimensional, there is a unique decomposition into indecomposables; hence, $R$ is a direct summand of $N^h$ for some $h \in N_t$. It follows that $\dim(R_t) \leq 1$. $\square$

This lemma is particularly useful, as it immediately shows that for certain orientations of Dynkin quivers (see Section 1.5) that admit a terminal object, the free functor is not additively surjective. Indeed, if an indecomposable representation of such a category lies in the additive image of the free functor (i.e., is additively set-realizable), then its dimension at the terminal object must be at most 1. This provides a quick obstruction to additive realizability in many cases.

**Example 1.3.6.** *Let us consider the following quiver, which is a Dynkin quiver of type $D_4$ (see Definition 1.5.2).*

$$\begin{tikzcd} & 4 \arrow[d] & \\ 1 \arrow[r] & 2 & 3 \arrow[l] \end{tikzcd}$$

*By the lemma above, the following indecomposable representation is not additively set realizable because the vector space on the terminal object (the vertex* 2*) has dimension* 2*. Thus, we deduce that for this quiver* free *is not additive surjective.*

$$\begin{tikzcd} & \mathbb{F} \arrow[d, "{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}"'] & \\ \mathbb{F} \arrow[r, "{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}"'] & \mathbb{F}^2 & \mathbb{F} \arrow[l, "{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}"] \end{tikzcd}$$

## 1.4 Left Kan extensions

In this section, we provide a brief overview of the left Kan extension, starting with the definition and concluding with some properties related to additive set realizability. The main references for this section are: [5, Sections 6.1 and 6.2] for the part concerning Kan extensions, and [20, Section 6.1] for the discussion on the connection with representations and their additive set realizability.

### 1.4.1 Left Kan extension and some properties

**Definition 1.4.1.** *Let $C, D, E$ three categories and $R : C \to E$ and $K : C \to D$ two functors, a **left Kan extension** of $R$ along $K$ is a pair made by a functor $Lan_K R : D \to E$*

*and a natural transformation $\eta : R \implies Lan_K R \circ K$ such that for any other such pair ($G : D \to E$, $\gamma : R \implies G \circ K$) $\gamma$ factors uniquely through $\eta$. In the picture the first diagram is the left Kan extension, the second represents a couple $G, \gamma$ and the third is how $\gamma$ must factor.*

$$
\begin{tikzcd}
C \arrow[rr, "R"] \arrow[dr, "K"'] & \arrow[d, phantom, "\Downarrow\eta"] & E \\
& D \arrow[ur, dashed, "Lan_K R"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
C \arrow[rr, "R"] \arrow[dr, "K"'] & \arrow[d, phantom, "\Downarrow\gamma"] & E \\
& D \arrow[ur, "G"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
C \arrow[rr, "R"] \arrow[dr, "K"'] & \arrow[d, phantom, "\Downarrow\eta"] & E \\
& D \arrow[ur, "Lan_K R"] \arrow[ur, bend right, "G"'] \arrow[ur, phantom, bend right=15, "\Downarrow\exists!"] &
\end{tikzcd}
$$

Unfolding the diagrams of the above definition, for any pair of objects $x, y$ and morphism $f : x \to y$ in $C$, there is a unique pair $\beta(x), \beta(y)$ such that the following commutes.

$$
\begin{tikzcd}
R(x) \arrow[r, "R(f)"] \arrow[d, "\eta(x)"'] \arrow[dd, bend right=60, "\gamma(x)"'] & R(y) \arrow[d, "\eta(y)"] \arrow[dd, bend left=60, "\gamma(y)"] \\
Lan_K R \circ K(x) \arrow[r, "Lan_K R \circ K(f)"] \arrow[d, dashed, "\beta(x)"'] & Lan_K R \circ K(y) \arrow[d, dashed, "\beta(y)"] \\
G \circ K(x) \arrow[r, "G(f)"'] & G \circ K(y)
\end{tikzcd}
\tag{1.2}
$$

For our purposes, we will only focus on the left Kan extension, but an analogous notion of right Kan extension exists and can be developed similarly.

Let us fix the setting in which we will use the left Kan extension. Let $D$ be a small category, and $C$ a full subcategory of $D$, with $K : C \hookrightarrow D$ the canonical inclusion functor. The target category $E$ in most cases will be $\mathrm{Vect}_{\mathbb{F}}$. Hence, as functor $R$ we consider a representation $R \in \mathrm{Rep}(C, )$. To illustrate this, let us look at an example.

**Example 1.4.2.** *Define $C$ and $D$ to be the two following quivers seen as categories.*

$$
C = \quad 1 \qquad 2 \;\text{----}\; n-1 \qquad\qquad
D = \quad
\begin{tikzcd}
& & n & \\
1 \arrow[urr] & 2 \arrow[ur] \arrow[rr, dash, dashed] & & n-1 \arrow[ul]
\end{tikzcd}
$$

*Consider $K$ the natural inclusion of $C$ in $D$ and $R$ a representation of $C$ on* $\mathrm{Vect}_{\mathbb{F}}$*, which* *is simply a set of vector spaces* $\{R_i\}_{1\le i\le n-1}$. *Let us see that*

$$Lan_K R = \begin{tikzcd} & & \bigoplus_{1\le i\le n-1} R_i & \\ R_1 \arrow[urr, "i_1"] & R_2 \arrow[ur, "i_2"] \arrow[rr, dashed, no head] & & R_{n-1} \arrow[ul, "i_{n-1}"'] \end{tikzcd}$$

*where the maps* $i_j$ *are the natural embeddings and* $\eta$ *the identity, is a Left Kan extension.*
*Take a couple* $(G,\gamma)$ *with* $G$ *a representation of* $D$ *on* $Vect_F$ *and* $\gamma$ *a representation homomorphism from* $R$ *to* $G\circ K$ *(i.e. a set* $\{\gamma_i : R_i \to G_i$ *for* $1 \le i \le n\}$*). We need to check that there exists a unique* $\beta : Lan_K R \to G$ *such that the diagram in Eq. (1.2) commutes. Define*

$$\beta(i) = \gamma_i \quad \text{if} \quad 1 \le i \le n-1 \text{ and } \beta(n) = \begin{bmatrix} g_1\circ\gamma_1 & 0 & 0 & \dots & 0 \\ 0 & g_2\circ\gamma_2 & 0 & \dots & 0 \\ 0 & 0 & g_3\circ\gamma_3 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & g_{n-1}\gamma_{n-1} \end{bmatrix}$$

*It is very easy to check that it works. The uniqueness of* $\beta$ *follows from the universal property of the direct sum.*

In the previous example, it is not a coincidence that $Lan_K R$ turns out to be a colimit. This is, in fact, a general phenomenon, provided that the codomain of $K$ admits certain colimits. Before stating the theorem, let us recall the definition of the **comma category** $K \downarrow d$, where $K : C \to D$ is a functor and $d \in D$ is an object. The objects of $K \downarrow d$ are pairs $(c,h)$, where $c \in C$ and $h \in \mathrm{Hom}_D(K(c),d)$. A morphism from $(c,h)$ to $(c',h')$ is a morphism $f \in \mathrm{Hom}_C(c,c')$ such that $h' \circ K(f) = h$. We denote by $\pi_d : K \downarrow d \to C$ the canonical projection functor, which sends an object $(c,h)$ to $c$, and a morphism $f$ to itself.

**Theorem 1.4.3.** *([5, Theorem 6.2.1]) Let* $R : C \to E$ *and* $K : C \to D$ *be two functors. If, for every* $d \in D$*, the colimit*

$$\mathrm{Lan}_K R(d) := \mathrm{colim}\big(K \downarrow d \xrightarrow{\pi_d} C \xrightarrow{R} E\big) \tag{1.3}$$

*exists in* $E$*, then these colimits assemble into a functor* $\mathrm{Lan}_K R : D \to E$*, which defines* *the left Kan extension of $R$ along $K$.*

*Moreover, the natural transformation $\eta : R \Rightarrow \mathrm{Lan}_K R \circ K$ is given by the universal morphisms associated to these colimits, i.e., by the components of the colimit cones.*

*Sketch.* The first step is to extend the data in Eq. (1.3) to a functor $\mathrm{Lan}_K : D \to E$. Namely, we have to define $\mathrm{Lan}_K(g)$ for each morphism $g$ in $D$. For $g : d \to d'$ we define

$$g_* : K \downarrow d \to K \downarrow d' \qquad (c,\phi) \mapsto (c, g\circ\phi).$$

Notice that the following diagram commutes.

$$\begin{tikzcd} K\downarrow d \arrow[rr, "g_*"] \arrow[dr, "\pi_d"'] & & K\downarrow d' \arrow[dl, "\pi_{d'}"] \\ & C & \end{tikzcd}$$

Thus, the colimit cone $\Lambda^{d'} : R\circ\pi_{d'} \to \mathrm{Lan}_K K(d')$ that defines $\mathrm{Lan}_K K(d')$ can be restricted along $g_*$ and reindexed to define a cone under the diagram $R\circ\pi_d = R\circ\pi_{d'}\circ g_*$. By the univeral property of the colimit $\mathrm{Lan}_K R(d)$, there is a unique morphism $h$ such that the following commutes.

$$\begin{tikzcd} & R(c) \arrow[dl, "\lambda^d_{f:K(c)\to d}"'] \arrow[dr, "\lambda^{d'}_{f:K(c)\to d'}"] & \\ \mathrm{Lan}_K R(d) \arrow[rr, dashed, "h"'] & & \mathrm{Lan}_K R(d') \end{tikzcd}$$

We define $Lan_K R(g) := h$ and the natural trasformation $\eta$ associated to the left Kan extension as

$$\eta_c := \lambda^K_{\mathrm{Id}_{K(c)}}(c) : F(c) \to Lan_K R(K(c)).$$

It remains to prove that the pair $(\mathrm{Lan}_K R, \eta)$ has the universal property in Definition 1.4.1. Thus, we consider a pair $(G,\gamma)$ as in the second picture of the definition and we look for the unique factorization of $\alpha : \mathrm{Lan}_K R \to G$ of $\gamma$ through $\mathrm{Lan}_K R \to G$. We define $\alpha_d : Lan_K R(d) \to G(d)$ to be the unique factorization of all function $G(f)\circ\gamma_c$ (for $f : K(c) \to d$) through $\mathrm{Lan}_K R$, i.e. the unique morphism, given by the universal property of $\mathrm{Lan}_K R(d)$ such that the following diagram commutes for any pair

$(c, f) \in K \downarrow d$.

$$\begin{tikzcd} R(c) \arrow[r, "\lambda^d_{f:K(c)\to d}"] \arrow[d, "\gamma_c"] & Lan_K R(d) \arrow[d, dashed, "\alpha_d"] \\ G(K(c)) \arrow[r, "G(f)"] & G(d) \end{tikzcd}$$

For the verification of the naturality of $\eta$ and $\alpha$, as well as the uniqueness of $\alpha$, we refer to [5]. □

Looking at our setting, where $D$ is a small category, $C$ is one of its full subcategories, and $K$ is the inclusion of $C$ into $D$, it follows from the previous theorem that for any $R \in \mathrm{Rep}(C, E)$, the Left Kan extension $\mathrm{Lan}_K R$ exists for any cocomplete category $E$.

By the next theorem, it readily follows that

$$\mathrm{Lan}_K : \mathrm{Rep}(C, E) \to \mathrm{Rep}(D, E)$$

is the left adjoint to

$$\mathrm{res} : \mathrm{Rep}(D, E) \to \mathrm{Rep}(C, E),$$

the functor obtained which is obtained by precomposing a representation on $D$ with $K$.

**Theorem 1.4.4.** *( [5, Proposition 6.1.5]) Let $C, D, E$ be three categories and $K : C \to D$ a functor such that left Kan extension exists for any $R : C \to E$, then it defines a left adjoint to the precomposition functor $K^* : Fun(D, E) \to Fun(C, E)$.*

*Proof.* By Definition 1.4.1, we have that for any $R \in Fun(C, E)$ and $S \in \mathrm{Fun}(D, E)$, there is natural isomorphism

$$\mathrm{Hom}_{\mathrm{Fun}(D,E)}(\mathrm{Lan}_K R, S) \cong \mathrm{Hom}_{\mathrm{Fun}(C,E)}(R, K^*(S)).$$

This is because, for any $\gamma \in \mathrm{Hom}_{\mathrm{Fun}(C,E)}(R, K^*(S))$ there is a unique

$$\beta \in \mathrm{Hom}_{\mathrm{Fun}(D,E)}(\mathrm{Lan}_{KR}, S) \text{ such that } \gamma = \beta \circ \eta$$

and for any $\beta \in \mathrm{Hom}_{\mathrm{Fun}(D,E)}(\mathrm{Lan}_K R, S)$, we have $\gamma = \beta(K) \circ \eta$, where $\eta$ is the natural transformation associated to the $\mathrm{Lan}_K R$. For $R, R', R'' \in \mathrm{Fun}(C, E)$, $f : R' \to R$ and, we denote $f^*$ the pre-composition from $\mathrm{Hom}_{\mathrm{Fun}(C,E)}(R, R'')$ to $\mathrm{Hom}_{\mathrm{Fun}(C,E)}((R', R''')$. By the naturality of the isomorphism above,

$$\mathrm{Hom}_{\mathrm{Fun}(D,E)}(\mathrm{Lan}_K R, S) \cong \mathrm{Hom}_{\mathrm{Fun}(C,E)}(R, K^*(S)) \xrightarrow{f^*} \mathrm{Hom}_{\mathrm{Fun}(C,E)}(R', K^*(S)) \cong$$

$$\cong \mathrm{Hom}_{Fun(D,E)}(\mathrm{Lan}_K R', S)$$

defines a natural trasformation from $\mathrm{Hom}_{\mathrm{Fun}(D,E)}(\mathrm{Lan}_K R, -)$ to $\mathrm{Hom}(\mathrm{Lan}_K R', -)$, which by the Yoneda lemma must equal pre-composition by a unique morphism $\mathrm{Lan}_K f : \mathrm{Lan}_K R \to \mathrm{Lan}_K R'$. This uniqueness implies that the assignment $\mathrm{Lan}_K : \mathrm{Hom}_{\mathrm{Fun}(C,E)} \to \mathrm{Hom}_{\mathrm{Fun}(D,E)}$ is functorial. Moreover, by its universal property, $\eta$ is the unit of the adjointness $\mathrm{Lan}_K \dashv K^*$. □

### 1.4.2 From Lan to the additive realizability

The aim of this subsection is, using the left Kan extension, to obtain the following theorem.

**Theorem 1.4.5.** *([20, Corollary 6.3]) Assume that $D$ is a small category and $C$ is one of its full subcategories.*

- *If all indecomposable additively Set-realizable representations of $D$ are indicator representations, then it holds for $C$ too.*
- *If $D$ has finitely many indecomposable additively Set-realizable representations, then it holds for $C$ too.*
- *If each representations of $D$ on* $\mathrm{Vect}_{\mathbb{F}}$ *is additively Set-realizable, the it holds for $C$ too.*

The result will immediately follow by the next lemma and the next proposition. Before the following lemma, we recall that the free functor is the left adjoint of the functor forget and that the left adjoint of a composition is the composition of the left adjoints.

**Lemma 1.4.6.** *([20, Lemma 6.1]) Let $D$ be a small category and let $C$ be one of its subcategories. Let $K : C \to D$ be the canonical inclusion functor. Both the restriction functor* $\mathrm{res}\colon \mathrm{Rep}(X, \mathrm{Set}) \to \mathrm{Rep}(S, \mathrm{Set})$ *and its left adjoint* $\mathrm{Lan}_K \colon \mathrm{Rep}(S, \mathrm{Set}) \to \mathrm{Rep}(X, \mathrm{Set})$ *commute with the functor* free.

*Proof.* By their definition, it easily follow the equalities

$$\mathrm{res} \circ \mathrm{free} = \mathrm{free} \circ \mathrm{res} \text{ and } \mathrm{forget} \circ \mathrm{res} = \mathrm{res} \circ \mathrm{forget}$$

respectively from $\mathrm{Rep}(D,\mathrm{Set})$ to $\mathrm{Rep}(C,\mathrm{Vect}_{\mathbb{F}})$ and from $\mathrm{Rep}(D,\mathrm{Vect}_{\mathbb{F}})$ to $\mathrm{Rep}(C,\mathrm{Set})$. By the second equality and using that:

- the free functor is the left adjoint of the functor forget;
- the left adjoint of a composition is the composition of the left adjoints;
- $\mathrm{Lan}_K R$ is the left adjoint of the functor res;

it turns out that $\mathrm{Lan}_K \circ \mathrm{free} = \mathrm{free} \circ \mathrm{Lan}_K$ as functor from $\mathrm{Rep}(C,\mathrm{Set})$ to $\mathrm{Rep}(D,\mathrm{Vect}_{\mathbb{F}})$. □

**Proposition 1.4.7.** *[20, Proposition 6.2] A representation $R : C \to \mathrm{Vect}_{\mathbb{F}}$ is additively* Set*-realizable if and only if* $\mathrm{Lan}_K R : X \to \mathrm{Vect}_{\mathbb{F}}$ *is additively* Set*-realizable.*

*Proof.* Suppose that $R$ is a direct summand of $\mathrm{free}(T)$ with $T \in \mathrm{Rep}(C,\mathrm{Set})$. As left adjoints preserve colimits (and hence coproducts), it follows that $\mathrm{Lan}_K R$ is a direct summand of $\mathrm{Lan}_K(\mathrm{free}(T))$. Hence, by Lemma 1.4.6, $\mathrm{Lan}_K R$ is a direct summand of $\mathrm{free}(\mathrm{Lan}_K T)$.
Suppose now that $\mathrm{Lan}_K R$ is a direct summand of $\mathrm{free}(T)$ with $T \in \mathrm{Rep}(D,\mathrm{Set})$. Then $R = \mathrm{res}(\mathrm{Lan}_K R)$ is a direct summand of $\mathrm{res}(\mathrm{free}(T))$, which equals $\mathrm{free}(\mathrm{res}(T))$ by Lemma 1.4.6. □

Theorem 1.4.5 is an immediate consequence of the above proposition.

## 1.5 Dynkin Quivers

In this section we will briefly recall what simply laced Dynkin diagrams are and which is their role in representation theory. At the end of the section we will present also some property of the Dynkin quivers (see Definition 1.5.2) related to the additive set realizability. The main reference used in this section is [15].

**Definition 1.5.1.** *A graph is said to be **simply laced Dynkin diagram** if it is of type $A, D$ or $E$. Types $A$ and $D$ are two infinite series while $E_6, E_7$ and $E_8$ are the exceptional diagrams. These types of diagrams are shown Fig. 1.2.*

**Definition 1.5.2.** *A **Dynkin quiver** is a quiver which underlying graph, i.e. the same graph without any direction for the arrows, is a simply laced Dynkin diagram.*

The strong interest for Dynkin quivers is principally due to the following theorem.

$$\begin{tikzcd}[column sep=small]
A_n & 1 \arrow[r, dash] & 2 \arrow[r, dash] & 3 \arrow[r, dash] & \cdots \arrow[r, dash] & n-1 \arrow[r, dash] & n
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
 & & & & & n-1 \\
D_n & 1 \arrow[r, dash] & 2 \arrow[r, dash] & 3 \arrow[r, dash, dashed] & n-2 \arrow[ur, dash] \arrow[dr] & \\
 & & & & & n
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
E_6 & & & 4 \arrow[d, dash] & & \\
 & 1 \arrow[r, dash] & 2 \arrow[r, dash] & 3 \arrow[r, dash] & 5 \arrow[r, dash] & 6
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
E_7 & & & 4 \arrow[d, dash] & & & \\
 & 1 \arrow[r, dash] & 2 \arrow[r, dash] & 3 \arrow[r, dash] & 5 \arrow[r, dash] & 6 \arrow[r, dash] & 7
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
E_8 & & & 4 \arrow[d, dash] & & & & \\
 & 1 \arrow[r, dash] & 2 \arrow[r, dash] & 3 \arrow[r, dash] & 5 \arrow[r, dash] & 6 \arrow[r, dash] & 7 \arrow[r, dash] & 8
\end{tikzcd}$$

Figure 1.2: Simply laced Dynkin diagrams

**Theorem 1.5.3.** ***(Gabriel's Theorem)*** *Let $Q$ be a connected quiver. The category* $\mathrm{Rep}(Q, \mathrm{Vect}_{\mathbb{F}})$ *has, up to isomorphism, only finitely many indecomposable objects if and only if $Q$ is a Dynkin quiver.*

Gabriel's theorem also provides a structure to build these indecomposable representations. The complete construction is in [15, Chapter 8]. Another good reference for this theorem is the thesis [18].

For any quiver $Q$, there is the quadratic form

$$q : \mathbb{Z}^n \to \mathbb{Z} \quad q(x) = \sum_{i \in Q_0} x_i^2 - \sum_{\alpha \in Q_1} x_{s(\alpha)} x_{t(\alpha)}.$$

The positive real roots are the vectors $x \in \mathbb{Z}^n$ such that $x_i > 0 \quad \forall 0 \leq i \leq n$ and $q(x) = 1$, where $n$ is the cardinality of $Q_0$. By the proof of Gabriel's theorem, for Dynkin quivers these positive real roots are finite and there is the following one-to-one correspondence from positive real roots to indecomposable representations.

Moreover, this correspondence has the property that $(x_1,\dots,x_n)$ goes to a representation $R$ such that $R_i = \mathbb{F}^{x_i}$. Through this thesis, we will use [15, Section 8.1-8.5] for the lists of all real positive roots of Dynkin quivers.

**Remark 1.5.4.** ***Indecomposable representations of*** $A_n$
*From the classification of the positive roots of* $A_n$*, one sees that for any positive root* $x$*, we have* $x_i \le 1$ *for all* $i$*, and the set of vertices* $i$ *such that* $x_i = 1$ *must form a connected subquiver of* $Q$*. In particular, every indecomposable representation of an* $A_n$ *quiver is thin.*

*This property, besides following from Gabriel's theorem, can also be derived directly using linear algebra techniques (see [4]). From this classification, and by Proposition 1.3.4, we deduce that the free functor*

$$\mathrm{free}\colon \mathrm{Rep}(A_n, \mathrm{Set}) \longrightarrow \mathrm{Rep}(A_n, \mathrm{Vect}_{\mathbb{F}})$$

*is additively surjective for any* $A_n$ *quiver and any field* $\mathbb{F}$.

We now move beyond the finite Dynkin case and consider a natural generalization: the quivers of type $\tilde{A}_n$. These are examples of a class called Euclidean quivers. We will consider also generalizations involving infinite quivers.

**Definition 1.5.5.** *A quiver of type* $\tilde{A}_n$ *is a quiver whose underlying graph is a cycle with* $n+1$ *vertices:*

$$\begin{tikzcd}
 & & n+1 \arrow[dll, no head] \arrow[dr, no head] & \\
1 \arrow[r, no head] & 2 \arrow[r, no head] & 3 \arrow[r, dashed, no head] & n
\end{tikzcd}$$

**Definition 1.5.6.** *An infinite quiver of type A has one of the following two underlying graphs:* $A_\infty$ *or* $A_\infty^\infty$*. A graph of type* $A_\infty^\infty$ *has the following shape and the vertices are indexable on* $\mathbb{Z}$

$$\begin{tikzcd}
\cdots \arrow[r, dashed, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, dashed, no head] & \cdots
\end{tikzcd}$$

*A graph of type* $A_\infty$ *has the following shape and the vertices are indexable on* $\mathbb{N}$

$$\begin{tikzcd}
\bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, no head] & \bullet \arrow[r, dashed, no head] & \cdots
\end{tikzcd}$$

As shown in [9, Theorem 1.7], for the quivers $A_\infty$ and $A_\infty^\infty$ we have the following result, which is the analogue of Remark 1.5.4 for $A_n$. However, in this case we must restrict to pointwise finite-dimensional representations, i.e., representations in $\mathrm{Rep}(Q, \mathrm{vect}_\mathbb{F})$.

**Theorem 1.5.7.** *([9, Theorem 1.7]) Any indecomposable pointwise finite-dimensional representation of $A_\infty$ and $A_\infty^\infty$ quivers is an indicator representation.*

As a direct consequence, and using again Proposition 1.3.4,

$$\mathrm{free} : \mathrm{Rep}(Q, \mathrm{set}) \to \mathrm{Rep}(Q, \mathrm{vect}_\mathbb{F})$$

is additive surjective.

We end this digression about Dynkin quivers illustrating some properties about $D_4$ and the additive realizability which will be crucial in several proofs.

**Remark 1.5.8.** *As one can sees in the list in [15], $D_4$ has one positive root with a coordinate $> 1$ (hence not thin indecomposable representations) and it has coordinate $2$ in the vertex $2$ (agree with Fig. 1.2) and $1$ in the other vertices.*

**Proposition 1.5.9.** *([20, Propositio 8.3]) For $D_4$ the* free *functor is additive surjective if and only if not all arrows are ingoing.*

*Proof.* By the previous remark, and since indicator representations are always additively set realizable, it is enough to focus on the additive set realizability of the unique indecomposable representation that is not thin. According to the list of positive roots, the pointwise vector space dimensions of this representation are given by the numbers in the diagram below, where each number corresponds to the dimension at the respective vertex.

$$\begin{tikzcd} & 1 \arrow[d, no head] & \\ 1 \arrow[r, no head] & 2 \arrow[r, no head] & 1 \end{tikzcd}$$

By Lemma 1.3.5, it follows that the above representation is not additive set realizable if the arrows are all ingoing. On the other hand, it is very easy to check that for all the other arrows orientations, the indecomposable representation $R$, with the above dimensions, has a multiplicative basis and hence we conclude by Proposition 1.3.3. For example, for the orientation with all arrows going outwards, the indecomposable representation with such pointwise dimensions can be written as follows:

$$\begin{tikzcd} & \mathbb{F} \arrow[d] & \\ \mathbb{F} \arrow[r, "\begin{bmatrix} 0 \\ 1 \end{bmatrix}"] & \mathbb{F}^2 \arrow[r, "\begin{bmatrix} 1 \\ 0 \end{bmatrix}", "\begin{bmatrix} 1 & 1 \end{bmatrix}"'] & \mathbb{F} \end{tikzcd}.$$

The standard bases of the vector spaces in this representation assemble into a multiplicative basis. Indeed, for the left arrow, the element 1 is mapped to $e_1$; for the vertical arrow, 1 is mapped to $e_2$; and for the right arrow, both $e_1$ and $e_2$ are mapped to 1.

$\square$

# Chapter 2

# Thread Quivers

The aim of this chapter is to show that for certain infinite small categories, studying all indecomposable representations is the same as studying all indecomposable representations of some special full subcategories. The main tool of this chapter is thread quivers ( see Definition 2.1.9); similar and more in-depth results about these structures can be found in [2], but here they are developed independently. Let us start with some definitions and examples.

## 2.1 Thread quivers: definitions and examples

.

**Definition 2.1.1.** *Let $P$ be a small category, $f \in \mathrm{Hom}_P(p, p')$ is an* ***indecomposable morphism*** *of $P$ if*

$$f = g \circ h \Longrightarrow (g = \mathrm{id} \textit{ and } h = f) \textit{ or } (h = \mathrm{id} \textit{ and } g = f).$$

*We denote the set of indecomposable morphisms of $P$ as $\mathrm{Ind}_P$.*

**Definition 2.1.2.** *A small category $P$ is* Ind***-complete*** *if any non-identity morphism in $P$ is in $\mathrm{Ind}_P$ or is a finite composition of morphisms in $\mathrm{Ind}_P$.*

For an Ind-complete category, using the language of quivers, for any $\alpha \in \mathrm{Ind}_P$, we denote $s(\alpha)$ the source (or domain) of $\alpha$ and $t(\alpha)$ the target (or codomain). To describe Ind-complete categories precisely, we need the following notions from quiver theory.

**Definition 2.1.3.** *1. A* ***relation*** *in a quiver $Q$ is a pair of paths $(p, p')$ such that $s(p) = s(p')$ and $t(p) = t(p')$, where at least one of $p, p'$ has length $\geq 2$. A set of relations is denoted by $\mathscr{R}$.*

2. *A* ***quiver with relations*** *is a pair $(Q, \mathscr{R})$ where $Q$ is a quiver and $\mathscr{R}$ is a set of relations on $Q$.*

3. *The* ***category of paths of*** $Q$ ***modulo relations*** $\mathscr{R}$*, denoted $Q/\mathscr{R}$, is a category defined as follows:*

   - *The objects are the vertices $Q_0$ of the quiver.*
   - *The morphisms are equivalence classes of paths in $Q$. Two paths $p, p'$ from $u$ to $v$ are equivalent if $p = p'$ or if $p$ can be transformed into $p'$ by a sequence of replacements based on the relations in $\mathscr{R}$. Specifically, for any relation $(r_1, r_2) \in \mathscr{R}$, we consider any path of the form $xr_1y$ to be equivalent to $xr_2y$, where $x$ and $y$ are paths (possibly empty) such that the concatenations are valid.*
   - *Composition of morphisms is defined by concatenating representative paths and then taking their equivalence class.*

**Remark 2.1.4.** *Every small* Ind*-complete category can be represented as a quiver with relations. Indeed, given such a category $P$, we can construct a quiver $Q$ as follows:*

- *the* ***vertices*** *$Q_0$ of $Q$ are the objects of $P$.*
- *the* ***arrows*** *$Q_1$ of $Q$ correspond to the indecomposable morphisms in $P$. An arrow $\alpha : s(\alpha) \to t(\alpha)$ in $Q$ corresponds to an indecomposable morphism $f \in \mathrm{Ind}_P$ with source $s(\alpha)$ and target $t(\alpha)$.*

*Since every morphism in $P$ can be written as a finite composition of indecomposable morphisms, any morphism in $P$ corresponds to a path in $Q$. However, different paths in $Q$ may represent the same morphism in $P$. To account for this, we introduce a set of relations $\mathscr{R}$. These relations specify when two distinct paths in $Q$ must be considered as the same morphism in $P$. Therefore, the category $P$ is equivalent to the category of paths of $Q$ modulo the relations in $\mathscr{R}$, i.e., $P \cong Q/\mathscr{R}$.*

Let us consider some examples to illustrate the concept of Ind-complete categories.

**Example 2.1.5.** *The category representing the total order of natural numbers $(\mathbb{N}, \leq)$ is* Ind*-complete. Indeed, it can be represented as an $A_\infty$ quiver with all arrows in the same direction and no relations.*

**Example 2.1.6.** *Consider the poset $P = \{a, b, c, d\}$ with the following Hasse diagram.*

$$\begin{tikzcd}
& d & \\
b \arrow[ur, "\beta"] & & c \arrow[ul, "\gamma"'] \\
& a \arrow[ul, "\alpha"] \arrow[ur, "\delta"'] &
\end{tikzcd}$$

*In this poset, the indecomposable morphisms correspond to the arrows in the Hasse diagram: $\alpha : a \to b$, $\beta : b \to d$, $\delta : a \to c$, $\gamma : c \to d$. The relation $a < d$ can be expressed by two different paths of indecomposable morphisms: $\beta \circ \alpha$ (i.e., $a \to b \to d$) and $\gamma \circ \delta$ (i.e., $a \to c \to d$). Since both paths represent the same morphism $f_{a,d} : a \to d$ in the poset, this implies a relation between paths in the associated quiver. Specifically, the category of this poset is equivalent to the category of paths of the quiver above modulo the relation $(\beta\alpha, \gamma\delta)$. This is an* Ind*-complete category, as every morphism (e.g., $f_{a,d}$) is a composition of indecomposable morphisms.*

**Example 2.1.7.** *On the other hand, the category representing the total order of rational numbers $(\mathbb{Q}, \leq)$ is not* Ind*-complete.*

- *The objects are the rational numbers $q \in \mathbb{Q}$.*
- *The morphisms are relations $q \leq r$.*

*Apart from the identity morphisms ($q \to q$), no other morphism $f_{q,r} : q \to r$ (where $q < r$) is indecomposable. This is because for any two distinct rational numbers $q < r$, there always exists another rational number $s$ such that $q < s < r$. Therefore, any morphism $f_{q,r}$ can be decomposed as $f_{s,r} \circ f_{q,s}$, meaning it is always decomposable. Since there are no indecomposable morphisms other than identities, and no non-identity morphism can be expressed as a finite composition indecomposable morphisms, this category is not* Ind*-complete.*

**Notation 2.1.8.** *In the following,* Totord *denotes the class of all total orders considered as categories. Totally ordered sets can be seen as categories where a unique morphism $a \to b$ exists if and only if $a \leq b$ in the poset, as further explained in Remark 1.1.9.*

**Definition 2.1.9.** *(compare with [2, Definition 1.1]) Let $P$ be an* Ind*-complete category. A **thread** of $P$ is a category $X$ constructed as follows:*

- ***Objects of*** *$X$: For each indecomposable morphism $\alpha \in \mathrm{Ind}_P$, let $T_\alpha \in \mathrm{Totord}$. The requirement for $T_\alpha$ to have a minimum element $m_\alpha$ is imposed only if the source vertex $s(\alpha)$ is also the source of another indecomposable morphism or if it is the target of another indecomposable morphism. Similarly, $T_\alpha$ is required to have a maximum element $M_\alpha$ only if the target vertex $t(\alpha)$ is also the target of another indecomposable morphism, or if it is the source of another indecomposable morphism. We consider each $T_\alpha$ as a category where a unique morphism exists from $x$ to $y$ if and only if $x \leq y$.*

  *The objects of $X$ are elements of the disjoint union $\bigsqcup_{\alpha\in\mathrm{Ind}_P} T_\alpha$ modulo the following equivalence relation $\sim$:*

  *For any object $p \in \mathrm{Obj}(P)$, all elements in the set*

$$S_p = \{m_\alpha \mid s(\alpha) = p, \alpha \in \mathrm{Ind}_P\} \cup \{M_\alpha \mid t(\alpha) = p, \alpha \in \mathrm{Ind}_P\}$$

  *are declared equivalent. That is, for any $x, y \in S_p$, we have $x \sim y$. The objects of $X$ are the equivalence classes $[x]$ under this relation.*

- ***Morphisms of*** *$X$: For any two objects $[x], [y] \in \mathrm{Obj}(X)$, a morphism from $[x]$ to $[y]$ in $X$ is an equivalence class of paths formed by composing morphisms. A path consists of a sequence of morphisms $f_k \circ \cdots \circ f_1$, where each $f_i$ is a morphism within some $T_{\alpha_i}$ (i.e., $f_i \in \mathrm{Hom}_{T_{\alpha_i}}(x_i, y_i)$), and the target of $f_i$ is identified with the source of $f_{i+1}$ under the equivalence relation $\sim$.*

  *Specifically, for $[x], [y] \in \mathrm{Obj}(X)$, $\mathrm{Hom}_X([x],[y])$ consists of:*

  1. *Morphisms from $\mathrm{Hom}_{T_\alpha}(x', y')$ if $x' \in [x]$, $y' \in [y]$, and $x', y' \in T_\alpha$ for some $\alpha \in \mathrm{Ind}_P$.*
  2. *Compositions of such morphisms, where the composition $g \circ f$ is valid if the target of $f$ is equivalent to the source of $g$ according to $\sim$.*

  *Note that this construction can lead to directed cycles in $X$ if the underlying quiver of $P$ contains cycles.*

*We denote **S(P)** the class of all threads of $P$.*

The above definition admits a natural interpretation within the structure of threads. Given an Ind-complete category $P$, its thread may be simply viewed as $P$ in which each arrow is replaced by a totally ordered set. Let us illustrate the construction of a thread with an example. Other examples of such construction, like $G$-posets and $B$-posets, will be discussed in the following chapter.

**Example 2.1.10.** *Consider the quiver $P$ representing the category $A_2$:*

$$1 \xrightarrow{\alpha} 2$$

*Here,* $\mathrm{Obj}(P) = \{1,2\}$ *and* $\mathrm{Ind}_P = \{\alpha\}$. *We have* $s(\alpha) = 1$ *and* $t(\alpha) = 2$.

*To construct a **thread** $X$ of $P$, we choose a total order $T_\alpha$ for the arrow $\alpha$. Let $T_\alpha = [0,10]$ with the standard ordering of real numbers, where $m_\alpha = 0$ (minimum of $T_\alpha$) and $M_\alpha = 10$ (maximum of $T_\alpha$). This thread $X$ is essentially the category corresponding to the total order* $[0,10]$.

**Definition 2.1.11.** *We define $FS(P)$, the subclass of $S(P)$ consisting of all threads built choosing for each indecomposable morphism $\alpha$, $T_\alpha$ a finite total order.*

**Definition 2.1.12.** *Let $P$ be an* Ind*-complete category. Consider the following:*

- *$X \in S(P)$ is a **thread** of $P$.*
- *$C$ is an arbitrary category.*
- *$R : X \to C$ is a **representation** (or functor) from $X$ to $C$.*
- *$X' \in S(P)$ and $X'$ is a full subcategory of $X$.*

*We say that $R$ is **well described** by $X'$ if for every object $x \in \mathrm{Obj}(X) \setminus \mathrm{Obj}(X')$, there exists an object $x' \in \mathrm{Obj}(X')$ such that $x$ and $x'$ are **strongly connected** and $x, x'$ are in the same total order $T_\alpha$. Additionally, for every morphism $\alpha : x \to x'$ and every morphism $\beta : x' \to x$ within $X$ that establishes this strong connection, the image under the representation, $R(\alpha)$ and $R(\beta)$, must be an isomorphism in $C$.*

**Definition 2.1.13.** *Let $R \in \mathrm{Rep}(X,C)$ be well described by a full subcategory of $X' \subset X$. A map $\phi : \mathrm{Obj}(X) \setminus \mathrm{Obj}(X') \to \mathrm{Obj}(X')$ is a **description map** of $R$ through $X'$ if for any $x \in \mathrm{Obj}(X) \setminus \mathrm{Obj}(X')$, $\phi(x)$ is an object in $X'$ with the property in Definition 2.1.12.*

The definitions of a **well-described representation** and a **description map** are key to simplifying how we study representations, especially for large or even infinite threads. While analyzing representations of such vast categories can be challenging, we already have a strong understanding of finite categories. Our goal is to bridge this gap: a representation is **well-described** when the parts of the thread outside of a subcategory $X'$ do not add new information. This happens because these objects are related to objects already in $Y$ such that, under the representation $R$, they are connected with isomorphisms(meaning they preserve all essential information). The **description map** then provides a practical way to find these matching objects, mapping those outside $X'$ to their counterparts within $X'$. This framework allows us to reduce the study of complex representations to how they behave on simpler, finite subcategories. This is a powerful simplification, and we will show that many fundamental representations of threads can indeed be understood by looking only at their finite parts.

## 2.2 Indecomposable representations of thread quivers

**Theorem 2.2.1.** *(compare with [2, Corollary 4.6]) Let $P$ be a connected* Ind*-complete category without directed cycles (i.e.* $\mathrm{Hom}_P(x,x) = \{\mathrm{Id}\}\ \forall x \in P$*). Consider $X \in S(P)$ and $R \in \mathrm{Rep}(X, \mathrm{vect}_{\mathbb{F}})$. Then $R$ is well described by a connected category $X' \in FS(P)$. Moreover, if $R$ is indecomposable, then so is $R|_{X'}$.*

Lemma 2.2.4 will outline our strategy for **discretizing a thread** of $P$. This step is crucial for proving Theorem 2.2.1. Before we delve into the lemma, let us consider a simple yet important observation.

**Remark 2.2.2.** *It is easy to see that if in the following diagram in a category $C$, the squares $\alpha$ and $\beta$ commute, then $f_3 \circ f_2 \circ f_1 = g_3 \circ g_2 \circ g_1$ .*

$$\begin{tikzcd}
\bullet \arrow[r, "f_1"] \arrow[d, "g_1"'] \arrow[dr, phantom, "\alpha"] & \bullet \arrow[r, "f_2"'] \arrow[d] \arrow[dr, phantom, "\beta"] & \bullet \arrow[d, "f_3"] \\
\bullet \arrow[r, "g_2"'] & \bullet \arrow[r, "g_3"'] & \bullet
\end{tikzcd}$$

For each $p \in P$, we introduce the $\mathrm{Cmp}(p)$ set.

**Definition 2.2.3.** *Given a poset $P$, for any $p \in P$,$\mathrm{Cmp}(p)$ is the set of all elements of $P$ comparable with $p$.*

**Lemma 2.2.4.** *Let $R$ be an indecomposable representation on* $\mathrm{Vect}_{\mathbb{F}}$ *of a poset $P$ considered as category. Suppose that there is an element $r \in P$ such that:*

- *$T := \mathrm{Cmp}(r)$ is a total order-category;*
- *There exists an interval $I \subseteq T$ containing $r$ such that for any $s, s' \in I$ with $s \leq s'$, the map $R_{ss'}$ is an isomorphism*
- *There is an element $r' \in \mathrm{Obj}(P) \setminus I$ such that $r' \leq s$ for each $s \in I$ and there is an element $r'' \in \mathrm{Obj}(P) \setminus I$ such that $r'' \geq s$ for each $s \in I$.*

*Let $P'$ be the full subcategory of $P$ with objects $Obj(P') = Obj(P) \setminus (I \setminus \{r\})$. Then the restriction $R|_{P'}$ is indecomposable.*

*Proof.* Let $R'$ denote the restriction of $R$ to $P'$. Suppose, by contradiction, that $R'$ is decomposable, i.e., there exists a non-trivial decomposition $R' = S' \oplus T'$, where $S'$ and $T'$ are non-zero subrepresentations of $R'$. We will construct a non-trivial decomposition $R = S \oplus T$ for $R$, leading to a contradiction.

We define $S$ as follows:

$$\begin{cases} S_p = S'_p & \forall p \in p' \\ S_p = S'_r & \forall p \in I \end{cases}$$

and for all pairs of comparable elements

$$\begin{cases} S_{qp} = S'_{qp} & \text{if } q, p \in P' \\ S_{qp} = \mathrm{Id} & \text{if } q, p \in I \\ S_{qp} = S'_{rp} \circ \mathrm{Id} & \text{if } q \in I, p \in P' \\ S_{qp} = \mathrm{Id} \circ S'_{qr} & \text{if } q \in P', p \in I \end{cases}$$

where Id is the identity from $S_q$ to $S_r$ in the third case and the identity from $S_r$ to $S_p$ in the fourth case. The representation $T$ is defined in the same way using $T'$ instead of $S'$. Let us observe that $S$ and $T$ are well defined representations because we have just added identities between the elements of $I$ and $r$ while other maps are ones induced by $S'$ and $T'$.

Now, it is enough to construct a representation isomorphism $\phi : R \to S \oplus T$ using a representation isomorphism $\psi : R' \to S' \oplus T'$. We define

$$\begin{cases} \phi_p = \psi_p & \text{if } p \in P' \\ \phi_p = \psi_r \circ R^{-1}_{rp} & \text{if } p \in I \text{ and } p > r \\ \phi_p = \psi_r \circ R_{pr} & \text{if } p \in I \text{ and } p < r. \end{cases}$$

Since $\psi_p$ is a vector space isomorphism for any $p \in \mathrm{Obj}(P')$, and $R_{r\to p}$ (or $R_{p\to r}$) are isomorphisms for any $p \in I$ (by hypothesis), $\phi_p$ is a vector space isomorphism for every $p \in P$. Thus, if $\phi$ is a representation homomorphism, it is a representation isomorphism. This isomorphism would then induce a decomposition $R \cong S' \oplus T'$, which can be written as $R = S \oplus T$. This means that we just need to check that for any pair $p < p' \in P$, $\phi_{p'} \circ R_{pp'} = (S \oplus T)_{pp'} \circ \phi_p$. It is enough to check the following six cases:

1) $p \notin I$ and $p' < r$, $p' \in I$
2) $p, p' \in I$, $pp' < r$
3) $p \in I$, $p < r$ and $p' = r$
4) $p' \notin I$ and $p, p' > r$, $p \in I$
5) $p, p' \in I$, $p, p' > r$
6) $p' \in I$, $p' > r$, and $p = r$

In fact, other cases are straightforward consequences of the well definition of $\psi$, like if $p, p' \in P'$, or follow from the composition of morphisms in $P$ using the Remark 2.2.2.Note that for certain cases, such as when $p, p' \notin I$ with $p < r < p'$, the verification requires multiple compositional steps.

1.

$$\begin{tikzcd} R_p \arrow[r, "R_{pp'}"'] \arrow[d, "\psi_p"] & R_{p'} \arrow[d, "\psi_r \circ R_{p'r}"] \\ S_p \oplus T_p \arrow[r, "\mathrm{Id} \circ (S_{pr} \oplus T_{pr})"'] & S_{p'} \oplus T_{p'} \end{tikzcd}$$

$$\mathrm{Id} \circ (S_{pr} \oplus T_{pr}) \circ \psi_p = (S_{pr} \oplus T_{pr}) \circ \psi_p = \psi_r \circ R_{pr} = \psi_r \circ R_{p'r} \circ R_{pp'}$$

2.

$$\begin{tikzcd} R_p \arrow[r, "R_{pp'}"'] \arrow[d, "\psi_r \circ R_{pr}"] & R_{p'} \arrow[d, "\psi_r \circ R_{p'r}"] \\ S_p \oplus T_p \arrow[r, "\mathrm{Id}"'] & S_{p'} \oplus T_{p'} \end{tikzcd}$$

$$\psi_r \circ R_{pr} = \psi_r \circ R_{p'r} \circ R_{pp'}$$

3.

$$\begin{tikzcd} R_p \arrow[r, "R_{pr}"'] \arrow[d, "\psi_r \circ R_{pr}"] & R_r \arrow[d, "\psi_r"] \\ S_p \oplus T_p \arrow[r, "\mathrm{Id}"'] & S_r \oplus T_r \end{tikzcd}$$

It is evident.

4.

$$\begin{tikzcd} R_p \arrow[r, "R_{pp'}"'] \arrow[d, "\psi_r \circ R_{rp}^{-1}"] & R_{p'} \arrow[d, "\psi_{p'}"] \\ S_p \oplus T_p \arrow[r, "(S_{rp'} \oplus T_{rp'}) \circ \mathrm{Id}"'] & S_{p'} \oplus T_{p'} \end{tikzcd}$$

$$(S_{rp'} \oplus T_{rp'}) \circ \mathrm{Id} \circ \psi_r \circ R_{rp}^{-1} = (S_{rp'} \oplus T_{rp'}) \circ \psi_r \circ R_{rp}^{-1} =$$

$$= \psi_{p'} \circ R_{rp'} \circ R_{rp}^{-1} = \psi_{p'} \circ R_{pp'} \circ R_{rp} \circ R_{rp}^{-1} = \psi_{p'} \circ R_{pp'}$$

5.

$$\begin{tikzcd} R_p \arrow[r, "R_{pp'}"'] \arrow[d, "\psi_r \circ R_{rp}^{-1}"] & R_{p'} \arrow[d, "\psi_r \circ R_{rp'}^{-1}"] \\ S_p \oplus T_p \arrow[r, "\mathrm{Id}"'] & S_{p'} \oplus T_{p'} \end{tikzcd}$$

$$\psi_r \circ R_{rp'}^{-1} \circ R_{pp'} = \psi_r \circ (R_{pp'} \circ R_{rp})^{-1} = \psi_r \circ R_{rp}^{-1} \circ R_{pp'}^{-1} \circ R_{pp'} = \psi_r \circ R_{rp}^{-1}$$

6.

$$\begin{tikzcd} R_r \arrow[r, "R_{rp'}"'] \arrow[d, "\psi_r"] & R_{p'} \arrow[d, "\psi_r \circ R_{rp'}^{-1}"] \\ S_r \oplus T_r \arrow[r, "\mathrm{Id}"'] & S_{p'} \oplus T_{p'} \end{tikzcd}$$

It is evident.

□

Before the proof of Theorem 2.2.1, we need to recall a general result about pointwise finite-dimensional representations of totally ordered sets.

**Theorem 2.2.5.** *([8, Theorem 1.2]) Pointwise finite-dimensional representations of a totally ordered set decompose into interval representations.*

*Proof. Theorem 2.2.1.* Using the terminology of Definition 2.1.9, we refer to the total order that replaces the indecomposable morphism $\alpha$ as $T_\alpha$.

If for any $\alpha \in \mathrm{Ind}_P$, the corresponding $T_\alpha$ is finite, then $X \in FS(P)$ and the thesis follows trivially by taking $X' = X$.

Hence, we suppose that there is at least one $\alpha \in \mathrm{Ind}_P$ such that $T_\alpha$ is not finite. Let's choose such an $\alpha$ and consider the restriction of $R$ to $T_\alpha$ ($R|_{T_\alpha}$). By Theorem 2.2.5, we have the following decomposition into interval representations:

$$R|_{T_\alpha} = \bigoplus_{i \in J} \mathbb{F}_{I_i}$$

where $I_i$ is an interval of $T_\alpha$. The infimum and supremum of these intervals are not necessarily elements of $T_\alpha$ itself (consider for example the interval $(\sqrt{2}, \sqrt{3})$ in $\mathbb{Q}$).

Therefore, we shall consider $T_\alpha$ as a subposet of a complete total order $\overline{T}_\alpha$ (e.g. $\mathbb{R}$ is a complete total order that contains $\mathbb{Q}$). Consequently, each $I_i$ takes one of the following forms:

$$[x_i, y_i], \quad [x_i, y_i), \quad (x_i, y_i], \quad (x_i, y_i)$$

Here, $x_i$ and $y_i$ represent these endpoints within $\overline{T}_\alpha$. If an interval is unbounded on one side, that endpoint is represented by $-\infty$ or $+\infty$ (e.g., $(-\infty, y_i]$ or $[x_i, +\infty)$).

Let us observe that if an interval $I_i$ (the support of $\mathbb{F}_{I_i}$) does not contain the maximum of $T_\alpha$ ($M_\alpha$) or the minimum $m_\alpha$, $\mathbb{F}_{I_i}$ can be factored out by $R$. Since $R$ is indecomposable by hypothesis, this would imply that $\mathbb{F}_{I_i}$ must be equal to $R$. However, if $R = \mathbb{F}_{I_i}$, then $R$ is well described by the full subcategory $X'$ of $X'$ whose objects are:

- an object in $I_i$ ( for all other elements of $I_i$ the maps from/to it are identities)
- an object $x'$ in $T_\alpha \setminus I_i$ such that $x' < x \;\; \forall x \in I_i$ and an object $x'' > x \;\; \forall x \in I_i$, in this way each $x \in T_\alpha \setminus I_i$ is strongly connected to $x'$ or $x''$ and the homomorphisms between them are the zero homomorphism (in this case the identity).
- maximum and minimum of each $T_\beta$ for each $\beta \in \mathrm{Ind}_P$, in this way each $x \in T_\beta$ is strongly connected to $m_\beta$ and the homomorphisms between them are the zero homomorphism (in this case the identity). Moreover taking both maximum and minimum $X'$ is connected.

Hence we would have the thesis because $X' \in FS(P)$, i.e. it is a thread such that all total ordered sets chosen to built it are finite.

Suppose indeed, that an interval $I_i$ is strictly contained within the interior of $T_\alpha$. This implies that $I_i$ does not contain $m_\alpha$ or $M_\alpha$. We have to show that the injection from $\mathbb{F}_{I_i}$ to $R|_{T_\alpha}$ and the projection from $R|_{T_\alpha}$ to $\mathbb{F}_{I_i}$ can be extended to an injection $\phi$ from $\mathbb{F}_{I_i}$ to $R$ and a projection $\pi$ from $R$ to $\mathbb{F}_{I_i}$. The interval representation $\mathbb{F}_{I_i}$ assigns the 0 vector spaces to objects outside $T_\alpha$, hence $\pi$ and $\phi$ outside $T_\alpha$ are trivially extended with 0 homomorphisms.

To show that this extension of $\phi$ works, we need to prove that for each $f : x \to y$ in $X$, $\phi_y \circ \mathbb{F}_{I_i}(f) = R_f \circ \phi_x$.

- If $x, y \in T_\alpha$, the commutativity holds trivially, as $\phi$ is defined as the original injection within $T_\alpha$.
- If $x$ and $y$ are both not in $I_i$ both sides of the equation are zero because $\phi$ objects outside $I_i$ to 0 homomorphisms because $\mathbb{F}_{I_i}$ is zero on such objects.

- The case where one of $x$ or $y$ is in $I_i$ while the other is not requires more careful consideration. By the construction of a thread category $X$, any morphism $f : x \to y$ in $X$ where $x \in I_i$ and $y \notin T_\alpha$ must necessarily factors through the maximum $M_\alpha$ of $T_\alpha$. Thus, $f$ can be written as $f = g \circ h$ where $h : x \to M_\alpha$ and $g : M_\alpha \to y$. Furthermore, as $\mathbb{F}_{I_i}$ has non-zero values only for objects within $I_i$, any morphism $f : x \to y$ where $x \in I_i$ and $y \notin I_i$ implies that $\mathbb{F}_{I_i}(f)$ is the zero map. Thus, the left side of the commutativity equation, $\phi_y \circ \mathbb{F}_{I_i}(f)$, is 0.

  Now consider the right side, $R_f \circ \phi_x$. We have that $R_f = R_g \circ R_h \circ \phi_x = R_g \circ \phi(M_\alpha) \circ \mathbb{F}_{I_i}(h)$ because $h \in T_\alpha$ and we know that $\phi$ is well defined inside $T_\alpha$. Due to the structure of $\phi$ (being zero homomorphisms outside $T_\alpha$ ), $\phi(M_\alpha) = 0$, hence also the right side is 0.

  If $y \in I_i$ and $x \notin I_i$ the proof is the same with the difference that $f$ factorizes through the minimum $m_\alpha$ instead of the maximum.

The proof for $\pi$ is entirely analogous.

Thus, we can assume that each interval $I_i$ has an endpoint equal to $m_\alpha$ or $M_\alpha$. It follows that any such interval takes one of the following forms:

$$[m_\alpha, y_i], \quad [m_\alpha, y_i), \quad (x_i, M_\alpha], \quad (x_i, M_\alpha)$$

where $x_i$ and $y_i$ are elements of $\overline{T}_\alpha$.

Furthermore, since $R$ is **pointwise finite-dimensional**, the vector spaces $R_{M_\alpha}$ and $R_{m_\alpha}$ must have finite dimensions. Given that every interval $I_i$ contains either $m_\alpha$ or $M_\alpha$, the index set $J$ of all intervals in the decomposition of $R|_{T_\alpha}$ must be finite. If $J$ were infinite, it would imply an infinite direct sum contributing to either $R_{m_\alpha}$ or $R_{M_\alpha}$, leading to an infinite dimension, which contradicts $R$ being pointwise finite-dimensional.

Let us define

$$\begin{aligned} S_\alpha = \{s \in \overline{T}_\alpha \mid & \text{ there is an interval } I_j \text{ in the decomposition} \\ & \text{of } R|_{T_\alpha} \text{ such that } s \text{ is an endpoint of } I_j\} \end{aligned}$$

The set $S_\alpha$ is finite by the finiteness of $J$. We consider the total order on $S_\alpha$ induced by $\overline{T}_\alpha$ and enumerate these as $m_\alpha = s_0, s_1, \ldots, s_v = M_\alpha$.

For any pair of consecutive elements $s_i, s_{i+1}$ in this ordered sequence, let $J_i = (s_i, s_{i+1})$ denote the open interval between them within $T_\alpha$. An important observation is that if we pick any two objects $t, u \in J_i$, the map $R_{tu}$ (or $R_{ut}$ if $u \le t$) is an isomorphism.

For each non-empty interval $J_i$, we choose a representative object $t_i \in J_i$. Now, define a new subcategory $X'_\alpha$ whose objects are the objects in $\left(X \setminus \bigcup_{i=1}^{v-1}(s_i, s_{i+1})\right) \bigcup \{t_1, .., t_{v-1}\}$. By applying Lemma 2.2.4, and utilizing the fact that $R$ is indecomposable, it follows that $R|_{X'_\alpha}$ is indecomposable. Moreover, from the construction of the objects $t_i$, it follows that $R$ is well-described by $X'_\alpha$. This is because, for any object $x \in J_i \setminus \{t_i\}$, $x$ is strongly connected to $t_i$, and $R$ maps any morphism between $x$ and $t_i$ to an isomorphism.

This process applies to each $\alpha \in \mathrm{Ind}_P$. Let $X'$ be the full subcategory of $X$ whose object set is $\mathrm{Obj}(X') = \bigcap_{\alpha \in \mathrm{Ind}_P} \mathrm{Obj}(X'_\alpha)$. Since each $X'_\alpha \cap T_\alpha$ effectively reduces the finite vertices (the endpoints $S_\alpha \cap T_\alpha$ plus the chosen $t_i$), the intersection $X'$ will also be in $FS(P)$.

By repeatedly applying the **well-described** property established for $R$ with respect to each $X'_\alpha$ (which is specifically constructed to capture $R$'s behavior within each $T_\alpha$), $R$ is ultimately well-described by this subcategory $X'$, which is also connected because it contains maximum and minimum of each $T_\alpha$.

Furthermore, the construction of each $X'_\alpha$ (involving the selection of $t_i$ based on the properties of $R$ on the intervals $J_i = (s_i, s_{i+1})$) is precisely what allows us to apply Lemma 2.2.4 to each thread. This application guarantees that the indecomposability of $R$ as a representation of $X$ is preserved when restricted to $X'$, meaning $R|_{X'}$ remains indecomposable. $\square$

Let us see why this theorem is so relevant. Let us choose a quiver (for example $D_4$ with the orientation in the figure below), choose an arrow ($h$) and replace it with a total order as follows, creating a thread. We denote this new category $X$.

$$\begin{tikzcd} & \bullet & \\ \bullet \arrow[r] & \bullet \arrow[u] & \bullet \arrow[l, "h"'] \end{tikzcd} \quad \longrightarrow \quad \begin{tikzcd} & \bullet & \\ \bullet \arrow[r] & \bullet \arrow[u] & \bullet \arrow[l, "T_h"'] \end{tikzcd}$$

If $T_h$ has an infinite number of vertices and we pick an indecomposable representation $R$ of this new category $X$, $R$ is well described by a finite subcategory $X'$. This means there exists an appropriate finite full subcategory $X'$ of $X$ and $\mathrm{res}|_{X'}(R)$, for any $x \in \mathrm{Obj}(X) \setminus \mathrm{Obj}(X')$, there is an $x' \in \mathrm{Obj}(X')$ such that $x, x'$ are strongly connected and $R_\alpha$ is an isomorphism for any $\alpha$ from $x$ to $x'$ (or from $x'$ to $x$). Thus, $R$ being well-described by $X'$ means it is completely understood by its restriction to an appropriate full subcategory $X'$ and a description map $\phi : \mathrm{Obj}(X) \setminus \mathrm{Obj}(X') \to \mathrm{Obj}(X')$.

**Proposition 2.2.6.** *Let $R \in \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$ be well described by a subcategory $X'$ of $X$ and let ,$\phi : \mathrm{Obj}(X) \setminus \mathrm{Obj}(X') \to \mathrm{Obj}(X')$ be the relative description map. Then, $R$ is additive-set realizable if and only if* $\mathrm{res}|_{X'}(R)$ *is additive set realizable.*

Before the proof of this proposition we need to recall the following sufficient and a necessary condition for a representation to be additive set realizable.

**Theorem 2.2.7.** *([20, Corollary 4.3]) Let $X$ be a small category, and $R \in \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$. Then $R$ is in the additive image of the functor* $\mathrm{free}_* : \mathrm{Rep}(X, \mathrm{Set}) \to \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$ *if and only if the natural epimorphism* $\mathrm{free}_*(R_{\textit{as pointed set}}) \to R$ *splits.*

The notation $R_{\text{as pointed set}}$ refers to the representation $R$ considered as a representation into the category of **pointed sets** $\mathrm{Set}_*$, where each vector space $R(x)$ is viewed as a set with its zero vector as the distinguished point.

*Proof.* Proposition 2.2.6 **(⇒)** If $R$ is additive set realizable, this means there exists a set representation $S$ of $X$ such that $\mathrm{free}(S) = R \oplus T$ for some $T \in \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$. The restriction of $\mathrm{free}(S)$ to $X'$ is $\mathrm{free}(\mathrm{res}|_{X'}(S))$. Also, $\mathrm{res}|_{X'}(R \oplus T) = \mathrm{res}|_{X'}(R) \oplus \mathrm{res}|_{X'}(T)$. Thus, $\mathrm{free}(\mathrm{res}|_{X'}(S)) = \mathrm{res}|_{X'}(R) \oplus \mathrm{res}|_{X'}(T)$. Since $\mathrm{res}|_{X'}(S)$ is a set representation of $X'$ and $\mathrm{res}|_{X'}(T)$ is a vector space representation of $X'$, $\mathrm{res}|_{X'}(R)$ is additive set realizable.

**(⇒)** To prove this direction, we will use the condition in Theorem 2.2.7. To not overload the notation, we define:

$$S = \mathrm{free}_*(R_{\text{as pointed set}}), \qquad R' := R|_{X'} \quad \text{and} \quad S' := \mathrm{free}_*(R'_{\text{as pointed set}})$$

Moreover we denote $\psi$ the natural epimorphism from $S$ to $R$ and $\psi'$ the natural epimorphism from $S'$ to $R'$. Using the criterion stated in Theorem 2.2.7, have to prove that if there exists a $\gamma' : R' \to S'$ such that $\psi' \circ \gamma' = \mathrm{Id}_{R'}$, there exists a $\gamma : R \to S$ such that $\psi \circ \gamma = \mathrm{Id}_R$.

For each pair $(x, \phi(x))$, we pick an arbitrary $\alpha$ from $x$ to $\phi(x)$ or from $\phi(x)$ to $x$. Observe that, by the well described property, $R_\alpha$ is invertible which implies that $S_\alpha$ is also invertible. Now we define $\gamma$ for the cases $\alpha : x \to \phi(x)$ and for the case $\alpha : \phi(x) \to x$..

1. $\gamma_x = S_\alpha^{-1} \circ \gamma'_{\phi(x)} \circ R_\alpha \qquad \text{if } \alpha : x \to \phi(x)$
2. $\gamma_x = S_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1} \qquad \text{if } \alpha : \phi(x) \to x$

Moreover, we assume $\gamma_x = \gamma'_x$ if $x \in \mathrm{Obj}(X')$.

We prove now that $\psi_x \circ \gamma_x = \mathrm{Id}_{\mathrm{R_x}}$ if $\gamma_x$ is defined as in the case 1 and if the $\gamma_x$ is defined as in the case 2.

1. Since $R_\alpha \circ \psi_x = \psi_{\phi(x)} \circ S_\alpha$, we have $\psi_x = R_\alpha^{-1} \circ \psi_{\phi(x)} \circ S_\alpha$. Substituting this into $\psi_x \circ \gamma_x$, we get $\psi_x \circ \gamma_x = R_\alpha^{-1} \circ \psi_{\phi(x)} \circ S_\alpha \circ S_\alpha^{-1} \circ \gamma'_{\phi(x)} \circ R_\alpha = R_\alpha^{-1} \circ \psi'_{\phi(x)} \circ \gamma'_{\phi(x)} \circ R_\alpha = R_\alpha^{-1} \circ \mathrm{Id}_{\phi(x)} \circ R_\alpha = \mathrm{Id}_x$.

2. Since $R_\alpha \circ \psi_{\phi(x)} = \psi_x \circ S_\alpha$, we have $\psi_x = R_\alpha \circ \psi_{\phi(x)} \circ S_\alpha^{-1}$. Substituting this into $\psi_x \circ \gamma_x$, we get $\psi_x \circ \gamma_x = R_\alpha \circ \psi_{\phi(x)} \circ S_\alpha^{-1} \circ S_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha = R_\alpha \circ \psi'_{\phi(x)} \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1} = R_\alpha \circ \mathrm{Id}_{\phi(x)} \circ R_\alpha^{-1} = \mathrm{Id}_x$.

Hence, we obtained that $\psi_x \circ \gamma_x = \mathrm{Id}_{R_x}$ in both cases.

Now, it remains to prove the naturality of $\gamma$ . This means that we have to prove that for each morphism $\beta : x \to x'$ in $X$, $\gamma_{x'} \circ R_\beta = S_\beta \circ \gamma_x$. We divide all possibility in 4 cases:

1. $\gamma_x$ is defined with an $\alpha$ from $x$ to $\phi(x)$ and $\gamma_{x'}$ is defined with an $\alpha'$ from $x'$ to $\phi(x')$;

2. $\gamma_x$ is defined with an $\alpha$ from $\phi(x)$ to $x$ and $\gamma_{x'}$ is defined with an $\alpha'$ from $x'$ to $\phi(x')$;

3. $\gamma_x$ is defined with an $\alpha$ from $\phi(x)$ to $x$ and $\gamma_{x'}$ is defined with an $\alpha'$ from $\phi(x')$ to $x'$;

4. $\gamma_x$ is defined with an $\alpha$ from $x$ to $\phi(x)$ and $\gamma_{x'}$ is defined with an $\alpha'$ from $\phi(x')$ to $x'$.

- **Case 1:** In this case $\gamma_x = S_\alpha^{-1} \circ \gamma'_{\phi(x)} \circ R_\alpha$ and $\gamma_{x'} = S_{\alpha'}^{-1} \circ \gamma'_{\phi(x')} \circ R_{\alpha'}$. Recall that we have the following maps $\alpha, \alpha'$ and $\beta$ between $x, x', \phi(x)$ and $\phi(x')$:

$$\begin{tikzcd} x \arrow[rr, "\beta"] \arrow[dr, "\alpha"'] & & x' \arrow[r, "\alpha'"] & \phi(x') \\ & \phi(x) \arrow[ur, dashed, "\delta"'] & & \end{tikzcd}.$$

  By the thread construction (specifically by the fact that $\phi(x)$ is in the same total order $T_\alpha$ of $x$), this implies that that there is a map $\delta : \phi(x) \to x'$ such that $\beta = \delta \circ \alpha$. It follows that $R_\beta = R_{\alpha'}^{-1} \circ R_{\alpha' \circ \delta} \circ R_\alpha$ and that $S_\delta = S_\beta \circ S_\alpha^{-1}$. Hence,

$$\gamma_{x'} \circ R_\beta = S_{\alpha'}^{-1} \circ \gamma'_{\phi(x')} \circ R_{\alpha'} \circ R_\beta = S_{\alpha'}^{-1} \circ \gamma'_{\phi(x')} \circ R_{\alpha'} \circ R_{\alpha'}^{-1} \circ R_{\alpha' \circ \delta} \circ R_\alpha = S_{\alpha'}^{-1} \circ \gamma'_{\phi(x')} \circ R_{\alpha' \circ \delta} \circ R_\alpha.$$

Because $\delta \circ \alpha' \in X'$ and we have the naturality of $\gamma'$ by hypothesis,

$$S_{\alpha'}^{-1} \circ \gamma'_{\phi(x')} \circ R_{\alpha' \circ \delta} \circ R_\alpha = S_{\alpha'}^{-1} \circ S_{\alpha' \circ \delta} \circ \gamma'_{\phi(x)} \circ R_\alpha = S_{\alpha'}^{-1} \circ S_{\alpha'} \circ S_\delta \circ \gamma'_{\phi(x)} \circ R_\alpha.$$

By the property of $S_\delta$, the last composition is equal to $S_\beta \circ S_\alpha^{-1} \circ \gamma'_{\phi(x)} \circ R_\alpha$ that (by the definition of $\gamma_x$) is equal to$S_\beta \circ \gamma'_x$. Thus, we obtained that $\gamma_{x'} \circ R_\beta = S_\beta \circ \gamma_x$.

- **Case 2:** In this case $\gamma_x = S_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1}$ and $\gamma_{x'} = S_{\alpha'}^{-1} \circ \gamma'_{\phi(x')} \circ R_{\alpha'}$. It follows that

$$\gamma_{x'} \circ R_\beta = S_{\alpha'}^{-1} \circ \gamma'_{\phi(x')} \circ R_{\alpha'} \circ R_\beta \circ R_\alpha \circ R_\alpha^{-1} = S_{\alpha'}^{-1} \circ \gamma'_{\phi(x')} \circ R_{\alpha' \circ \beta \circ \alpha} \circ R_\alpha^{-1}.$$

Because $\alpha' \circ \beta \circ \alpha \in X'$ and we have the naturality of $\gamma'$ by hypothesis, the right side is equal to $S_{\alpha'}^{-1} \circ S_{\alpha' \circ \beta \circ \alpha} \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1}$, which is equal to

$$S_{\alpha'}^{-1} \circ S_{\alpha'} \circ S_\beta \circ S_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1} = S_\beta \circ S_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1} = S_\beta \circ \gamma_x.$$

Thus, we obtained that $\gamma_{x'} \circ R_\beta = S_\beta \circ \gamma_x$.

- **Case 3:** In this case $\gamma_x = S_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1}$ and $\gamma_{x'} = S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R_{\alpha'}^{-1}$. Recall that we have the following maps $\alpha, \alpha'$ and $\beta$ between $x, x', \phi(x)$ and $\phi(x')$:

$$\begin{tikzcd}
\phi(x) \arrow[r, "\alpha"'] & x \arrow[rr, "\beta"'] \arrow[dr, dashed, "\delta"'] & & x' \\
& & \phi(x') \arrow[ur, "\alpha'"'] &
\end{tikzcd}.$$

By the thread construction, this implies that that there is a map $\delta : x \to \phi(x')$ such that $\beta = \alpha' \circ \delta$.

It follows that

$$\gamma_{x'} \circ R_\beta = S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R_{\alpha'}^{-1} \circ R_\beta = S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R_{\alpha'}^{-1} \circ R_{\alpha'} \circ R_\delta \circ R_\alpha \circ R_\alpha^{-1} = S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R_{\delta \circ \alpha} \circ R_\alpha^{-1}$$

Because $\delta \circ \alpha \in X'$ and we have the naturality of $\gamma'$ by hypothesis, the last composition is equal to $S_{\alpha'} \circ S_{\delta \circ \alpha} \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1}$, which is equal to

$$S_{\alpha'} \circ S_\delta \circ S_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1} = S_\beta \circ S_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha^{-1} = S_\beta \circ \gamma_x.$$

Thus, we obtained that $\gamma_{x'} \circ R_\beta = S_\beta \circ \gamma_x$.

- **Case 4:** In this case $\gamma_x = S_\alpha^{-1} \circ \gamma'_{\phi(x)} \circ R_\alpha$ and $\gamma_{x'} = S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R_{\alpha'}^{-1}$.Recall that we have the following maps $\alpha, \alpha'$ and $\beta$ between $x, x', \phi(x)$ and $\phi(x')$:

$$\begin{tikzcd} x \arrow[rrr, "\beta"] \arrow[dr, "\alpha"'] & & & x' \\ & \phi(x) \arrow[r, dashed, "\delta"'] & \phi(x') \arrow[ur, "\alpha'"'] & \end{tikzcd}.$$

By the thread construction, this implies that that there is a map $\delta : \phi(x) \to \phi(x')$ such that $\beta = \alpha' \circ \delta \circ \alpha$. It follows that

$$\gamma_{x'} \circ R_\beta = S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R^{-1}_{\alpha'} \circ R_\beta = S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R^{-1}_{\alpha'} \circ R_{\alpha' \circ \delta \circ \alpha} =$$

$$= S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R^{-1}_{\alpha'} \circ R_{\alpha'} \circ R_\delta \circ R_\alpha = S_{\alpha'} \circ \gamma'_{\phi(x')} \circ R_\delta \circ R_\alpha.$$

Because $\delta \in X'$ and we have the naturality of $\gamma'$ by hypothesis, the last composition is equal to $S'_\alpha \circ S_\delta \circ \gamma'_{\phi(x)} \circ R_\alpha$ that is equal to $S'_\alpha \circ S_\delta \circ S_\alpha \circ S^{-1}_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha$. The last composition is equal to $S_\beta \circ S^{-1}_\alpha \circ \gamma'_{\phi(x)} \circ R_\alpha$ which (by the definition of $\gamma_x$) is equal to$S_\beta \circ \gamma'_x$. Thus, we obtained that $\gamma_{x'} \circ R_\beta = S_\beta \circ \gamma_x$.

□

**Remark 2.2.8.** *Let $X$ be a thread of $Q$. By Proposition 2.2.6 and Theorem 2.2.1, the additive set realizable representations of $X$ correspond to all additive set realizable representations of the categories in $\{Y \in FS(Q)$ such that $Y$ is a full subcategory of $X\}$ up to a description map $\phi$.*

*This characterization is even more powerful if we consider a thread $X$ of a finite rooted tree quiver $Q$, quivers with a maximum and which underlying graph is a tree( Definition 5.1.1). Indeed, such quivers, as shown in Theorem [20, Corollary 6.11] and independently in [17, Theorem C], admit only a finite number of additive set realizable representations. Hence, because the posets in $FS(Q)$ are still finite rooted tree quivers, in this case to classify all additive set realizable representation of $X$, one should still check all full subcategories of $X$ in $FS(Q)$ but for each of these categories, there will be just finitely many additive set realizable representations.*

In the final chapter, based on [17], we will provide a precise description of the indecomposable representations in $\mathrm{Rep}(Q, \mathrm{vect}_{\mathbb{F}})$, specifically when $Q$ is a rooted tree quiver.

# Chapter 3

# About indicator representations

The aim of this chapter is to prove Theorem 3.2.2, which is a generalization to infinite categories of the following theorem, proved in [20, Theorem 7.1]. It describes a necessary and sufficient condition of small categories to obtain that the indecomposable representations in the additive image of the free functor consist only of indicator representations.

**Theorem 3.0.1** ([20, Theorem 7.1])**.** *Let $X$ be a finite category. Then all indecomposable additively* Set*-realizable representations are indicator representations if and only if $X$ is equivalent to the poset category of a poset that is a disjoint union of*

- *posets with a Hasse diagram of type $A$;*
- *posets with a single maximum, such that the poset without the maximum is a disjoint union of posets with a Hasse diagram of type $A$ with unique minima.*

Another important result of this chapter is Theorem 3.2.1, which is a classification of small categories such that any indecomposable representation is an indicator representation. For both theorems, we will consider only pointwise finite-dimensional representations.

## 3.1 A-posets, G-posets and B-posets

Before stating and proving the two classification results, we need to introduce the notions of $A$-poset, $G$-poset, and $B$-poset, which are based on the definitions concerning switchbacks and the sets $\mathrm{Cmp}(p)$ (see Definition 2.2.3).

**Definition 3.1.1.** *Given a poset $P$, $p \in P$ is a **switchback** if it is a minimum or a maximum of* $\mathrm{Cmp}(p)$.

**Definition 3.1.2.** *An $A$-**poset** $X$ is a poset such that:*

- *$X$ is connected, i.e., every pair of points can be connected by a finite chain of comparable elements;*
- *If $x$ is not a switchback, then Cmp$(x)$ (see Definition 3.1.1) is totally ordered;*
- *If $x$ is a switchback, then Cmp$(x) \setminus \{x\}$ is either totally ordered, or a disjoint union of two totally ordered sets;*
- *$X \setminus \{x\}$ is not connected for each $x \in X$.*

**Remark 3.1.3.** *Notice that all quivers of type $A$ (finite and infinite) are $A$-posets. Moreover, this extends the notion of quivers of type $A$ to arbitrary cardinality by ensuring the exclusion of subcategories of type $D_4$ or $\tilde{A}$(finite or infinite) as follows.*

*From the second and third bullet points of the definitions, we see that Cmp$(x)$ contains at most two switchbacks different from $x$, hence it cannot contain $D_4$. On the other hand, by the last bullet point, it follows that $X$ cannot contain any $\tilde{A}_n$-quiver.*

**Definition 3.1.4.** *A connected poset $X$ is a glued poset (G-**poset**) if and only if it is a gluing of several total orders as in the Fig. 3.1. In detail:*

- *There exists $\{X_n\}_{n \in I}$ a set of totally ordered sets with $I \subset \bar{\mathbb{Z}} = \mathbb{Z} \cup \{+\infty, -\infty\}$ such that each $X_n$ is a full subcategory of $X$.*
- *Except for the first and last elements of $I$, if they exist, $X_n$ has both maximum $(M_n)$ and minimum $(m_n)$; while if $n$ is the minimum or the maximum of $I$, $X_n$ must have a maximum or a minimum.*
- *WLOG, we assume that for any $n$ in $I$, if $n$ is not a maximum on $I$, the consecutive element of $n$ in $I$ is $n+1$ and similarly, the previous is $n-1$ and that $0 \in I$.*
- *Except for the first and last elements of $I$, if they exist, $X_n$ has both maximum $(M_n)$ and minimum $(m_n)$.*
- *$Obj(X) = \sqcup_{n \in I} Obj(X_n)/\sim$, where for any the equivalence relation is the following.*
  *$\ldots m_{-3} \sim m_{-2}$, $M_{-2} \sim M_{-1}$, $m_{-1} \sim m_0$, $M_0 \sim M_1$, $m_1 \sim m_2$, $M_2 \sim M_3 \ldots$*
- *Apart from those induced by the equivalence relation, there are no morphisms between an element in $X_n$ and one in $X_{n'}$ if $n \neq n'$*

A $G$-poset can be seen as illustrated in Fig. 3.1, where any arrow represents a totally ordered set, which starts from its minimum and ends in its maximum.

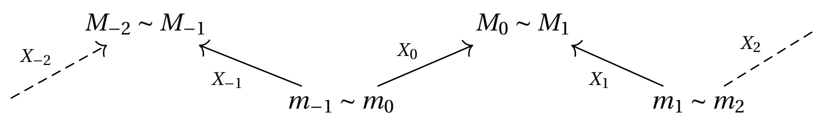



Figure 3.1: G-poset

This visualization induces a total order on the category, analogous to the lexicographic one. In detail:$x <_l y$ if

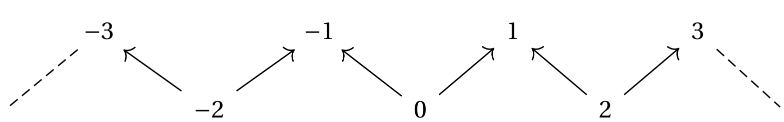
- $x \in X_n$ with $n$ even and $y \in X_n$ with $x < y$ or $y \in X_m$ with $m > n$
- $x \in X_n$ with $n$ odd and $y \in X_n$ with$x > y$ or $y \in X_m$ with $m > n$

The following proposition will be crucial in the proof of Theorem 3.2.2 because it will allow us to state that if a connected poset does not contain an $\tilde{A}$-quiver (finite or infinite), and it is not a $G$-poset, it contains $D_4$.

**Proposition 3.1.5.** *A category $X$ is a $G$-poset if and only if it is an $A$-poset.*

*Proof.* By the visualization in Fig. 3.1, it is clear that a $G$-poset is an $A-$poset. Thus, let us prove that if $X$ is an $A$-poset, then it is a $G$-poset.
**First claim: $X$ has countable many switchbacks.** Fix $x \in X$, and define $S_n(x)$ as the set of switchbacks reachable from $x$ with $n$ comparisons. Because any point is comparable with at most 2 switchbacks, then $S_{n-1}(x)$ and $S_n(x)$ differ by at most 2 elements. Since every switchback is in $S_n(x)$ for some $n$ because one can go from any point $x$ to $y$ by a finite number of comparable elements, the result follows.
**Second claim: $X$ is a $G$-poset.** Consider in $X$ the subposet $S$ given by only the switchbacks. We may assume that $S$ is non-empty, otherwise the category is totally ordered.

This poset $S$ is isomorphic to a (possibly infinite) zigzag, i.e. the following quiver.




To see this, let $f\colon S \to \mathbb{Z}$ be the following injective set map. It sends an arbitrary switchback $x$ to 0. Now, if $y \succ x$ and $x \prec y'$ are switchbacks, then let $f(y) = -1$ and

$f(y') = 1$. After, we can continue inductively on $S_n(x)$. For instance, in $S_2(x) \setminus S_1(x)$ we have at most two elements. If $z$ is connected to $y$, then let $f(z) = -2$, and if $z'$ is connected to $y'$, then $f(z') = 2$. Note that $z$ cannot be comparable with both $y$ and $y'$, because then $\mathrm{Cmp}(z) \setminus \{z\}$ would not be of the form as prescribed in the third bullet point of $A$-poset definition. By the second bullet point of the $A$-poset definition , if $x$ is not a switchback, $\mathrm{Cmp}(x)$ is a total order. It follows that if we consider two consecutive switchback $s, s'$ with $s < s'$ in the poset $S$, the elements $x \in \mathrm{Obj}(X)$ such that $s < x < s'$ form a total order. Now, it is enough to glue the totally ordered sets between the corresponding switchbacks to get a $G$-poset. □

**Definition 3.1.6.** *An $\tilde{A}$-**category** is a small category that can be constructed as follows. Start with a $G$-poset consisting of:*

- *a (finite or infinite) collection of total orders, equipped with a* first *total order $X_f$ and a* last *total order $X_l$;*
- *each total order has both a* maximum *and a* minimum *element.*

*This setup implies the existence of a first and last* switchback *(denoted $s_1$ and $s_0$, respectively); the $\tilde{A}$-category is obtained now establishing the equivalence relation $s_0 \sim s_1$.*

The last step before stating the two results of this chapter is the following definition.

**Definition 3.1.7.** *A connected poset $X$ is a $B$-poset if and only if it is a gluing of two total orders as in Fig. 3.2. In detail, there exists two totally ordered sets $X_0$ and $X_1$ , both with maximum (denoted as $M_0$ and $M_1$) and minimum (denoted as $m_0$ and $m_1$), such that :*

- *$X_0$ and $X_1$ are two full subcategories of $X$;*
- *$Obj(X) = Obj(X_0) \bigsqcup Obj(X_1) /_{m_0 \sim m_1,\, M_0 \sim M_1}$;*
- *the comparability relations are just those coming from the structure of $X_0$ and $X_1$.*

A $B$-poset can be seen as in the figure below, where the two arrows represent two totally ordered sets, which start from its minimum and end in its maximum.

$$\begin{tikzcd}
M_0 \sim M_1 \\
m_0 \sim m_1 \arrow[u, bend left, "X_0"] \arrow[u, bend right, "X_1"']
\end{tikzcd}$$

Figure 3.2: $B$-poset

## 3.2 Characterization theorems about indicator representations

We are now ready to state the two classifications. The first theorem is a classification of all small categories for which any indecomposable representation is thin, while the second one is the generalization to infinite categories of Theorem 3.0.1.

**Theorem 3.2.1.** *Given $X$ a small category, every pointwise finite-dimensional indecomposable representation of $X$ on* $\mathrm{vect}_{\mathbb{F}}$ *is an indicator representations if and only if $X$ is a disjoint union of $G$-posets and $B$-posets.*

**Theorem 3.2.2.** *Let $X$ be a small category. Then all pointwise finite-dimensional indecomposable additively **Set** realizable representations are indicator representations if and only if $X$ is equivalent to a poset category consisting of a disjoint union of*

1. *$G-posets$,*

2. *posets with a single maximum $M$, such that the poset without the maximum is a disjoint union of $G$-posets built with maximum two total orders. This means for every $G$-poset $C \subseteq X \setminus \{M\}$, if $C$ has a minimum, then the minimum is unique (i.e., $C$ is a total order, or $C = X_1 \cup X_2$ with $X_1$ and $X_2$ being total orders with the same minimum).*

### 3.2.1 Some preliminaries for the proofs

To proceed to the proof of the two theorems, we need some preliminary results. The first one concerns a classification of injective objects that we will use when we will deal with $B-$poset in the proof of Theorem 3.2.1.

**Theorem 3.2.3.** *(see [7, Proposition 3.2.5]) Let $P$ be a partially ordered set. Then for any $R \in \mathrm{Rep}(P, \mathrm{Vect}_{\mathbb{F}})$, the following are equivalent:*

1. *$R$ is injective and indecomposable.*

2. *$R \cong k_j$ for $j \subseteq P$ a subset which is a filtered ideal, that is a subset closed under smaller elements.*

The last things before the proofs are remarks about some specific quivers, which will be used in both classifications to exclude cases where $X$ has some specific full subcategory. The first one regards the Kronecker quiver, which is the following quiver.

$$\begin{tikzcd} a \\ b \arrow[u, bend left, "\beta"] \arrow[u, bend right, "\alpha"'] \end{tikzcd}$$

**Remark 3.2.4.** *Let us observe that the following representation $R$ of the Kronecker quiver is not an indicator representation and is additively-set realizable and indecomposable.*

$$\begin{tikzcd} F^2 \\ F \arrow[u, bend left, "\begin{bmatrix} 0 \\ 1 \end{bmatrix}"] \arrow[u, bend right, "\begin{bmatrix} 1 \\ 0 \end{bmatrix}"'] \end{tikzcd}$$

*Indecomposability: the sum of both the image of the maps in the representations is the entire $F^2$ and $R_b$ is one dimensional. Hence, if by contraddiction $R = S \oplus T$, $S_a = F$ and $T_a = 0$. By surjectivity of $R_\alpha \oplus R_\beta$, $S_b = F^2$, hence $T = 0$.*

*Additive set realizability: $R$ is the image of the following set representation through the* free*-functor.*

$$\begin{tikzcd} \{b, c\} \\ \{a\} \arrow[u, bend left, "a \to b"] \arrow[u, bend right, "a \to c"'] \end{tikzcd}$$

We will use this property of the Kronecker quiver in the proof of both theorems to say that for both classifications we have to consider just posets.

Moreover, notice that this representation can be extended to $\tilde{A}$-categories which are not a cycle. Let $X$ be a such $\tilde{A}$-category. We choose a sink $t$ and a source $s$. We define $R$ to be $F^2$ on $t$, $F$ on all other verteces and all the maps between 1-dimensional spaces the identities. Now, let us divide the elements in $X \setminus \{t\}$ which are comparable with $t$ in two sets $W$ and $Y$ such that in each set all elements are pairwise comparable.

For any element $w \in W$ and for any element $y \in Y$,

$$R_{wt} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } R_{yt} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

**Remark 3.2.5.** *This remark deals with the following poset.*

$$\begin{tikzcd}
\bullet & & \bullet & \\
& \bullet \arrow[ul] \arrow[ur] & \bullet \arrow[u] & \bullet \arrow[ul]
\end{tikzcd}$$

*It has an indecomposable non indicator additively Set-realizable representation, an example is*

$$\begin{tikzcd}
F & & F^2 & F \arrow[l, "\begin{bmatrix} 0 \\ 1 \end{bmatrix}"'] \\
& F^2 \arrow[ul, "\begin{bmatrix} 1 & 1 \end{bmatrix}"'] \arrow[ur, "Id"] & & F \arrow[ul, "\begin{bmatrix} 1 \\ 0 \end{bmatrix}"]
\end{tikzcd}$$

*Indeed the representation above is the image of the following set representation through the* free *functor.*

$$\begin{tikzcd}
\{c\} & & \{a,b\} & \{f\} \arrow[l, "f\to b"'] \\
& \{a,b\} \arrow[ul, "\substack{a\to c \\ b\to c}"'] \arrow[ur, "Id"] & & \{g\} \arrow[ul, "f\to a"]
\end{tikzcd}$$

The last result we need for the proofs is the classification of the indecomposable representations of $\tilde{A}_n$- quivers.

**Theorem 3.2.6.** *([20, Theorem 8.4]) Let $Q$ be an $\tilde{A}_n$-quiver and $R \in \mathrm{Rep}(Q, \mathrm{vect}_{\mathbb{F}})$. Then $R$ is indecomposable if and only if $R$ is of one of the following form.*

- ***String**: let $\gamma : \{1, 2, \ldots, m\} \to Q_0$ be an undirected path moving clockwise around $Q$. (Explicitly, $\gamma$ is given by adding a constant and considering the result modulo $n+1$.) Let*

  $$S_\gamma(v) = \gamma^{-1}(v) \cup \{*\} \in \mathrm{set},$$

  *which is a representation by saying that for $1 \le l < m$ we have $l \mapsto l+1$ or $l+1 \mapsto l$ (depending on the direction of the arrow $\alpha_{\gamma(l)}$, and interpreting the vertex and arrow indices cyclically). At the ends of the path we declare that if the arrow*

*$\alpha_{\gamma(1)-1}$ is oriented outward from $v_{\gamma(1)}$, then $S(\alpha_{\gamma(1)-1})(1) = *$, and symmetrically if the arrow $\alpha_{\gamma(m)}$ is oriented outward from $v_{\gamma(m)}$ then $S(\alpha_{\gamma(m)})(m) = *$.*

- ***Band:***

$$R \cong \begin{tikzcd} & V \arrow[r, dash, "id"'] & V \arrow[dd, dash, dotted, bend left] \\ V \arrow[ur, dash, "id"] \arrow[dr, dash, "\phi"'] & & \\ & V \arrow[r, dash, "id"] & V \end{tikzcd}$$

*where $\phi : V \to V$ is an isomorphism such that the following representation of the 1-loop quiver is indecomposable.*

$$\begin{tikzcd} V \arrow[loop right, "\phi"] \end{tikzcd}$$

*Equivalently, $V = F[X]/(p(X)^m)$ for a monic irreducible polynomial $p(X)$ and $\phi(v) = X \cdot v$.*

### 3.2.2 Proofs of the characterization theorems about indicator representations

**Proof of Theorem 3.2.1**

*Proof of the "$\Leftarrow$" direction.* Suppose that any indecomposable representation of $X$ is an indicator representation. By Remark 3.2.4, it is easy to check that $X$ has to be a poset. Indeed, if there are $x, y \in X$ such that $|\mathrm{Hom}(x, y)| \geq 2$, then the Kronecker quiver can be embedded in $X$. Let incl denote the inclusion functor from $\mathrm{Rep}(\text{Kronecker quiver}, \mathrm{vect}_{\mathbb{F}})$ to $\mathrm{Rep}(X, \mathrm{vect}_{\mathbb{F}})$ and $R' = \mathrm{incl}(R)$ where $R$ is the one in Remark 3.2.4. The representation $\mathrm{incl}(R)$ is indecomposable because otherwise an eventual decomposition would hold also for $R$. By the same argument, by the classifications of $D_4$ and $\tilde{A}_n$ indecomposable representations (Remark 1.5.8 and Theorem 3.2.6) $D_4$ and $\tilde{A}\_n$(with any orientation) cannot be a full subcategory of $X$. Analyzing $D_4$, it means that:

- No $x$ is smaller than 3 or more pairwise incomparable elements of $X$.
- If $x \in X$ is smaller than 2 incomparable elements then $x$ is minimal.

- If $x$ is greater than 2 incomparable elements then $x$ is maximal.
- No $x$ is greater than 3 or more pairwise incomparable elements of $X$.

Suppose that $X$ is not a totally ordered set (which is a G-poset). If there is an $M$ greater than two incomparable point, it is a maximum and it is the maximum of two totally ordered sets.

Suppose that these two totally ordered sets have another intersection apart $M$; by the third bullet point, it is the minimum of both. Hence we obtain two totally ordered sets ($X_0$ and $X_1$) with same maximum and same minimum, hence a $B$-poset. By the first and the last bullet point, there can not be other objects in $X$

Suppose that $X_0$ and $X_1$ have no other intersections apart from $M$. If they do not have a minimum, $X$ is a G-poset with $|I| = 2$. Otherwise, suppose that one of them (WLOG $X_1$) has a minimum, then, by bullet points above, it is the minimum of $X_1$ and, if it is comparable with other elements, it is the maximu of another totally ordered set ($X_2$). $X_1$ and $X_2$ cannot intersect in any other point, otherwise there would be a point lower than 2 incomparable elements which is not a minimum. Proceeding iteratively in this way with an eventual minimum of $X_2$ and an eventual maximum of $X_0$ and excluding the presence of other morphisms apart ones coming from the mentioned total orders we obtain that $X$ is a G-poset. Starting by supposing that there exists an $M$ which is greater than two incomparable elements, the procedure is analogous.

*Proof of the "$\Rightarrow$" direction.* Let $X$ be a G-poset and let $R$ be one of its indecomposable representations. Notice that a $G$-poset is a thread of $P$ with $P$ a quiver of type $A_n$, $A_\infty$ or $A_\infty^\infty$ (depending on how many total orders the $G$-poset is made of). It is evident that $FS(P)$ consist of quiver of type $A_n$, $A_\infty$ or $A_\infty^\infty$. Recall that the indecomposable pointwise finite-dimensional representations of such quivers are indicator representations. It follows, by Theorem 2.2.1 that $R$ is well described by a quiver of type $A_n$, $A_\infty$ or $A_\infty^\infty$ and the restriction of $R$ to such quiver is indecomposable. This means that $R$ is an indicator representation.

Suppose now that $X$ is a $B$-poset and $R$ one of its indecomposable representations. To prove that $R$ is thin, we first present a direct argument and then the approach analogous to the $G$-poset case.

Let $m$ and $M$ denote, respectively, the minimum and the maximum of $X$. Let us consider two cases: $R_{mM} \neq 0$ and $R_{mM} = 0$. If $R_{mM} \neq 0$, then we can embed $k_X$ (the indicator representation of the entire category $X$) in $R$. To see this monomorphism, it is enough to choose a vector $v \in R_m$ such that $R_{mM}(v) \neq 0$ and map the generator $1 \in k_X(x)$ to $R_{mx}(v)$ for any $x \in X$. Because $k_X$ is injective (by direct check or consequence of Theorem 3.2.3), the monomorphism splits and hence $k_X$ is a direct summand of $R$, because $R$ is indecomposable, $R = k_X$.

On the other hand, if $R_{mM} = 0$, we proceed as follows. Let us consider $\hat{X} := X \setminus \{M\}$: it is the following $G$−poset, where $X_0$ and $X_1$ are the total orders which compose $X$ and the maxima $M_0$ and $M_1$ on both sides are not included.

$$M_0 \xleftarrow[X_0]{} m \xrightarrow[X_1]{} M_1$$

Because we already completed the proof for $G$-posets, we can use it, obtaining that the restriction of $R$ to $\hat{X}$ is the direct sum of indicator representations $\bigoplus_{j\in J} k_j$ with $J$ a set of connected intervals of $\hat{X}$. If there is an interval $j \in J$, such that $M$ is not one of its endpoints, $k_j$ is a direct summand also of $R$. Hence, we suppose that, for any $j$, $j =< x, M_1)$ (type 1), $j = (M_0, x >$ (type 2) or $j = (M_0, M_1)$ (type 3) where $x \in \hat{X}$, $<$ means that the interval can be open or closed and $< x, y >$ denotes the interval made by all points between $x$ and $y$ referring to the lexicographicphic total order given by the diagram above representing $\hat{X}$. Now, it is enough to check that for any type of $j$ mentioned above, $k_j$ generates an interval representation which is a direct summand of $R$ (and hence equal to $R$). Notice that the first two cases are analogous; hence, we will analyze just the first one.

- Suppose that $j$ is of type 3. Because $R_{mM} = 0$, we can embed $k_j$ in $R$ using the analogous embedding we used for $k_X$ in the case $R_{mM} \neq 0$. As from Theorem 3.2.3 $j$ is injective, this embedding splits.

  The same argument also works if $j$ is of type 1 and $m \in j$.

- Suppose that $j =< x, M_1)$ is of type 1 and $m \notin j$. Consider $S$ the direct summand of $R|_{\hat{X}}$. If $R_{x'M}(v) = 0$ for each $v \in S_{x'}$ and $\forall x' \in j$, $k_j$ can be easily factored out by $R$ because this implies that the embedding $\iota : k_j \to R$ is well defined. The projection from $R$ to $k_j$ will also be well defined because, by the decomposition of $R|_{\hat{X}}$ for any $v \in \iota(k_j)_z$ with $z \in j$, $v \notin \mathrm{Im}(R_{yz})$ for any $y \notin j$.

- Suppose now that $j$ is still of type 1 but and $R_{xM}(v) \neq 0$ for some $v \in S_x$ ; this means that $R_{xM}(v) \neq 0$ for each $v \in S_x$ for each $x \in j$ because $S_x$ is one dimensional for each $x \in j$ and the maps are linear. This implies that $m \notin j$ (because $R_{mM} = 0$) and hence that $R^{-1}_{mx}(v) = \emptyset$ $\forall v \neq 0 \in S_x$ $\forall x \in j$. We consider now the decomposition of $R|_{\overline{X}}$ where $\overline{X}$ is the full subcategory of $X$ with objects $\mathrm{Obj}(X)\setminus\{m\}$. This is again a direct sum of interval representations. Among these interval representations, there is a representation $S'$ (with associated interval $j'$ such that $S'_x = S_x$ $\forall x \in j$ and because $S_{xM} \neq 0$ $\forall x \in j$, $S_M \neq 0$. Because

$R_{mM} = 0$, $R_{mx}^{-1}(v) = \emptyset\ \forall v \neq 0 \in x\ \forall x \in j'$. It follows that $S'$ can be split not just as a direct summand of $R|_{\overline{X}}$ but also as a direct summand of $R$. Hence $R = S'$, which is an interval representation.

□

**Alternative proof for "indecomposable representations of $B$-posets are thin"** Notice that a $B$-poset is a thread of a quiver $Q$ of type $\tilde{A}_2$ and that $FS(Q)$ consist of $\tilde{A}_n$-quivers hence, by Theorem 2.2.1, it is enough to analyze the indecomposable representations of $\tilde{A}_n$-quivers. By the classification Theorem 3.2.6 of indecomposable representations of $\tilde{A}_n$ quivers, their indecomposable representations are bands or strings. As our category is a poset, if the representation is a band, every map (including $\phi$) must be the identity. Thus, the restriction of $R$ to a subcategory of $X$ that well describes $R$ is an indicator representation; thus, $R$ is an indicator representation. On the other hand, string representations are poset representations if and only if they $m < n$, hence when they are thin. Thus, if the restriction of $R$ is a string, $R$ is an indicator representation. □

**Proof of Theorem 3.2.2**

*Proof of the "$\Leftarrow$" direction.* As one can see by reading the proof of [20, Theorem 7.1] in the finite case, the same implication—aside from some details concerning the existence of minima—can be carried out similarly in our (infinite) setting. However we will simplify it using Theorem 3.2.1.

By Theorem 3.2.1, if $X$ satisfies the condition 1, every indecomposable representation is thin and the result follows immediately. Let us suppose that $X$ satisfies the condition 2. Let $R$ be an indecomposable additively set-realizable representation and let $t$ denote the maximum point. By Lemma 1.3.5, $\dim(R_t) \leq 1$. If $\dim(R_t) = 0$, the result follows from the case of $G$-posets because the full subcategory made by $\{x \text{ such that } R_x \neq 0\}$ is a disjoint union of $G$-posets. Thus, we assume that $\dim(R_t) = 1$. Consider any connected component $C$ of $X \setminus \{t\}$ and $Y = C \cup \{t\}$. $C$ is a $G$-poset made by two total order with the same minimum or a total order. It follows that $Y$ is a total order or a $B$-poset. We claim that the restriction $R_{|Y}$ is indecomposable. Suppose, by contradiction, that $R_{|Y} = R' \oplus R''$ is a non-trivial decomposition and assume without loss of generality that $R'_t = 0$. Then, any split monomorphism $R' \hookrightarrow R_{|Y}$ extends to a split monomorphism $R' \hookrightarrow R$ by setting $R' = 0$ on all objects in $X \setminus C$. Hence $R_{|Y}$ is indecomposable for any connected component $C$. Hence, $R|_Y = R'$ and it is indecomposable, thus by Theorem 3.2.1, $R_{|Y}$ is an indicator representation for any connected component $C$. It follows that $R$ is an indicator representation. □

For the other implication, we need the following lemma.

**Lemma 3.2.7.** *([20, Lemma 7.2]) Let $X$ be a finite category with one object. Then all indecomposable additively Set-realizable representations are indicator representations only if the category has no morphisms apart from the identity.*

*Proof.* Let $M$ be a monoid. The regular representation $R$ in the set $M$ is

$$M \to \mathrm{Hom}_{\mathrm{Set}}(M, M) : m \mapsto (m \circ -).$$

In particular all summands of free$(R)$ are additively Set-realizable. On the other hand, because there is just one object, there is only one indicator representation, which maps every morphism to the identity. Sums of copies of this indicator representation will still map every morphism to the identity.

Thus, if every indecomposable additively Set-realizable representation is an indicator representation, then free$(R)$ is isomorphic to some vector space with all morphisms mapped to identity. However, base changes do not affect identities, and thus $R$ itself needs to map every morphism to the identity, i.e. $m \cdot n = n$ for all $m, n \in M$. Choosing $n = id$, it follows $m = id$, and thus $M = \{id\}$ as wanted. □

*Proof of the "⇒" direction.* By Remark 3.2.4 in conjuction with Theorem 1.4.5, we know that $X$ is a poset. By Lemma 3.2.7 and Theorem 1.4.5, the only morphism from an object to itself is the identity. By Proposition 1.5.9 except for the orientation where all arrows are ingoing. for any orientation of $D_4$, there is a non-indicator additive set-realizable representation. Hence, using again Theorem 1.4.5, it follows that

- No $x \in X$ is smaller than 3 or more pairwise incomparable elements of $X$.
- If $x \in X$ is smaller than 2 incomparable elements, then $x$ is minimal.
- If $x \in X$ is greater than 2 incomparable elements, then x is maximal.

By Remark 3.2.5 and using again Theorem 1.4.5, it follows that: if there is an element smaller than two different maxima, then $X$ is an $A$-poset or an $\tilde{A}$-category but the second possibility is excluded considering an extension of the representation in Remark 3.2.4 to $\tilde{A}_n$ as in the comment after the remark.

Suppose $X$ is not a $G$-poset, hence an $A$-category; by Remark 3.1.3, it follows that $X$ contains a quiver of type $D_4$, hence there is an element $M$ from which at least three chains of morphisms arrive/start. By the bullet points, this point is a maximum and, because there is not an element smaller than two different maxima, this maximum is unique. By Theorem 1.4.5, also for $X' = X \setminus \{M\}$ holds that if there is a maximum it is unique. By the third bullet point, we obtain that in $X'$ all objects smaller than a given point are linearly ordered. By the first bullet point, at most two chains of descending elements from distinct maxima in $X'$ can intersect, and by the second bullet point, the intersection point must be minimum. Hence we obtain the thesis. $\square$

**Some visualization.** We now illustrate the possible configurations of a connected component $C \subset X'$ when it contains infinitely many objects and it is discrete. If it contains an intersection of two linearly ordered chains, it is the union of two totally ordered set which intersect in the minimum, hence it is isomorphic to a $G$-poset with exactly 2 total orders (if it is discrete it is of type $A_\infty$ if a chain is finite and the other not and of type $A_\infty^\infty$ otherwise). If $C$ does not contain such intersection, it is a total order (it is of type $A_\infty$ with the arrows as in the third configuration below and if it has a minimum as in the fourth one).

From the finite case, for the second case, it changes that in the maximum point can have infinitely many incoming morphism and that for any component $C$, if it is discrete, we have the following four more configurations where the arrows in the infinite arms continues always in the same direction. If the vertices are not discrete the difference is that any arrow in the four configurations below must be thought as total orders and not morphism.

$$- - - - \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \bullet$$

$$- - - - \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \bullet - - - -$$

$$\bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet - - - -$$

$$- - - - \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \bullet$$

# Chapter 4

# Additive set realizability and Dynkin quivers

The aim of this chapter is to take another step toward proving that, for Dynkin quivers, the orientations that make the free additive functor surjective do not depend on the field. We already know this for $A_n$, since in that case all indecomposable representations are indicator representations and hence are additively set-realizable. By [20, Proposition 8.3], we also know it for $D_n$-quivers. Thus, it remains to prove it for $E_6$, $E_7$, and $E_8$. The aim of this chapter is to prove it for $E_6$.

Thus, in Theorem 4.1.1, we will generalize the classification proved in [20, Section 8.1.3] for the field $\mathbb{F}_2$ to any field.

## 4.1 Additive set realizability and $E_6$: classification of its orientations

**Theorem 4.1.1.** *If $X$ is a quiver which underlying graph is $E_6$, the functor* free *is additively surjective if and only if $X$ has one of the following structures:*

- *One arm of length two is completely oriented outwards,*
- *the arm of length one is oriented outwards, and at least one additional arrow is oriented outwards.*

*A complete list specifying all orientations of $E_6$ for which the functor* free *is additively surjective is provided in the Appendix.*

In the next remark we will see how most of the orientations for which free is not additive surjective can be interpreted in terms of hereditary sets, subsets of posets which are closed under greater elements.

**Remark 4.1.2.** *Recall that in a quiver $Q$, $H \subset Q_0$ is* ***hereditary*** *if for each $\alpha \in Q_1$, if the source of $\alpha$ is in $H$, then also its target is in $H$ (see [13, Definition 5.1]). It turns out that all the orientations of $E_6$ for which the functor free is not additively surjective—except for orientation 4 in the appendix (which corresponds to a poset with unique maximum at vertex 4)—are such that proper hereditary subsets of $Q_0$ characterize them. The diagrams below illustrate these cases: each one encodes a group of orientations, because the arrows labelled $\alpha$ and $\beta$ can have any direction.*

- $H := \{3\}$ hereditary

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r, dash, "\alpha"] & 2 \arrow[r] & 3 & 5 \arrow[l] \arrow[r, dash, "\beta"] & 6 \end{tikzcd}$$

- $H := \{3,5\}$ hereditary and $\{3\}$ not hereditary

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r, dash, "\beta"] & 2 \arrow[r] & 3 \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

- $H := \{2,3\}$ hereditary and $\{3\}$ not hereditary

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] & 5 \arrow[l] \arrow[r, dash, "\beta"'] & 6 \end{tikzcd}$$

- $H := \{2,3,5\}$ hereditary and $\{3\}$, $\{2,3\}$ and $\{3,5\}$ not hereditary

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

## 4.2 Some preliminaries before the proof

To prove Theorem 4.1.1, we need to recall two results from [20]. For the first one, concerning the edge contraction of a quiver, we refer to [20, Section 6.3].

**Definition 4.2.1.** *Let $X$ be a quiver and $\alpha$ an arrow between two distinct objects in $X$ (i.e., not a loop). The edge contraction of $X$ at $\alpha$ is given by the quiver $Y$ obtained from $X$ by removing the arrow $\alpha$ and identifying its two endpoints.*

**Theorem 4.2.2.** *If $c : X \to Y$ is an edge contraction, the restriction functor*

$$\mathrm{Rep}(Y, \mathrm{Vect}_{\mathbb{F}}) \to \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}}) \quad R \mapsto R \circ c$$

*preserves and reflects additively set realizability. In particular if* $\mathrm{free} : \mathrm{Rep}(Y, \mathrm{Set}) \to \mathrm{Rep}(Y, \mathrm{Vect}_{\mathbb{F}})$ *is not additively surjective, neither* $\mathrm{free} : \mathrm{Rep}(X, \mathrm{Set}) \to \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$ *is additvely surjective.*

The second result we need concerns a result about reversing arrows from a source with the property that it is the source of maximum two arrows, for which we refer to [20, Section 6.4].

**Definition 4.2.3.** *Let $d$ be a source of a quiver $Q$. We denote by $\mu_d(Q)$ the quiver obtained from $Q$ by reversing all arrows starting in $d$.*

**Theorem 4.2.4.** *Let $Q$ be a quiver and $d$ a source of $Q$ with at most two arrows starting in it. If the functor* $\mathrm{free}_* : \mathrm{Rep}(Q, \mathrm{Set}_*) \to \mathrm{Rep}(Q, \mathrm{Vect}_{\mathbb{F}})$ *is essentially surjective (additively surjective), then so is the functor* $\mathrm{free}_* : \mathrm{Rep}(\mu_d(Q), \mathrm{Set}_*) \to \mathrm{Rep}(\mu_d(Q), \mathrm{Vect}_{\mathbb{F}})$.

Because the additive images of $\mathrm{free}_*$ and free are the same, we will use Theorem 4.2.4 to reduce our study to few orientations of $E_6$ (because the others are obtainable from them by applying some $\mu_d$ with $d$ sources with degree $\leq 2$).

## 4.3 Proof of the classification theorem of $E_6$ orientation

**Proof of Theorem 4.1.1**

*Proof of the "$\Rightarrow$" direction.* Suppose that for the orientation of $X$, free is additively surjective. Assume, by contradiction, that $X$ has no arm completely oriented outward. Then the arm of length one, as well as at least one arrow in each arm of length two, must be oriented inward. Thus, we are considering the class of orientations that can be described via hereditary subsets of vertices (as detailed in Remark 4.1.2). For each such orientation, consider the edge contraction of $X$ at the arrows $\alpha$ and $\beta$ (as labeled in the diagrams of the remark). The resulting quiver is of type $D_4$ with all arrows oriented inwards. According to Theorem 4.2.2 and the non-surjectivity of free in this

$D_4$ configuration (details in Proposition 1.5.9), we obtain a contradiction. Suppose now that $X$ has the following orientation.

$$\begin{tikzcd} & & t & & \\ \bullet \arrow[r] & \bullet \arrow[r] & \bullet \arrow[u] & \bullet \arrow[l] & \bullet \arrow[l] \end{tikzcd}$$

By positive roots of $E_6$ ([15, pag 217]) and their correspondence with dimensions of indecomposable representations illustrated in the first chapter, we can notice that there exists an indecomposable representation $R$ of $X$ such that $R_t$ has dimension two. The contradiction now follows because $t$ is a terminal object of $X$, and by Lemma 1.3.5, any indecomposable representation assigning dimension greater than 1 to a terminal object cannot lie in the additive image of the free functor. Thus, $X$ can have just the orientations stated in the theorem. □

*Proof of the "$\Leftarrow$" direction.* **First step: setting the stage of the proof**

In this proof, we refer to the list of all possible orientations of the quiver $E_6$ provided in the appendix, where for each orientation, we indicate whether the corresponding free functor is additively surjective. We use this list to assign a label to each orientation of $E_6$ that we examine, which corresponds to the number in the list.

According to the enumeration in the appendix and the statement of the theorem, we have to prove that for all the 22 following orientations of $X$ free is additively surjective: 1, 2, 3, 5, 9, 10, 11, 12, 13, 17, 18, 19, 20, 21, 25, 26, 27, 28, 29, 30, 31, 32.

**Second step: reduction to four orientations**

In this step, applying many times Theorem 4.2.4 and some symmetry argument, we will obtain that it is enough to consider the orientations: 5, 11, 20 and 31. We resume the entire process in the table below, which has the following structure:

- in the first column are listed all orientations of $E_6$ for which we want to prove that free is additively surjective;
- the second column lists an orientation from which the desired property can be inferred for the orientation in that row;
- the third column explain why proving that free is additively surjective for the orientation in the second column is sufficient: **symmetry** means that the orientation are symmetric around the arrow between 3 and 4 and **reflecting on** $s$ means that we are using Theorem 4.2.4 with $s$ as source;
- the last column is the link between the orientation 5, 11, 20 or 31 and the orientation we are analyzing.

| **Orientations** | **Implied by** | **way** | **chain from 5, 11, 20 or 31** |
|---|---|---|---|
| 1 | 5 | reflecting on 4 | $5 \Longrightarrow 1$ |
| 2 | 3 | reflecting on 5 | $20 \Longrightarrow 3 \Longrightarrow 2$ |
| 3 | 20 | symmetry | $20 \Longrightarrow 3$ |
| 5 | / | / | / |
| 9 | 10 | reflecting on 6 | $11 \Longrightarrow 10 \Longrightarrow 9$ |
| 10 | 11 | reflecting on 5 | $11 \Longrightarrow 10$ |
| 11 | / | / | / |
| 12 | 20 | reflecting on 2 | $20 \Longrightarrow 12$ |
| 13 | 30 | symmetry | $31 \Longrightarrow 30 \Longrightarrow 13$ |
| 17 | 21 | reflecting on 4 | $5 \Longrightarrow 21 \Longrightarrow 17$ |
| 18 | 19 | reflecting on 2 | $20 \Longrightarrow 19 \Longrightarrow 18$ |
| 19 | 20 | reflecting on 6 | $20 \Longrightarrow 19$ |
| 20 | / | / | / |
| 21 | 5 | reflecting on 1 | $5 \Longrightarrow 21$ |
| 25 | 29 | reflecting on 4 | $31 \Longrightarrow 30 \Longrightarrow 29 \Longrightarrow 25$ |
| 26 | 27 | reflecting on 5 | $31 \Longrightarrow 27 \Longrightarrow 26$ |
| 27 | 31 | reflecting on 4 | $31 \Longrightarrow 27$ |
| 28 | 1 | symmetry | $5 \Longrightarrow 1 \Longrightarrow 28$ |
| 29 | 30 | reflecting on 6 | $31 \Longrightarrow 30 \Longrightarrow 29$ |
| 30 | 31 | reflecting on 5 | $31 \Longrightarrow 30$ |
| 31 | / | / | / |
| 32 | 5 | symmetry | $5 \Longrightarrow 32$ |

Table 4.1: Reduction to four orientations

**Third step: Proof that for the orientation 5,11,20 and 31 free is additively surjective**

Now, let us look again at the positive roots of $E_6$, to prove that for the orientations **5**,**11**,**20** and **31** free is additive surjective. By the list in [15, pag 217], apart from those arising from subquivers of type $A_n$ and $D_n$ the positive roots (and hence the dimensions of the indecomposable representations) of $E_6$ are the following:

$$\begin{matrix} 11111 & 11211 & 12211 & 11221 & 12221 & 12321 & 12321 \\ 1 & 1 & 1 & 1 & 1 & 1 & 2. \end{matrix} \tag{4.1}$$

We already know that representations arising from $A_n$-quivers are additively set realizable because they are thin. Ones from $D_n$ (specifically $D_4$ and $D_5$) are also additively set realizable because it is easy to check that, for these orientations, they have a multiplicative basis and hence we can apply Proposition 1.3.3 (see [20, Proposition 8.3]). For the representations corresponding to the first five roots listed above, we can apply the same argument as they are just extensions of representations of $D_5$ through some isomorphisms. For the last two representations, computer experiments confirm the existence of a multiplicative basis. The results of the computations are reported below: for each orientation labeled by $n$, the representation $n.1$ corresponds to the indecomposable representation associated with the sixth positive root in Eq. (4.1), while $n.2$ corresponds to the one associated with the last one.

## 5.1

$$\begin{tikzcd}
 & & \mathbb{F} \arrow[d, swap, "\begin{bmatrix}0\\0\\1\end{bmatrix}"] & & \\
\mathbb{F} \arrow[r, swap, "\begin{bmatrix}1\\0\end{bmatrix}"] & \mathbb{F}^2 \arrow[r, swap, "\begin{bmatrix}1&0\\0&1\\0&0\end{bmatrix}"] & \mathbb{F}^3 \arrow[r, swap, "\begin{bmatrix}1&0&0\\0&1&1\end{bmatrix}"] & \mathbb{F}^2 \arrow[r, swap, "\begin{bmatrix}1&1\end{bmatrix}"] & \mathbb{F}
\end{tikzcd}$$

## 5.2

$$\begin{tikzcd}
 & & \mathbb{F} \arrow[d, swap, "\begin{bmatrix}1&0\\0&0\\0&1\end{bmatrix}"] & & \\
\mathbb{F} \arrow[r, swap, "\begin{bmatrix}0\\1\end{bmatrix}"] & \mathbb{F}^2 \arrow[r, swap, "\begin{bmatrix}1&0\\0&1\\0&0\end{bmatrix}"] & \mathbb{F}^3 \arrow[r, swap, "\begin{bmatrix}1&0&0\\0&1&1\end{bmatrix}"] & \mathbb{F}^2 \arrow[r, swap, "\begin{bmatrix}1&1\end{bmatrix}"] & \mathbb{F}
\end{tikzcd}$$

## 11.1

$$\begin{tikzcd}
 & & \mathbb{F} & & \\
\mathbb{F} \arrow[r, swap, "\begin{bmatrix}0\\1\end{bmatrix}"] & \mathbb{F}^2 & \mathbb{F}^3 \arrow[l, "\begin{bmatrix}0&1&0\\1&0&1\end{bmatrix}"] \arrow[u, "\begin{bmatrix}0&0&1\end{bmatrix}"] & \mathbb{F}^2 \arrow[l, "\begin{bmatrix}0&0\\1&0\\0&1\end{bmatrix}"] \arrow[r, swap, "\begin{bmatrix}1&1\end{bmatrix}"] & \mathbb{F}
\end{tikzcd}$$

**11.2**

$$\begin{tikzcd}
 & & \mathbb{F}^2 & & \\
\mathbb{F} \arrow[r, "{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}"'] & \mathbb{F}^2 & \mathbb{F}^3 \arrow[l, "{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}}"'] \arrow[u, "{\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}}"] & \mathbb{F}^2 \arrow[l, "{\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}"'] \arrow[r, "{\begin{bmatrix} 0 & 1 \end{bmatrix}}"'] & \mathbb{F}
\end{tikzcd}$$

**20.1**

$$\begin{tikzcd}
 & & \mathbb{F} & & \\
\mathbb{F} & \mathbb{F}^2 \arrow[l, "{\begin{bmatrix} 1 & 0 \end{bmatrix}}"'] \arrow[r, "{\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}"'] & \mathbb{F}^3 \arrow[u, "{\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}}"] & \mathbb{F}^2 \arrow[l, "{\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}}"'] & \mathbb{F} \arrow[l, "{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}"']
\end{tikzcd}$$

**20.2**

$$\begin{tikzcd}
 & & \mathbb{F}^2 & & \\
\mathbb{F} & \mathbb{F}^2 \arrow[l, "{\begin{bmatrix} 1 & 1 \end{bmatrix}}"'] \arrow[r, "{\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}}"'] & \mathbb{F}^3 \arrow[u, "{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}}"] & \mathbb{F}^2 \arrow[l, "{\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}}"'] & \mathbb{F} \arrow[l, "{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}"']
\end{tikzcd}$$

**31.1**

$$\begin{tikzcd}
 & & \mathbb{F} \arrow[d, "{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}"'] & & \\
\mathbb{F} & \mathbb{F}^2 \arrow[l, "{\begin{bmatrix} 1 & 0 \end{bmatrix}}"'] & \mathbb{F}^3 \arrow[l, "{\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}}"'] & \mathbb{F}^2 \arrow[l, "{\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}}"'] \arrow[r, "{\begin{bmatrix} 1 & 1 \end{bmatrix}}"'] & \mathbb{F}
\end{tikzcd}$$

### 31.2

$$
\begin{tikzcd}
 & & & \mathbb{F}^2 \arrow[d, "\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}"'] & & \\
\mathbb{F} & \mathbb{F}^2 \arrow[l, "\begin{bmatrix} 1 & 1 \end{bmatrix}"] & & \mathbb{F}^3 \arrow[ll, "\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}"] & \mathbb{F}^2 \arrow[l, "\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}"] \arrow[r, "\begin{bmatrix} 1 & 0 \end{bmatrix}"'] & \mathbb{F}
\end{tikzcd}
$$

We can immediately notice that the standard basis of these representations is multiplicative, as each column of every matrix has all entries equal to 0 except for at most one entry equal to 1. This implies that each vector in the canonical basis of every vector space in the representation is mapped to another vector in the canonical basis. The GAP code used for these experiments is available at the following repository: `https://github.com/enry15/Thesis-coding`. Given the dimension vector of the target representation, the program executes a loop in which it generates random representations with the prescribed dimensions and endowed with a multiplicative basis. The process ends as soon as an indecomposable representation is found. The code works with the field of rational numbers but, checking that the endomorphism ring is $\mathbb{F}$, one can see that the representations found are indecomposable over any field. Here we see just the case of the representation 31.1.

An endomorphism of the representation 31.1 is a set of 6 vector space homomorphisms $\alpha, \gamma, \beta : \mathbb{F} \to \mathbb{F}$, $A, C : \mathbb{F}^2 \to \mathbb{F}^2$ and $B : \mathbb{F}^3 \to \mathbb{F}^3$ such that

- $\begin{bmatrix} 1 & 0 \end{bmatrix} A = \alpha \begin{bmatrix} 1 & 0 \end{bmatrix}$;
- $\begin{bmatrix} 1 & 1 \end{bmatrix} C = \beta \begin{bmatrix} 1 & 1 \end{bmatrix}$;
- $A \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} B$;
- $\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} C = B \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$;
- $B \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \gamma$,

where we identified the vector space homomorphism and the corresponding matrix. By the first bullet point, the first row of $A$ is $(\alpha, 0)$ and, by the last one, the central column of $B$ is $(0,\gamma,0)$. By the second bullet point, the second row of $C$ is $(\beta - c_{1,1}, \beta - c_{1,2})$. Hence, the third bullet point become

$$\begin{bmatrix} \alpha & 0 \\ a_{1,1} & a_{1,2} \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} b_{1,1} & 0 & b_{1,3} \\ b_{2,1} & \gamma & b_{2,3} \\ b_{3,1} & 0 & b_{3,3} \end{bmatrix}$$

Computing both products, it turns out that $b_{2,1} + b_{3,1} = 0$, $\gamma = \alpha = b_{2,3} + b_{3,3}$, $a_{2,2} = b_{1,1}$ and $0 = a_{2,1} = b_{1,3}$. Thus,

$$A = \begin{bmatrix} \alpha & 0 \\ 0 & a_{2,2,} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} a_{2,2} & 0 & 0 \\ b_{2,1} & \alpha & b_{2,3} \\ -b_{2,1} & 0 & \alpha - b_{2,3} \end{bmatrix}.$$

Thus, the fourth bullet point became

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} c_{1,1} & c_{1,2} \\ \beta - c_{1,1} & \beta - c_{1,2} \end{bmatrix} = \begin{bmatrix} a_{2,2} & 0 & 0 \\ b_{2,1} & \alpha & b_{2,3} \\ -b_{2,1} & 0 & \alpha - b_{2,3} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Computing both products, it turns out that $b_{2,1} = b_{2,3} = 0, c_{1,2} = 0$ and $c_{1,1} = a_{2,2} = \beta = \alpha$. Hence, $\alpha = \beta = \gamma$, $A = C = \alpha \mathrm{Id}_{\mathbb{F}^2}$ and $B = \alpha \mathrm{Id}_{\mathbb{F}^3}$. Thus, the endomorphism ring is isomorphic to $\mathbb{F}$ as wanted. □

### 4.3.1 Deeper explanation for the orientation 5

Although we have completed the classification of the $E_6$ orientations for which the free functor is additively surjective using computational arguments, we also provide, in the specific case of orientation 5, an explicit and deeper explanation of the result based on geometric arguments. We present this alternative in the next chapter (in Proposition 5.1.15). It relies on the theory of rooted tree quivers and on related tools introduced in [17] last year, which we will explore in detail in the next chapter.

# Chapter 5

# Posets with a terminal object

In this chapter, we describe the indecomposable additive set-realizable representations of posets with a terminal object. In the first part, we focus on finite posets, while in the second part we present a construction that preserves additive realizability even in the case of infinite posets.

## 5.1 Rooted Tree Quivers

### 5.1.1 Rooted Tree Quivers: Definitions and Basic Properties

**Definition 5.1.1.** *Let **Rtree** be the (not full) subcategory of Quiv whose objects are **rooted tree quivers** that is, quivers whose underlying graph is a tree with a distinguished terminal vertex (called the root).*

*A morphism $f\colon T \to T'$ in **Rtree** is a morphism of quivers in **Quiv** that maps the root of $T$ to the root of $T'$.*

*We sometimes write a rooted tree quiver as a pair $(Q,\sigma)$, where $Q$ is the quiver and $\sigma$ its root.*

**Remark 5.1.2.** *The quiver consisting of a single vertex and no arrows is an initial object in* Rtree.

**Definition 5.1.3.** *Let $Q$ be a rooted tree quiver. Define the category* $\mathrm{Rtree}_Q$ *as follows:*

*An object in* $\mathrm{Rtree}_Q$ *is a morphism $f\colon T \to Q$, where $T$ is a rooted tree quiver and $f \in \mathrm{Hom}_{\mathrm{Rtree}}(T,Q)$. For two such objects $f\colon T \to Q$ and $f'\colon T' \to Q$, a morphism $g\colon T \to T'$ in* $\mathrm{Rtree}_Q$ *is a morphism in* $\mathrm{Hom}_{\mathrm{Rtree}}(T,T')$ *such that $f' \circ g = f$. By slight* *abuse of notation, we often denote the object $(T, f\colon T \to Q)$ simply by $T$, and refer to it as a **rooted quiver over** $Q$.*

To reach our goal and identify the indecomposables of $\mathrm{Rep}(Q, \mathrm{Vect}_{\mathbb{F}})$, with $Q \in$ Rtree, we need the following equivalent definition of rooted tree quivers.

**Definition 5.1.4.** *An **inductive rooted tree quiver** is:*

- *Base case: either the quiver $*$ with one vertex and no edges;*
- *Inductive case: or a quiver of the form $G(Q_1, \dots, Q_k)$, where $Q_1, \dots, Q_k$ are inductive rooted tree quivers, and where $G(Q_1, \dots, Q_k)$ is constructed by taking the disjoint union of $Q_1, \dots, Q_k$, adjoining a new vertex $\sigma$, and adding a single edge from the root of $Q_i$ to $\sigma$, for each $1 \le i \le k$.*

We refer to $G$ as the gluing operation. In Fig. 5.1, there is a couple of examples of inductive rooted tree quivers, specifying from which $Q_i$ we are starting and the added vertex $\sigma$.

$$Q: \quad \begin{tikzcd} \bullet \arrow[r] & \bullet \\ \bullet \arrow[ur] & \end{tikzcd} \qquad\qquad Q_1: \quad \begin{tikzcd} \bullet \arrow[r] & \bullet \arrow[r] & \bullet \\ \bullet \arrow[ur] & & \end{tikzcd}$$

$$G(Q): \quad \begin{tikzcd} \bullet \arrow[r] & \bullet \arrow[r] & \sigma \\ \bullet \arrow[ur] & & \end{tikzcd} \qquad\qquad Q_2: \quad \begin{tikzcd} \bullet \arrow[r] & \bullet \end{tikzcd}$$

$$G(Q_1, Q_2): \quad \begin{tikzcd} \bullet \arrow[r] & \bullet \arrow[r] & \bullet \arrow[dr] & \\ \bullet \arrow[ur] & \bullet \arrow[r] & \bullet \arrow[r] & \sigma \end{tikzcd}$$

Figure 5.1: Examples of inductive tree quivers

By [17, Lemma 2.10], every rooted tree quiver is isomorphic to an inductive rooted tree quiver. Similarly, by [17, Lemma 2.12] every rooted tree quiver $T$ over a quiver $Q$ is equivalent to an inductive rooted tree quiver over $Q$ (following definition).

**Definition 5.1.5.** *([17, Definition 2.11]) Let $(Q, \sigma)$ be an inductive rooted tree quiver. An **inductive rooted tree quiver over** $Q$ is a pair $(T, f_T : T \to Q)$ such that:*

- *Base case: either $T = *$ and $f_T = u_* : * \to Q$;*

- *Inductive case: or* $Q = G(Q_1, \dots, Q_k)$, $T = G(T_1^\bullet, \dots, T_k^\bullet)$, *and* $f = G(f_1^\bullet, \dots, f_k^\bullet)$, *where, for each* $1 \le i \le k$, *we have that* $T_i^\bullet = \{T_i^1, \dots, T_i^{\ell_i}\}$ *is a (possibly empty) list of rooted tree quivers, and* $f_i^\bullet = \{f_i^1 : T_i^1 \to Q_i, \dots, f_i^{\ell_i} : T_i^{\ell_i} \to Q_i\}$ *is a (possibly empty) list of rooted tree quiver morphisms. The rooted tree quiver* $T$ *is constructed using the same operation G as in Definition 5.1.4, that is*

$$G(T_1^\bullet, \dots, T_k^\bullet) := G(T_1^1, \dots, T_1^{\ell_1}, \dots, T_k^1, \dots, T_k^{\ell_k}),$$

  *resulting in a rooted tree quiver with root* $\tau$. *The morphism* $G(f_1^\bullet, \dots, f_k^\bullet)$ *is uniquely characterized by mapping* $\tau$ *to* $\sigma$, *and by coinciding with* $f_i^j$ *when restricted to* $T_i^j$ *for every* $1 \le i \le k$ *and* $1 \le j \le \ell_i$.

**Definition 5.1.6.** *Let* $Q$ *be a quiver and let* $(T, f_T)$ *be a quiver over* $Q$. *It induces a set representation* $R$ *of* $Q$ *such that for each vertex* $x$ *of* $Q$

$$R_x = \{y \in T \text{ such that } y \in f_0^{-1}(x)\}$$

*and for each arrow* $\alpha$ *of* $Q$,

$$R_\alpha = \Sigma_{\beta \in f^{-1}(\alpha)} \beta.$$

*The linearization* $L(T)$ *of* $T$ *is* $\mathrm{free}(R)$.

**Definition 5.1.7.** *Let* $Q$ *be a quiver and let* $R$ *be one of its representations on* $\mathrm{vect}_{\mathbb{F}}$. $R$ *is a linearized rooted tree quiver over* $Q$ *if it is isomorphic to* $L(T)$ *with* $T$ *a rooted tree quiver over* $Q$.

The Fig. 5.2 is an example of linearization of a rooted tree quiver $T$ over a rooted tree quiver $Q$. Before the next definition, let us recall that a **preorder** is a set equipped with a reflexive and transitive relation. It differs from a partially ordered set (poset) in that it does not require antisymmetry; thus, $x \le y$ and $y \le x$ does not imply $x = y$.

**Definition 5.1.8.** *([17, compare with Proposition 4.1 and Definition 2.14]) Given a rooted tree quiver* $(Q, \sigma)$, *let us define the following preorder in* $\mathrm{Rtree}_{/Q}$. *Given a rooted tree quiver* $Q$ *and a couple of rooted tree quivers* $S, T$ *over* $Q$, $S \le_Q T$ *if one of the following equivalent conditions holds:*

- $\mathrm{Hom}_{\mathrm{Rtree}_{/Q}}(S, T) \neq \emptyset$;
- *There exists a morphism* $\phi$ *from* $L(S)$ *to* $L(T)$ *such that* $\phi_\sigma : L(S)_\sigma \to L(T)_\sigma$ *is not* 0.

$$T: \quad \begin{tikzcd} & 2 \arrow[r] & 3 \\ & 2 \arrow[ur] & \end{tikzcd} \qquad L(T): \quad \begin{tikzcd} 0 \arrow[r, "0"] & \mathbb{F}^2 \arrow[r, "{\begin{bmatrix}1 & 1\end{bmatrix}}"] & \mathbb{F} \end{tikzcd}$$

$$Q: \quad \begin{tikzcd} 1 \arrow[r] & 2 \arrow[r] & 3 \end{tikzcd}$$

Figure 5.2: Example of linearization of a rooted tree quiver over a rooted tree quiver. The quiver $T$ is shown on the top left, with its vertices labeled by their image in $Q$ under the structure morphism $f: T \to Q$ and the quiver $Q$ is shown below. The linearization $L(T)$ is depicted on the right: to each vertex $i$ of $Q$, it assigns a vector space whose dimension equals the number of vertices in $T$ mapping to $i$; the arrows of $Q$ induce linear maps between these vector spaces, defined by identity inclusions. We will adopt the convention of labeling the vertices of a rooted quiver over $Q$ by their image in $Q$ throughout the rest of the text

By the first condition of Definition 5.1.8, it follows that if $(S, f_S : S \to Q), (T; f_T : T \to Q)$ are two rooted quivers over $Q$ such that there exist $x, y \in Q_0$ such that $x \in \operatorname{im}(f_T), x \notin \operatorname{im}(f_S), y \in \operatorname{im}(f_S)$ and $y \notin \operatorname{im}(f_T)$, then $S$ and $T$ are incomparable.

**Definition 5.1.9.** *([17, Definition 2.15]) A rooted tree quiver $T$ over a rooted tree quiver $Q$ is* reduced *if:*

- *Base case: either $T = *$;*
- *Inductive case: or the following holds. We have $Q = G(Q_1, \dots, Q_k)$, $T = G(T_1^\bullet, \dots, T_k^\bullet)$, with $T_i^\bullet = \{T_i^1, \dots, T_i^{\ell_i}\}$ a (possibly empty) list of reduced rooted tree quivers over $Q_i$ for every $1 \le i \le k$. And, moreover, for every $1 \le i \le k$ and every $1 \le j, j' \le \ell_i$, if $j \neq j'$, the rooted tree quivers $T_i^j$ and $T_i^{j'}$ over $Q_i$ are incomparable with respect to $\preceq_{Q_i}$.*

In the following, provided a rooted tree quiver $Q$ and $x \in Q_0$, $Q_{\le x}$ is the full subquiver of $Q$ consisting of all elements lower than $x$. Notice that $(Q_x, x)$ is a rooted tree quiver. Given a representation $R$ of $Q_{\le x}$, $\overline{R}$ is the following representation of $X$ induced by the inclusion of $Q_{\le x}$ in $Q$: $\overline{R}_y = R_y$ if $y \le x$ and $\overline{R}_y = 0$ otherwise; $\overline{R}_\alpha = R_\alpha$ if $\alpha$ is in $Q_{\le x}$ and $\overline{R}_\alpha = 0$ otherwise.

**Definition 5.1.10.** *(see [17, Definition 2.22,2.23 and Lemma 2.24]) Let $Q$ be a rooted tree quiver; $R \in \operatorname{Rep}(Q, \mathrm{Vect}_{\mathbb{F}})$ is a rooted tree module if there exists an $x \in Q_0$ such that $(T, f_T : T \to Q_{\le x})$ a rooted tree quiver over $Q_{\le x}$ such that $R \cong \overline{L(T)}$.*

We now introduce the reduced rooted tree modules, which will be the modules that determine the additive image of a rooted tree quiver.

**Definition 5.1.11.** *([17, Definition 2.25]) Let $Q$ be a rooted tree quiver, $R \in \mathrm{Rep}(Q, \mathrm{Vect}_{\mathbb{F}})$ is a reduced rooted tree module if there exists an $x \in Q_0$ and a reduced rooted tree quiver $(T, f_T : T \to Q_{\leq x})$ over $Q_{\leq x}$ such that $R \cong \overline{L(T)}$.*

### 5.1.2 Additive image of free for rooted tree quivers

The next theorem is a combination of [17, Theorem A], [17, Theorem B] and [17, Corollary C] with the fact that the 0−homology functor and the free functor have the same essential image (and hence also additive image). Moreover, we consider free as functor from $\mathrm{Rep}(Q, \mathrm{set})$ to$\mathrm{Rep}(Q, \mathrm{vect}_{\mathbb{F}})$, i.e. working on finite-set and pointwise finite-dimensional representations.

**Theorem 5.1.12.** *(see [17, Theorem A], [17, Theorem B] and [17, Corollary C]) Let $Q$ be a finite rooted tree quiver. Then:*

- *every rooted tree module over $Q$ decomposes as a direct sum of reduced tree modules over $Q$;*
- *the essential image of* free *consists of the finite direct sum of linearizations of rooted tree quivers over $Q$;*
- *the additive image of* free *consists of the finite direct sum of rooted tree modules over $Q$;*
- *the additive image of* free *is of finite type (i.e. it contains a finite number of indecomposable) and all the indecomposables are the reduced rooted tree modules over $Q$.*

**Corollary 5.1.13.** *Let $Q$ be a rooted finite tree quiver.* free *is essentially (additive) surjective if all the indecomposable in* $\mathrm{Rep}(Q, \mathrm{vect}_{\mathbb{F}})$ *are linearizations of reduced rooted tree quiver ( reduced rooted tree modules).*

By this corollary, it follows readily that the unique rooted tree quiver such that free is essentially surjective is the trivial quiver with $*$ one point and no arrow. Indeed as soon as we have another point $x$ apart from the root, we can consider reduced rooted tree quiver over $Q_{\leq x}$, which correspond to reduced rooted tree module, that are not reduced rooted quiver, over $Q$. The following corollary and the next section present some key consequences of Theorem 5.1.12.

**Corollary 5.1.14.** *Let $Q$ be a finite rooted tree quiver. The essential and additive images of* $\mathrm{free}: \mathrm{Rep}(Q,\mathrm{set}) \to \mathrm{Rep}(Q,\mathrm{vect}_{\mathbb{F}})$ *do not depend on the field* $\mathbb{F}$.

*Proof.* By Theorem 5.1.12, the essential image (additive image) consists of all finite directs sums of linearizations of reduced rooted tree quiver (all finite direct sums of all reduced rooted tree modules). Because the reduced rooted tree modules are in 1 to 1 correspondence with the reduced rooted tree quiver over $Q_{\leq x}$ for all $x \in Q_0$, it is enough to observe that for any rooted tree quiver, the reduced tree quiver over do not depend on the field, which is clear from Definition 5.1.9 and Definition 5.1.8. □

### 5.1.3 Explicit construction of reduced rooted tree quivers

A rooted tree quiver, by its inductive definition, is the result of $G(Q_1,\dots,Q_k)$ for some rooted tree quivers $Q_1,\dots,Q_k$. This means that, by knowing the indecomposables in the additive image of free for $Q_1,\dots,Q_k$, we can understand which ones occur for $G(Q_1,\dots,Q_k)$.

To better understand the indecomposables in the additive image, we now construct them explicitly. Let $T_i^\bullet$ be the set of all the reduced rooted tree modules over $Q_i$ and $S_i^\bullet$ be the set of all reduced rooted tree quivers over $Q_i$. Notice that, for any $S_i^j \in S_i^\bullet$, $L(S_i^j) \in T_i^\bullet$. An indecomposable representation in the additive image of $\mathrm{free}: \mathrm{Rep}(Q,\mathrm{set}) \to \mathrm{Rep}(Q,\mathrm{vect}_{\mathbb{F}})$ has one of the following shapes:

- either it is isomorphic to $\overline{T}_i^j$ for a $T_i^j \in T_i^\bullet$.
- or it is isomorphic to $L(G(S_1^1,\dots,S_1^{l_1},\dots,S_k^1,\dots,S_k^{l_k}))$ where $(S_1^1,\dots,S_1^{l_1})$ is a possibly empty list in $S_i^\bullet$ such that its elements are pairwise incomparable.

We now apply the classification to two concrete examples: a quiver of type $E_6$ and one of type $E_7$.

**Proposition 5.1.15.** *The functor* free *for the following rooted tree quivers is additive surjective for any field* $\mathbb{F}$.

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd} \qquad \begin{tikzcd} & & 4 \arrow[d] & & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \arrow[r] & 7 \end{tikzcd}$$

*Proof.* In this proof, we denote $E_6$ with the above orientation as $\overline{E}_6$ and $E_7$ with the above orientation as $\overline{E}_7$. As stated in [20, Subsection 8.1.3], for both $\overline{E}_6$ and $\overline{E}_7$, by computational experiments through the code in [11], one obtains that free is additive surjective for the field $\mathbb{F} = \mathbb{Z}/2\mathbb{Z}$. By Corollary 5.1.14, we obtain that it is additive surjective for any field.

For the quiver $\overline{E}_6$ we see also a direct approach, while we omit it for $E_7$ because it is the same story but with one more inductive step and much more cases (as $E_7$ has more positive roots than $E_6$.

By [15, Subsection 8.3.3], the positive roots of $E_6$ (and hence the dimensions of the indecomposable representations), which do not arise from subquivers of type $D_n$ or $A_n$, are

$$\begin{matrix} 11111 & 11211 & 12211 & 11221 & 12221 & 12321 & 12321 \\ 1 & 1 & 1 & 1 & 1 & 1 & 2 \end{matrix} \tag{5.1}$$

The representations corresponding to the positive roots rising from quivers of type $A_n$ are easily additive set realizable because they are indicator representations, those arising from $D_4$ are also additive set realizable by Proposition 1.5.9 because the $D_4$-subquivers of $\overline{E}_6$ do not have all ingoing arrows. For ones rising from $D_5$-subquivers, with the same argument used for $D_4$ in Proposition 1.5.9 (see [20, Proposition 8.3]), it turns out that for the $D_5$-subquivers of $\overline{E}_6$, free is additive surjective.

For the representations corrisponding to the positive roots in Eq. (5.1), we will see that can be described as in the second bullet point of the list before the proposition (hence they are linearizations of reduced rooted tree quiver); thus our aim is finding a reduced rooted tree quivers over $\overline{E}_6$ for any positive root, with the property that the cardinality of the image of each vertex is its coordinate in the positive root analyzed. Observe that $\overline{E}_6 = G(\overline{D}_5)$, where $\overline{D}_5$ is $D_5$ with the following orientation.

$$\begin{tikzcd} & & 4 \arrow[d] & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \end{tikzcd}.$$

Notice that the indecomposable representations corrisponding to the first three positive roots in Eq. (5.1) are simply $L(G(T))$ for some reduced rooted tree quiver $T$ over $\overline{D}_5$. For the last four reduced rooted tree quivers, we see the explicit construction; in the figures below, for any rooted tree quiver over $\overline{E}_6$ or $\overline{D}_5$ we label the vertices as their image like in Fig. 5.2.

**About the $4^{th}$ positive root:** The following is the reduced rooted tree quiver corresponding to this root:

$$\begin{tikzcd}
1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \\
 & 4 \arrow[r] & 3 \arrow[r] & 5 \arrow[ur] &
\end{tikzcd}$$

It is reduced because it is equal to $G(T_1, T_2)$ where $T_1$ and $T_2$ are the following incomparable reduced rooted tree quivers over $\overline{D}_5$.

$$T_1: \quad 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 5 \qquad\qquad T_2: \quad 4 \longrightarrow 3 \longrightarrow 5$$

They are reduced because each vertex of $\overline{D}_5$ has at most one vertex in its image. T hey are incomparable because the image of $T_1$ has a vertex over 1 and no vertices over 4 while $T_2$ has a vertex over 4 and no vertices over 1.

**About the $5^{th}$ positive root:** The following is the reduced rooted tree quiver corresponding to this root:

$$\begin{tikzcd}
1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \\
 & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[ur] & \\
 & & 4 \arrow[u] & &
\end{tikzcd}$$

It is reduced because it is equal to $G(T_1, T_2)$ where $T_1$ and $T_2$ are the following incomparable reduced rooted tree quivers over $\overline{D}_5$. $T_1$ and $T_2$ are reduced and incomparable for the same reason of the previous case.

$$T_1: \quad 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 5 \qquad\qquad T_2: \quad \begin{tikzcd}
2 \arrow[r] & 3 \arrow[r] & 5 \\
 & 4 \arrow[u] &
\end{tikzcd}$$

**About the $6^{th}$ positive root:** the following is the reduced rooted tree quiver corresponding to this root:

$$\begin{tikzcd}
1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \\
 & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[ur] & \\
 & 4 \arrow[r] & 3 \arrow[ur] & &
\end{tikzcd}$$

It is reduced because it is equal to $G(T_1, T_2)$ where $T_1$ and $T_2$ are the following incomparable reduced rooted tree quivers over $\overline{D}_5$.

$$T_1 : \quad 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 5 \qquad T_2 : \quad \begin{tikzcd} 2 \arrow[r] & 3 \arrow[r] & 5 \\ 4 \arrow[r] & 3 \arrow[ur] & \end{tikzcd}$$

$T_1$ and $T_2$ are incomparable for the same reason of the $4^{th}$ root-case. $T_1$ is still reduced as it is the same of previous cases. To see that $T_2$ is reduced we can see $\overline{D}_5$ as $G(\overline{A}_4)$ with $\overline{A}_4$ the $A_4$ with the orientation

$$1 \longrightarrow 2 \longrightarrow 3 \longleftarrow 4$$

and $T_2$ as $G(T_2', T_2'')$ with

$$T_2' : \quad 2 \longrightarrow 3 \qquad T_2'' : \quad 4 \longrightarrow 3$$

which are reduced and incomparable for the same reason of the previous pairs.

**About the $7^{th}$ positive root:**the following is the reduced rooted tree quiver corresponding to this root:

$$\begin{tikzcd} & 4 \arrow[dr] & & & \\ & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[ur] & \\ & 4 \arrow[r] & 3 \arrow[ur] & & \end{tikzcd}$$

]It is reduced because it is equal to $G(T_1, T_2)$ where $T_1$ and $T_2$ are the incomparable reduced rooted tree quivers over $\overline{D}_5$ below. They are incomparable because $\mathrm{Hom}_{\mathrm{Rtree}/Q}(T_1, T_2) = 0$ because 1 is in the image of $T_1$ and not in the image of $f_{T_2}$; $\mathrm{Hom}_{\mathrm{Rtree}/Q}(T_2, T_1) = 0$ because the unique map that would work for vertices does not preserves the arrows endpoints. $T_1$ is reduced for the same reason of previous cases and $T_2$ is reduced because of the same reason of $T_2$ in the $6^{th}$ root-case.

$$T_1 : \quad \begin{tikzcd} 2 \arrow[r] & 3 \arrow[r] & 5 \\ & 4 \arrow[u] & \end{tikzcd} \qquad T_2 : \quad \begin{tikzcd} 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \\ & 4 \arrow[r] & 3 \arrow[ur] & \end{tikzcd}$$

□

## 5.2 Joined representations

In this section we see a construction which glues two representations of posets with a maximum and preserves the additively set realizability.

**Definition 5.2.1.** *Consider two posets $X$ and $Y$, each with a terminal object ($t_X$ and $t_Y$). We define the* ***joined category*** *$J_{XY}$ as follows:*

- $\mathrm{Obj}(J_{XY}) = \big(\mathrm{Obj}(X) \sqcup \mathrm{Obj}(Y)\big) / \sim$ *with equivalence relation $t_X \sim t_Y$.*
- *the morphisms, and hence the composition, in $J_{XY}$ are the same as the categories $X$ and $Y$.*

*We denote $t_J$ the terminal object of the category $J_{XY}$.*

**Construction 5.2.2.** *Let us consider two posets $X$ and $Y$ with respective terminal objects ($t_X$ and $t_Y$) and a pair of representations $R \in \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$ and $S \in \mathrm{Rep}(Y, \mathrm{Vect}_{\mathbb{F}})$. Let us observe that they induce a* ***joined representation*** *$T \in \mathrm{Rep}(J_{XY}, \mathrm{Vect}_{\mathbb{F}})$. If $R_{t_X} = S_{t_Y}$, homomorphism and vector spaces are the same of $R$ and $S$ (i.e.$T_x := R_x$ for any $x \in \mathrm{Obj}(X)$, $T_y := S_y$ for any $y \in \mathrm{Obj}(Y)$, $T_\alpha = R\alpha$ if $\alpha$ is morphism in $X$ and $T_\alpha = S_\alpha$ if $\alpha$ is a morphism in $Y$).*
*On the other hand, if $R_{t_X} \neq S_{t_Y}$, by Lemma 1.3.5, one has dimension $1$ and other has dimension $0$. In this case, the unique difference is that, wlog if $R_{t_X} = 0$, $T_{xt_J} = 0$ but $T_{t_J}$ is one dimensional.*

*Going through the proof, it also turns out that if $R$ is direct summand of $\mathrm{free}(N)$, $N$ can be chosen such that $N_t$ is a singleton (set with just one element).*

**Proposition 5.2.3.** *Consider two posets $X$ and $Y$ as in Definition 5.2.1. Let $R \in \mathrm{Rep}(X, \mathrm{Vect}_{\mathbb{F}})$ and $S \in \mathrm{Rep}(Y, \mathrm{Vect}_{\mathbb{F}})$ be indecomposable additive Set-realizable. Then also the induced representation $T$ on $J_{XY}$ is additive Set-realizable.*

Before the proof, we need the following lemma.

**Lemma 5.2.4.** *Let $X$ be a poset, and suppose it has a terminal object $t$. Let $R, S : X \to \mathrm{Vec}_F$ be two representations such that $R_t$ and $S_t$ are one-dimensional vector spaces. Suppose that $R$ is a direct summand of $S$, i.e., there exists a split monomorphism $\phi : R \to S$.*

*Then, $\phi$ can be chosen so that $\phi_t : R_t \to S_t$ is the identity.*

*Proof.* ( *Proposition 5.2.3*) We pick $N$ and $M$ two set representations of $X$ and $Y$ with $N(t_X)$ and $N(t_Y)$ singletons such that $R$ is a direct summand of free$(N)$ and $S$ is a direct summand of free$(M)$. As we defined the representation $T$ in Construction 5.2.2, we can define a set-representation $P$ on $J_{XY}$ induced by $N$ and $M$. Namely $P_x := N_x$ for any $x \in \mathrm{Obj}(X)$, $P_y := M_y$ for any $y \in \mathrm{Obj}(Y)$, $P_\alpha = N_\alpha$ if $\alpha$ is morphism in $X$ and $P_\alpha = M_\alpha$ if $\alpha$ is a morphism in $Y$.

Now, It is enough to check that $T$ is a direct summand of free$(P)$. We consider $\gamma$ a split embedding of $R$ in free$(N)$ and $\delta$ a split embedding of $S$ in free$(M)$. Using the lemma we choose them to be the identity in the terminal objects $t_X$ and $t_Y$; it follows that also the left inverses of the embeddings are the identity in the terminal objects. We define $j$ to be an embedding of $T$ in free$(P)$ as $j_x = \gamma_x$ for any $x \in \mathrm{Obj}(X)$ and $j_y = \delta_y$ for any $y \in \mathrm{Obj}(Y)$. To check that $j$ splits, it is enough to define a projection $\pi$ in the same way we defined $j$ but starting from the left inverses of $\delta$ and $\gamma$. □

Let us see an example of application of this proposition.

**Example 5.2.5.** *Let us consider the following two posets with a terminal object: the first is a thread quiver of $D_4$, with the $T_\alpha = [0,1]$ for each arrow $\alpha$ and and the second is $E_7$ with the orientation as in Proposition 5.1.15. We label them respectively $D_4^t$ and $\overline{E}_7$.*

$$\begin{tikzcd} & 4' \arrow[d, "{[0,1]_\beta}"'] & \\ 1' \arrow[r, "{[0,1]_\alpha}"'] & 2' \arrow[r, "{[0,1]_\gamma}"'] & 3' \end{tikzcd} \qquad \begin{tikzcd} & & 4 \arrow[d] & & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \arrow[r] & 7 \end{tikzcd}$$

*Let us consider the following representation $R$ of $\overline{E}_7$ on* $\mathrm{vect}_{\mathbb{F}}$

$$R: \begin{tikzcd} & & \mathbb{F}^2 \arrow[d] \arrow[d, phantom, shift left=4ex, "\begin{bmatrix}1&0\\0&0\\0&1\end{bmatrix}"] & & & \\ \mathbb{F} \arrow[r, "\begin{bmatrix}1\\0\end{bmatrix}"'] & \mathbb{F}^2 \arrow[r, "\begin{bmatrix}1&0\\0&1\\0&0\end{bmatrix}"'] & \mathbb{F}^3 \arrow[r, "\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}"'] & \mathbb{F}^3 \arrow[r, "\begin{bmatrix}1&0&0\\0&1&1\end{bmatrix}"'] & \mathbb{F}^2 \arrow[r, "\begin{bmatrix}1&1\end{bmatrix}"'] & \mathbb{F} \end{tikzcd}$$

*and the representation $S$ on $D_4^t$ defined as follows.*

$$\begin{cases} S_x = \mathbb{F} & \text{if } x \in [0,0.5]_\alpha, [0,0.5]_\beta, \text{ or } [0.5,1]_\gamma \\ S_x = \mathbb{F}^2 & \text{if } x \in (0.5,1]_\alpha, (0.5,1]_\beta, \text{ or } [0,0.5)_\gamma \end{cases} \qquad \textit{and}$$

$$
\begin{cases}
S_{xy} = Id & \textit{if } S_x = S_y \\
S_{xy} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \textit{if } x \in [0, 0.5]_\alpha \textit{ and } S_y = \mathbb{F}^2 \\
S_{xy} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} & \textit{if } x \in [0, 0.5]_\beta \textit{ and } S_y = \mathbb{F}^2 \\
S_{xy} = \begin{bmatrix} 1 & 1 \end{bmatrix} & \textit{if } S_x = \mathbb{F}^2 \textit{ and } y \in [0.5, 1]_\gamma
\end{cases}
$$

*$R$ is indecomposable because is in the representation* 5.1 *in the proof of Theorem $E_6$ extended with an isomorphism from the vertex* 3 *to* 5 *and it is additively set realizable as we proved in Proposition 5.1.15 that* free *is additive surjective for $\overline{E}_7$. $S$ is indecomposable because it is well described by $D_4$ and the restriction of $S$ to $D_4$ is the following indecomposable representation.*

$$
\begin{tikzcd}
 & 4 \arrow[d, "\begin{bmatrix} 0 \\ 1 \end{bmatrix}"'] & \\
1 \arrow[r, "\begin{bmatrix} 1 \\ 0 \end{bmatrix}"'] & 2 \arrow[r, "\begin{bmatrix} 1 & 1 \end{bmatrix}"'] & 3
\end{tikzcd}
$$

*We already discussed the additive set realizability of this representation in Proposition 1.5.9. Thus, we can apply Proposition 5.2.3 and it follows that the representation $T$ of the thread quiver $J_{D_4^t \overline{E}_7}$ built as in Construction 5.2.2, is additively set realizable.*

$$
J_{D_4^t \overline{E}_7}: \quad
\begin{tikzcd}
 & & 4 \arrow[d] & & & & \\
1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \arrow[r] & 7 \sim 3' & \\
 & & & & 1' \arrow[r, "{[0,1]_\alpha}"'] & 2' \arrow[u, "{[0,1]_\gamma}"'] & 4' \arrow[l, "{[0,1]_\beta}"']
\end{tikzcd}
$$

# Further directions

## Further directions

In this thesis, we analyzed the additive image of the free functor from set valued to vector space valued representations, focusing on posets, Dynkin quivers, and establishing some general results applicable to all small categories. Across various chapters, we developed new constructions, and the results outline numerous opportunities for further study. Moreover, the fourth chapter naturally leads to the conjecture that for Dynkin quivers, the orientations for which the free functor from set representations to vector space representations is additively surjective are independent on the base field $\mathbb{F}$.

## About new constructions

In Chapter 2, we analyzed thread quivers and proved that, under certain conditions, indecomposable representations of thread quivers $X$ are *well described* by a subcategory $X'$ with a simpler structure. We also proved some results that rely on this property. Specifically, we demonstrated that a vector space representation $R$ of $X$ well described by $X'$, is indecomposable or additively set realizable if and only if its restriction $R|_{X'}$ has the same property. A further direction is to study whether the same preservation extends to other properties, such as being an exceptional module, which means that $\mathrm{Ext}^1(R,R) = 0$. The thread quivers are a very recent structure, introduced in October 2024 by Charles Paquette, Job Daisie Rock and Emine Yildirim in [2]. They focused heavily on hereditary categories and a significant research direction that they left is proving that $\mathrm{Rep}(X, \mathrm{vect}_{\mathbb{F}})$ is a hereditary category for each thread quiver $X$, meaning that $\mathrm{Ext}^2(R,S) = 0$ for each pair of objects in in the category of pointwise finite-dimensional representations of $X$.

In Chapter 5, we considered rooted tree quivers, which are posets with a unique terminal object. In this setting, we presented the geometric interpretation of the additively set realizable representations developed by Riju Bindua, Thomas Brüstle and Luis Scoccola in [17]. A natural extension would be to study whether similar geometric tools can be developed for other quivers, like quivers with cycles or with more than one terminal object.

In the same chapter, we also introduced the constructions of *joined categories* and the associated *joined representations* for small categories endowed with terminal objects. We observed that such representations preserve the additive set realizability. Here, an interesting direction would be to build a similar construction for general small categories (not possessing a terminal category) and determine conditions for such joined representation to preserve the additive set realizability. However, a potential obstacle for a more general construction lies in the fact that, in general, there is no canonical vertex, such as a terminal object, that can play the same pivotal role. This difficulty arises because, in our case, the construction relied on the fact that for any indecomposable additively set realizable representation $R$ on a category with a terminal object $t$, we have $\dim(R_t) \leq 1$.

## Dynkin quivers conjecture

The following conjecture arises from the observation that, for all Dynkin quivers for which there is a classification of the orientations for which free is additive surjective, this classification is independent of the choice of the base field $\mathbb{F}$.

In the case of $A_n$-quivers, this independence is true as all indecomposable representations are indicator representations and therefore are additively set realizable. For $D_n$-quivers, it is known (see [20, Proposition 8.3]) that free is additive surjective if and only if not all arms contain an ingoing arrow- a property again independent on the base field. In the case of $E_6$, we proved this in the present thesis (see Theorem 4.1.1). However, for $E_7$ and $E_8$ the question remains open and only answered for the field $\mathbb{F}_2$ in [20, Section 8.1.3].

**Conjecture.** For every Dynkin quiver $Q$, the set of orientations for which the free functor

$$\text{free}\colon \operatorname{Rep}(Q, \text{set}) \longrightarrow \operatorname{Rep}(Q, \text{vect}_{\mathbb{F}})$$

is additively surjective does not depend on the base field $\mathbb{F}$.

The natural continuation of Chapter 4 aims to prove the conjecture above. The idea for $E_7$ and $E_8$, is to extend the classifications present in [20] for the fields $\mathbb{F}_2$ to arbitrary fields as we did in this thesis for $E_6$. It is natural to attempt to prove it using the same strategy adopted in this work for $E_6$:

- prove the direction "if free is additive surjective, then the quivers must have specific orientations" using the bound for the dimension of $R_t$ (where $t$ is a terminal object) and using some edge contractions;
- prove the other implication reducing all cases for which we want to prove that free is additive surjective to a few orientations using that such property is preserved when reversing arrows from a source of at most degree 2, and employing symmetry arguments;
- demonstrate that a significant number of representations under analysis are additive set realizable by known classifications;
- for the largest roots, verify through computational experiments (adapting the code used here for $E_6$) that the correspondent representation can be written with a multiplicative basis, which is the same for all fields.

However, this last step may fail for $E_8$. Indeed, Ringel in [3, Section 4: Dynkin quivers] proved that for Dynkin quivers, an indecomposable representation has a thin vertex, i.e., a vertex $x$ such that $\dim(R_x) = 1$, if and only if the representation is a **radiation module**. Such modules constitutes a generalization of the rooted tree module. Similar to these, they possess a corresponding graphical interpretation (called the **radiation quiver**). Because this graphical interpretation does not depend on the field, one may complete the proof. The down side, however, is that, looking at the positive roots of $E_7$ and $E_8$ the largest root of $E_8$ does not contain a coordinate equal to 1. This means that there is a representation that does not have a thin vertex. Hence, this representation is not a radiation module. As a consequence, there may not exist a standard family of matrices that works uniformly over all fields, and the additive set realizability might terefore depend on the choice of the field.

To conclude, this thesis has delineated fundamental structural results for multiple categories and, in doing so, has charted new and promising research avenues, exploring the potential extensions and boundaries of the methodologies employed.

# Appendix

## List of $E_6$ orientations

In this appendix, we list the 32 possible orientations of the quiver $E_6$, indicating for each whether the corresponding free functor is additive surjective. The orientations are labeled according to the enumeration used in the proof of Theorem 4.1.1 and according to the order adopted in [10].

**1: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[u] \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd}$$

**2: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[u] \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

**3: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[u] & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**4: Not additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[u] & 5 \arrow[l] & 6 \arrow[l] \end{tikzcd}$$

**5: Additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd}$$

**6: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

**7: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**8: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 \arrow[r] & 3 & 5 \arrow[l] & 6 \arrow[l] \end{tikzcd}$$

**9: Additively surjcetive**

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] \arrow[u] \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd}$$

**10: Additively surjcetive**

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] \arrow[u] \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

**11: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] \arrow[u] & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**12: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] \arrow[u] & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**13: Additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd}$$

**14: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

**15: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**16: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 \arrow[r] & 2 & 3 \arrow[l] & 5 \arrow[l] & 6 \arrow[l] \end{tikzcd}$$

**17: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 & 2 \arrow[l] \arrow[r] & 3 \arrow[u] \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd}$$

**18: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 & 2 \arrow[l] \arrow[r] & 3 \arrow[u] \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

**19: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 & 2 \arrow[l] \arrow[r] & 3 \arrow[u] & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**20: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 & 2 \arrow[l] \arrow[r] & 3 \arrow[u] & 5 \arrow[l] & 6 \arrow[l] \end{tikzcd}$$

**21: Additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 & 2 \arrow[l] \arrow[r] & 3 \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd}$$

**22: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 & 2 \arrow[l] \arrow[r] & 3 \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

**23: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 & 2 \arrow[l] \arrow[r] & 3 & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**24: Not additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 & 2 \arrow[l] \arrow[r] & 3 & 5 \arrow[l] & 6 \arrow[l] \end{tikzcd}$$

**25: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 & 2 \arrow[l] & 3 \arrow[l] \arrow[u] \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd}$$

**26: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 & 2 \arrow[l] & 3 \arrow[l] \arrow[u] \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

**27: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 & 2 \arrow[l] & 3 \arrow[l] \arrow[u] & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**28: Additively surjective**

$$\begin{tikzcd} & & 4 & & \\ 1 & 2 \arrow[l] & 3 \arrow[l] \arrow[u] & 5 \arrow[l] & 6 \arrow[l] \end{tikzcd}$$

**29: Additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 & 2 \arrow[l] & 3 \arrow[l] \arrow[r] & 5 \arrow[r] & 6 \end{tikzcd}$$

**30: Additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 & 2 \arrow[l] & 3 \arrow[l] \arrow[r] & 5 & 6 \arrow[l] \end{tikzcd}$$

**31: Additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 & 2 \arrow[l] & 3 \arrow[l] & 5 \arrow[l] \arrow[r] & 6 \end{tikzcd}$$

**32: Additively surjective**

$$\begin{tikzcd} & & 4 \arrow[d] & & \\ 1 & 2 \arrow[l] & 3 \arrow[l] & 5 \arrow[l] & 6 \arrow[l] \end{tikzcd}$$

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